

KENYATTA UNIVERSITY
INSTITUTE OF OPEN LEARNING

SMA 335
ORDINARY DIFFERENTIAL EQUATIONS I

BY
D. SENGOTTAIYAN

DEPARTMENT OF MATHEMATICS

PREFACE

This module is designed primarily to provide the readers with the best preparation possible for the ordinary Differential Equations (SMA 335 ordinary Differential Equations I)

In its present form this module has developed from course given by the author over the last **thirty two** years in various universities to the audience of Mathematicians, Physicists and Engineers in the university of **Madras, Kenyatta University, University of Nairobi and Jomo Kenyatta University of Agriculture and Technology.**

This module, Ordinary Differential Equations I is compiled from the Author's Advanced Differential Equations, Oxford Publications, London and Nairobi. Most of the theory and problems are freely taken from the Author's Book for which the author has sole Copyright.

It is hoped that it will be of great interest to students of pure and Applied Mathematics following Ordinary Differential Equations I.

Each Chapter begins with a brief statement of definitions principles and important Theorems followed by a set of solved problems. Attention has been given to the Chapters on Applications of the theory.

The author is pleased to acknowledge Dr L. O. Odongo who worked through the entire manuscript and checked all the problems in each and every Chapter.

SENGOTTAIYAN

Kenyatta University

LESSON 1

Differential Equations and Their Solutions

1.1 Introduction

The subject of differential equations constitutes a very important and useful branch of modern mathematics. In this lesson we shall consider some definition of ordinary differential equations.

1.2 Objectives of the lesson

By the end of this lesson you should be able to.

- define a differential equation
- define the order and the degree of a differential equation.
- form the differential equation and solve simple differential equations.

1.3 Definition of ordinary differential equation

An equation containing any derivatives such as $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$, is called an ordinary differential equation.

Examples of differential equations

1. $\frac{dy}{dx} = 5x$
2. $\frac{dy}{dx} + 3x + 5 = 0$
3. $\frac{d^2y}{dx^2} + 8 \frac{dy}{dx} + 9y = 0$
4. $\frac{d^2y}{dx^2} + y = \sin x$

are ordinary differential equations.

1.4 Definition of Partial Differential Equations

A differential equation involving partial derivatives of a dependent variable, with respect to more than one independent variable, is called a partial differential equation.

For example,

$$\frac{\partial}{\partial x}(2x^2 + 5y^2 + 3xy^2) = 0$$

$$\frac{\partial}{\partial x}(\sin x + 5y \cos x + x^2y^3) = 0$$

$$\frac{\partial^2}{\partial y^2}(x^3 + 3x^2y + y^2) = 0 \text{ are called partial differential equations.}$$

This means when you differentiate a term with respect to x, y is treated as constant similarly x is treated as a constant when we differentiate a term with respect to y.

Then $\frac{\partial}{\partial x}(3y^2x^3) = 3y^2 \frac{\partial}{\partial x}x^3 = 9x^2y^2$

$\frac{\partial}{\partial y}(7y^2 \sin x) = 7 \sin x \frac{\partial}{\partial y}y^2 = 14y \sin x$

1.5 Meaning of $\frac{d^2y}{dx^2}$, $\frac{d^3y}{dx^3}$ $\frac{d^ny}{dx^n}$

$\frac{d^2y}{dx^2}$ means that we differentiate $\frac{dy}{dx}$ again with respect to x.

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left[\frac{dy}{dx} \right]$$

$$\frac{d^3y}{dx^3} = \frac{d}{dx} \left[\frac{d^2y}{dx^2} \right]$$

similarly $\frac{d^ny}{dx^n}$ is defined.

1.6 The order of a differential equation

The order of a differential equation is the order of the highest differential coefficient present in the equation.

For example the equation,

$$7 \frac{d^3y}{dx^3} + 8 \frac{d^2y}{dx^2} + 9 \frac{dy}{dx} + 8y = e^x$$

contains, $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$, $\frac{d^3y}{dx^3}$ the highest differential coefficient is $\frac{d^3y}{dx^3}$. Hence the order of the equation is three.

Consider $3 \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^{10} + 8y = 0$

The highest differential coefficient present in this equation is $\frac{d^2y}{dx^2}$. Hence the order of the equation is two.

1.7 The degree of a differential equation

The degree of a differential equation is the degree (or power) of the highest differential coefficient present, when the differential coefficient are free from radicals and fractional exponents.

For example consider the equation,

$$\frac{d^3 y}{dx^3} + \left(\frac{d^2 y}{dx^2} \right)^{10} + 3 \left(\frac{dy}{dx} \right)^7 + 8y = 0$$

The highest order present in this equation is $\frac{d^3 y}{dx^3}$. The degree of this $\frac{d^3 y}{dx^3}$ is one. Hence the degree of the differential equation is just one.

Example 1

Consider,

$$\left(\frac{d^3 y}{dx^3} \right)^5 + \left(\frac{d^2 y}{dx^2} \right)^{11} + \left(\frac{dy}{dx} \right)^{20} + y = x$$

Here the highest differential coefficient is $\frac{d^3 y}{dx^3}$ or the order of the equation is 3. The degree of this highest differential coefficient is five and hence the degree of this equation is five.

Example 2

Consider, $\left(\frac{d^2 y}{dx^2} \right)^{\frac{3}{2}} + 5 \frac{dy}{dx} + 3y = 0$

The highest differential coefficient present here is $\frac{d^2 y}{dx^2}$. First it should be free from radicals and fractional exponents.

$$\left(\frac{d^2 y}{dx^2} \right)^{\frac{3}{2}} = -5 \frac{dy}{dx} - 3y$$

$$\left[\frac{d^2 y}{dx^2} \right]^3 = \left[-5 \frac{dy}{dx} - 3y \right]^2$$

Then the degree of the equation is three.

Consider,

$$\sqrt{3 + \left(\frac{dy}{dx} \right)^2} = 2x$$

Removing the radical it is written as $3 + \left(\frac{dy}{dx} \right)^2 = 4x^2$.

Hence the degree of the equation is two.

1.8 The solution of a differential equation

The solution of a differential equation is an equation between x and y , which will satisfy the differential equation.

Example 3

- a) Show that $y = x^2$ is a solution of the equation, $3 \frac{d^2y}{dx^2} + 5 \frac{dy}{dx} = 10x + 6$
 b) Check whether $y = x^2 + c$ where c is any constant is also a solution of the equation.

Solution

$$\text{Let } y = x^2 \quad (1)$$

$$\frac{dy}{dx} = 2x \quad (2)$$

$$\frac{d^2y}{dx^2} = 2 \quad (3)$$

substituting (1), (2) and (3) in the given equation
 $3(2) + 5(2x) = 6 + 10x$ thus the equation is satisfied

1.9 General and particular solution of a differential equation.

$$\text{Consider } \frac{dy}{dx} = 2 \quad (1)$$

$$\text{Then } dy = 2dx$$

$$\text{Integrating, } \int dy = \int 2dx$$

$$y = 2x + c \text{ where } c \text{ is any arbitrary constant}$$

we say that $y = 2x + c$ is the general solution of the equation (1)

Geometrically, $y = 2x + c$ is a family of straight lines parallel to $y = 2x$

The solution $y = 2x + c$, represents a family of straight lines. This family $y = 2x + c$ is called **the general solution** of the equation $\frac{dy}{dx} = 2$.

When c takes a particular value (say $c = 3$) we get a particular member of the family, $y = 2x + 3$. It is a particular solution.

A particular solution is one where a value is given to c .

In the above example, the order of the equation $\frac{dy}{dx} = 2$, is one. Hence the general solution of this equation contains one arbitrary constant c .

In general, the solution of an n^{th} order differential equation will have n arbitrary constants. For example the general solution of a second order equation will have **two** arbitrary constants. When we give particular values for these constants we get particular solutions of the equation.

Example 4

Consider the equation $\frac{dy}{dx} = 3x^2$

$$dy = 3x^2 dx$$

integrating both sides,

$$\int dy = 3 \int x^2 dx$$

$$y = x^3 + c$$

The general solution of the equation is a family of curves. When c takes the value, say $c = 2$, $y = x^3 + 2$ is a particular curve of the family and it is called a particular solution of the equation

1.10 Forming Differential Equations

Differential Equations are often formed by expressing certain scientific laws or relationships in mathematical terms. We can also form them by eliminating the constants in an equation. If n constants are eliminated we will get an n^{th} order equations.

Example 5

The scientific law states that the acceleration of a particle during upward motion under gravity is $-g$. Write down this law as a differential equation.

Solution

Velocity of a particle is $\frac{dx}{dt}$

Acceleration of the particle is $\frac{d^2x}{dt^2}$ where x is distance traveled by the particle at time t from a fixed origin.

Then $\frac{d^2x}{dt^2} = -g$ is the required differential equation.

Can you account for the minus sign on the right hand side?

Example 6

Form the differential equation from

$y = cx^2 + \sin x$ by eliminating the constant c .

Solution

The expression

$$y = cx^2 + \sin x$$

contains only one constant c . Then we can get differential equation of first order by differentiating the expression once and eliminating the constant c .

$$\text{Let } y = cx^2 + \sin x \quad (1)$$

$$\frac{dy}{dx} = 2cx + \cos x \quad (2)$$

To eliminate one constant we need two equations (In general to eliminate n constants we need $n + 1$ equations).

From (1) we can find c and the value of c is substituted in (2)

From (1), $cx^2 = y - \sin x$

$$c = \frac{y - \sin x}{x^2}$$

putting this value of c in (2)

$$\frac{dy}{dx} = 2 \left[\frac{y - \sin x}{x^2} \right] x + \cos x$$

$\frac{dy}{dx} = \frac{2}{x}(y - \sin x) + \cos x$ is the required differential equation whose general solution is,

$$y = cx^2 + \sin x$$

Example 7

Find the differential equation whose solution is $y = A \cos 8x + B \sin 8x$.

Solution

To eliminate two constants A and B we need three equations. In addition to the given equation we need two more. These two equations are obtained by differentiating the given expression twice.

$$\text{Let } y = A \cos 8x + B \sin 8x \quad (1)$$

$$y' = \frac{dy}{dx} = -8A \sin 8x + 8B \cos 8x \quad (2)$$

$$y'' = \frac{d^2y}{dx^2} = -64A \cos 8x - 64B \sin 8x \quad (3)$$

$$\frac{d^2y}{dx^2} = -64(A \cos 8x + B \sin 8x)$$

$$\frac{d^2y}{dx^2} = -64y \text{ using (1) in (3)}$$

This is the required differential equation.

1.11 Solving simple differential equations

We are going to see several methods to solve a differential equation or finding solutions of differential equations. Before that we shall see here the method of finding solutions of some simple equations using integration.

Example 8

Solve the equation, $\frac{dy}{dx} = \sin x$.

Solution

Now $dy = \sin x \, dx$

Integrating both sides

$$\int dy = \int \sin x dx$$

then, $y = -\cos x + c$

$y + \cos x = c$ is the required general solution of the equation.

Example 9

The differential equation of a family of circles is given by,

$$\frac{dy}{dx} + \frac{x}{y} = 0$$

- Find the equation to the family of circles.
- If one of the family passes thro the point (1, 2) find its equation.

Solution

$$\text{Let } \frac{dy}{dx} + \frac{x}{y} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y} = 0$$

$$y dy = -x dx$$

Integrating both sides,

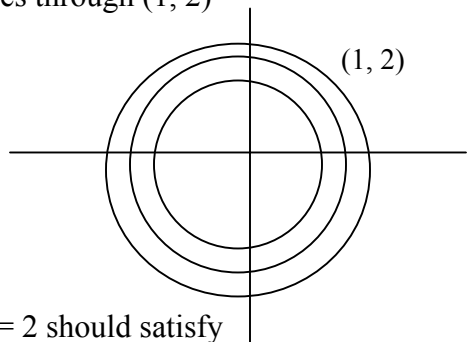
$$\int y dy = -\int x dx$$

$$\frac{y^2}{2} = \frac{-x^2}{2} + c$$

$$x^2 + y^2 = A \text{ (where } A = 2c)$$

This is the equation of all the circles of the family.

If it passes through (1, 2)



$x = 1, y = 2$ should satisfy

$$x^2 + y^2 = A$$

$$1^2 + 2^2 = A \text{ or } A = 5$$

Hence the particular solution or the equation of the particular circle is $x^2 + y^2 = 5$

Exercise 1

- Find the order and degree of the following differential equations.

$$\text{a) } \frac{d^2 y}{dx^2} + 7\left(\frac{dy}{dx}\right)^3 + 8y^2 = \sin x$$

b) $\left(\frac{d^2y}{dx^2}\right)^3 + 10\left(\frac{dy}{dx}\right)^8 + 9y = \cos x$

c) $\left(\frac{d^3y}{dx^3}\right)^{\frac{1}{2}} + \left(\frac{dy}{dx}\right)^2 + y = 0$

d) $\left(\frac{dy}{dx}\right)^3 + 4\frac{dy}{dx} + 6y = e^x$

e) $\left(\frac{dy}{dx}\right)^3 - 3\frac{d^2y}{dx^2} + 7y = e^{2x}$

f) $\frac{d^2v}{dx^2} \frac{dv}{dx} + x\left(\frac{dv}{dx}\right)^2 + v = 0$

2. Find the differential equation by eliminating the constant c in the following

a) $y = 3x^2 + 5x + c$

b) $y = \sin x + c$

c) $2y = 5x^2 + c$

d) $y = Ax$

e) $y = A \sin x$

3. Find the second order differential equation by eliminating the constants A and B .

a) $y = Ax + B$

b) $y = Ae^x + B$

4. Show that the differential equation associated with the primitive $y = Ax^2 + Bx + C$ is

$$\frac{d^3y}{dx^3} = 0$$

5. Obtain the differential equation associated with the primitive $y = Ae^{2x} + Be^x + c$

6. Show that the differential equation associated with $y = Ae^x + B$ is $\frac{d^2y}{dx^2} = \frac{dy}{dx}$.

7. Find

a) $\frac{\partial}{\partial x}(3x^2 + 5y^2 + 8xy)$

b) $\frac{\partial}{\partial y}(3x^2 + 5y^2 + 8xy)$

c) $\frac{\partial^2}{\partial x^2}(8x^3 + 9x^2 - 7y^2)$

d) $\frac{\partial^2}{\partial y^2}(8x^3 + 9x^2 - 7y^2)$

8. Find the general solution of the following equations

a) $\frac{dy}{dx} = 2x^2 + 3x + 5$

b) $\frac{dy}{dx} = 2e^x + \sin x$

c) $\frac{dy}{dx} = x^3 + 5x + 1$

9. Find the family of circles represented by $\frac{dy}{dx} + \frac{x}{y} = 0$

Find the particular circle of the family which passes through the point (2, -2).

Summary

You have learnt the following from this lesson:

1. Definitions of ordinary and particular differential equations.
2. The order and degree of a differential equation.
3. The meaning of the general and particular solutions of a differential equation.
4. Formation of differential equations by eliminating the arbitrary constants.
5. Finding the solution of simple differential equations.

Further Reading

1. Differential Equations

By Shepley L. Ross

John Wiley & sons

New York Toronto. Singapore.

2. Mathematics method for Science Students

By G. Stephenson

Addison Wesley Longman Limited

Edinburgh Gate, Harlow, England, 1973.

3. Differential equations

By Dr. D. Sengottaiyan

Oxford publications

London Nairobi.

LESSON 2

Equations of First order – Variables Separable and Homogeneous Equations

2.1. Introduction

There are various methods for solving a first order linear differential equations. The method of “**Separation of Variables**” is the most fundamental. If the differential equation is of the form,

$$\frac{dy}{dx} = F(x, y) = \frac{f(x)}{g(y)}$$

we say that the variables are separable and obtain the solution from

$$g(y)dy = f(x)dx$$

by integrating both sides

$$\text{Hence } \int g(y)dy = \int f(x)dx$$

In this method, we must manage to transform the given equation in the form

$$g(y)dy = f(x)dx$$

so that the variables $g(y)$ and dy are on one side and $f(x)$ and dx on the other side.

Not all functions $F(xy)$ are separable into the form $f(x) dx = g(y) dy$.

2.2. Objective of the lesson

By the end of this lesson you should be able to:

- i. solve the first order equation using the method of separation of variables
- ii. reduce a homogeneous equation to the variables separable form by using the substitution $y = vx$ and solve the equation.
- iii. Transform $\frac{dy}{dx} = \frac{ax + by + c}{px + qy + r}$ into a homogeneous equation of first order and solve this equation.

2.3. Method of separation of variables

As mentioned in the introduction, the first order linear equation

$$\frac{dx}{dy} = F(x, y)$$

is to be transformed in the form

$$g(y)dy = f(x)dx \tag{2}$$

Only certain problems can be split into this form (2) and in such case we integrate each side to obtain the general solution. We put one constant C at the end.

Example 1

Solve the equation $\frac{dy}{dx} = \frac{2x^2}{y^3}$

Solution

We can transform the equation.

$$\frac{dy}{dx} = \frac{2x^2}{y^3} \text{ into the form}$$

$$y^3 dy = 2x^2 dx$$

$$\text{Then } \int y^3 dy = \int 2x^2 dx$$

$$\frac{y^4}{4} = \frac{2x^3}{3} + C$$

$$\text{or } \frac{y^4}{4} = \frac{2x^3}{3} + C$$

$$\text{or } 3y^4 = 8x^3 + 3A, \text{ is the required solution}$$

Example 2

$$\text{Solve } \frac{dy}{dx} = \frac{xy^2 + x}{yx^2 + y}$$

Solution

$$\text{Let } \frac{dy}{dx} = \frac{xy^2 + x}{yx^2 + y}$$

We shall try to separate f(y) and dy on one side and f(x) and d(x) on the other side.

$$\frac{dy}{dx} = \frac{x(1+y^2)}{y(1+x^2)}$$

$$ydy = \frac{x(1+y^2)}{(1+x^2)} dx$$

$$\frac{ydy}{1+y^2} = \frac{x}{1+x^2} dx$$

$$\int \frac{ydy}{1+y^2} = \int \frac{xdx}{1+x^2} \quad (2)$$

we can make the integrals in the form of $\int \frac{u'}{u} dx$ whose integration is $\ln u$. (2) becomes

$$\frac{1}{2} \int \frac{ydy}{1+y^2} = \frac{1}{2} \int \frac{xdx}{1+x^2}$$

$$\frac{1}{2} \ln(1+y^2) = \frac{1}{2} \ln(1+x^2) + C$$

$$\ln(1+y^2) = \ln(1+x^2) + C_1, \quad (C_1 = 2C)$$

Whenever $\ln$ appears after integration we always call the constant as $\ln A$ and put A inside the logarithm to reduce the number of terms.

$$\ln(1+y^2) = \ln A(1+x^2)$$

Raising both sides to the base e,

$$e^{\ln(1+y^2)} = e^{\ln A(1+x^2)}$$

Then $(1 + y^2) = A(1 + x^2)$ is the required solution.

2.4. Solving homogeneous differential equation of first order.

First we shall define a homogeneous equation of first order and then we shall see that the homogeneous equation can be solved using the method of separation of variables when we use the substitution $y = vx$

2.5. Definition of homogeneous equation of first order

Consider $\frac{dy}{dx} = f(x, y)$

Suppose we put $x = kx$ and $y = ky$. If $f(x, y)$ becomes $k^n f(x, y)$ then $f(x, y)$ is called a homogeneous function in x and y . In this section we consider homogeneous equation of order 0 so that $k^n = k^0 = 1$. In other words we consider $f(kx, ky) = f(x, y)$ as the particular homogeneous equation.

For example

$$\begin{aligned} f(x, y) &= \frac{xy}{x^2 + y^2} \\ f(kx, ky) &= \frac{kx.ky}{k^2x^2 + k^2y^2} \\ &= \frac{k^2}{k^2} \left(\frac{xy}{x^2 + y^2} \right) \\ &= \frac{xy}{x^2 + y^2} \\ &= f(x, y) \end{aligned}$$

Here $f(kx, ky) = k^0 \frac{xy}{x^2 + y^2} = f(x, y)$. Hence $\frac{xy}{x^2 + y^2}$ is a homogeneous function of degree 0.

2.6. Method of solving $\frac{dy}{dx} = f(x, y)$ when $f(x, y)$ is a homogeneous function of x

and y

$$\text{Let } \frac{dy}{dx} = f(x, y)$$

Put the substitution $y = vx$

$$\text{So that } \frac{dy}{dx} = v + x \frac{dv}{dx}.$$

As we shall see in the following examples the given homogeneous function will become

$$v + x \frac{dv}{dx} = f(v)$$

$$x \frac{dv}{dx} = f(v) - v$$

$$\frac{x}{dx} = \frac{f(v) - v}{dv}$$

or
$$\frac{dv}{f(v) - v} = \frac{dx}{x}$$

Then
$$\int \frac{dv}{f(v) - v} = \int \frac{dx}{x}$$

Thus the variables v and x are separated and the equation is solved.

Example 3

a. Show that $\frac{xy}{x^2 + y^2}$ is a homogeneous function in x and y .

b. Using the substitution $y = vx$ transform the equation,

$$\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$$

into an equation containing v and x only.

c. Hence solve the resulting equation using the method of separation of

variables. $\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$

Solution

a. Let
$$f(x, y) = \frac{xy}{x^2 + y^2}$$

$$\begin{aligned} f(kx, ky) &= \frac{k^2(xy)}{k^2(x^2 + y^2)} \\ &= f(x, y) \end{aligned}$$

Hence $\frac{xy}{x^2 + y^2}$ is a homogeneous equation.

b. If we let $y = vx$, then $\frac{dy}{dx} = v + x \frac{dv}{dx}$.

Substitution of this in the given equation, we get

$$v + x \frac{dv}{dx} = \frac{xy}{x^2 + y^2} \text{ becomes}$$

$$v + x \frac{dv}{dx} = \frac{x \cdot xv}{x^2 + v^2 x^2}$$

$$v + x \frac{dv}{dx} = \frac{v}{1 + v^2}$$

$$x \frac{dv}{dx} = \frac{v}{1 + v^2} - v$$

$$x \frac{dv}{dx} = \frac{v - v(1 + v^2)}{1 + v^2}$$

$$= \frac{v - v - v^3}{1 + v^2}$$

$$x \frac{dv}{dx} = \frac{-v^3}{1 + v^2}$$

$$\frac{x}{dx} = \frac{-v^3}{(1 + v^2)dv}$$

$$\frac{dx}{x} = -\frac{(1 + v^2)dv}{v^3} \quad (\text{variables are separated})$$

$$\int \frac{dx}{x} = -\int \left[\frac{1}{v^3} + \frac{v^2}{v^3} \right] dv$$

$$\ln A = -\int \left[v^{-3} + \frac{1}{v} \right] dv$$

$$= \frac{v^{-2}}{2} - \ln v$$

$$\ln Ax - \ln v = \frac{1}{2v^2}$$

$$\ln \frac{Ax}{v} = \frac{1}{2v^2} \quad \text{where } v = \frac{y}{x}$$

$$\ln \left(\frac{Ax^2}{y} \right) = \frac{x^2}{2y^2}$$

This is the required solution.

Example 4

Solve $(x^3 + y^3)dx - 3xy^2 dy = 0$

Solution

$$\text{Let } (x^3 + y^3)dx - 3xy^2 dy = 0$$

$$(x^3 + y^3)dx = 3xy^2 dy$$

$$\frac{dy}{dx} = \frac{x^3 + y^3}{3xy^2},$$

$\frac{x^3 + y^3}{3xy^2}$ is a homogeneous function of degree 0.

Let $y = vx$ be the substitution.

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

then

$$v + x \frac{dv}{dx} = \frac{x^2 + v^3 x^3}{3x \cdot v^2 x^2}$$

$$v + x \frac{dv}{dx} = \frac{1 + v^3}{3v^2}$$

$$x \frac{dv}{dx} = \frac{1 + v^3}{3v^2} - v$$

$$= \frac{1 + v^3 - v(3v^2)}{3v^2}$$

$$\frac{x}{dx} = \frac{1 - 2v^3}{3v^2 dv}$$

therefore

$$\frac{dx}{x} = \frac{3v^2 dv}{1 - 2v^3}$$

$$\int \frac{dx}{x} = \int \frac{3v^2 dv}{1 - 2v^3}$$

$$\ln(Ax) = \frac{1}{-2} \int \frac{-6v^2 dv}{1 - 2v^3}$$

$$\ln(Ax) = -\frac{1}{2} \ln(1 - 2v^3) \quad (\text{using the substitution } u = 1 - 2v^3)$$

$$\ln(Ax) = \ln(1 - 2v^3)^{-\frac{1}{2}}$$

$$Ax = (1 - 2v^3)^{-\frac{1}{2}} \quad \text{where } v = \frac{y}{x}$$

$$Ax = \left(1 - \frac{2y^3}{x^3}\right)^{-\frac{1}{2}}$$

$$A^2 x^2 = \left(1 - \frac{2y^3}{x^3}\right)^{-1}$$

$$A^2 x^2 = \frac{x^3}{x^3 - 2y^3}$$

$$A^2 = \frac{x}{x^3 - 2y^3}$$

or

$$x = c(x^3 - 2y^3)$$

is the required solution.

Example 5

Solve the equation $(x^2 - 3y^2)dx + 2xydy = 0$

Solution

Let $(x^2 - 3y^2)dx + 2xydy = 0$

or
$$\frac{dy}{dx} = -\frac{(x^2 - 3y^2)}{2xy}$$

$$\frac{dy}{dx} = \frac{3y^2 - x^2}{2xy}$$

if
$$f(x, y) = \frac{3y^2 - x^2}{2xy}$$

$$\begin{aligned} f(kx, ky) &= \frac{3k^2y^2 - k^2x^2}{2kxky} \\ &= \frac{k^2(3y^2 - x^2)}{k^2(2xy)} \\ &= f(x, y) \end{aligned}$$

Hence the given equation is homogeneous.

Letting $y = vx$, so that $\frac{dy}{dx} = v + x \frac{dv}{dx}$, the given equation becomes

$$\begin{aligned} v + x \frac{dv}{dx} &= \frac{3v^2y^2 - x^2}{2xvx} \\ &= \frac{3v^2 - 1}{2v} \end{aligned}$$

$$x \frac{dv}{dx} = \frac{3v^2 - 1}{2v} - v$$

$$x \frac{dv}{dx} = \frac{v^2 - 1}{2v}$$

$$\frac{dx}{x} = \frac{2v}{v^2 - 1} dv$$

Integrating, we find $\ln(v^2 - 1) = \ln|cx|$

Hence $v^2 - 1 = cx$

$$\frac{y^2}{x^2} - 1 = cx$$

$$y^2 - x^2 = cx^3$$

which is the required solution for the given equation.

2.7. First order equation reducible to homogeneous form.

Equation of the form

$$\frac{dy}{dx} = \frac{ax + by + c}{px + qy + r} \quad (1)$$

$$\text{or } (ax + by + c)dx + (px + qy + r)dy = 0 \quad (2)$$

where a, b, c, p, q, r are constants.

can be transformed into homogeneous equations of first order by a suitable transformation and solved.

Case 1. If $\frac{a}{p} \neq \frac{b}{q}$ then the transformation

$$x = X + h$$

$$y = Y + k$$

where (h, k) is the solution of the system

$$ah + bk + c = 0$$

$$ph + qk + r = 0$$

reduces the equation to the homogeneous equation

$$(aX + bY)dx + (pX + qY)dy = 0 \text{ in the new variables } X \text{ and } Y$$

Case II. If $\frac{a}{p} = \frac{b}{q}$ then the transformation

$z = ax + by$ reduces the equation to a **separable equation** in the variables x and z.

Example 6

Solve the equation

$$(3x + 2y - 5)dx + (2x + 3y - 5)dy = 0$$

Solution

$$\begin{array}{ll} \text{Taking} & x = X + h \quad dx = dX \\ & y = Y + k \quad dy = dY \end{array} \quad (1)$$

$$\frac{dy}{dx} = \frac{3x + 2y - 5}{2x + 3y - 5} \text{ becomes}$$

$$\frac{dY}{dX} = -\frac{3X + 2Y + 3h + 2k - 5}{2X + 3Y + 2h + 3k - 5}$$

choose h, k such that

$$\left. \begin{array}{l} 3h + 2k - 5 = 0 \\ 2h + 3k - 5 = 0 \end{array} \right] \quad (2)$$

solving the simultaneous equations (2) we have h = 1 and k = 1

$$\text{Hence} \quad \frac{dY}{dX} = -\frac{3X + 2Y}{2X + Y}$$

$$\text{Putting} \quad Y = vX$$

$$\frac{dY}{dX} = v + X \frac{dv}{dX}$$

$$X \frac{dv}{dX} = -\frac{3+2v}{2+3v} - v$$

or

$$= -\frac{3+4v+3v^2}{3v+2}$$

or

$$-\frac{dX}{X} = -\frac{(3v+2)dv}{3v^2+4v+3}$$

Integrating

$$\ln(3v^2+4v+3) = -2\ln X + \ln C$$

or

$$3v^2+4v+3 = \frac{C}{X^2}$$

$$\frac{3Y^2}{X^2} + \frac{4Y}{X} + 3 = \frac{C}{X^2}$$

or

$$3Y^2 + 4XY + 3X^2 = C$$

or

$$3(y-1)^2 + 4(x-1)(y-1) + 3(x-1)^2 = c$$

or

$$3(x^2+y^2) + 4xy - 10(x+y) = c$$

Example 7

Solve $\frac{dy}{dx} = -\frac{x-y+3}{2x-2y+5}$

Solution

Here Ratio of coefficients of $x = \frac{1}{2}$ and ratio of coefficients of $y = \frac{-1}{-2} = \frac{1}{2}$. Hence we

put

$$x - y = z.$$

Then $\frac{dx}{dx} - \frac{dy}{dx} = \frac{dz}{dx}$

Or $1 - \frac{dy}{dx} = \frac{dz}{dx}$ or $\frac{dy}{dx} = 1 - \frac{dz}{dx}$

Then

$$1 - \frac{dz}{dx} = \frac{z-3}{2z+5}$$

or

$$\begin{aligned} \frac{dz}{dx} &= 1 - \frac{z-3}{2z+5} \\ &= \frac{2z+5-z-3}{2z+5} \end{aligned}$$

$$\frac{dz}{dx} = 1 - \frac{z+2}{2z+5}$$

or
$$dx = 1 - \frac{2z+5}{z+2}$$

$$= \frac{2z+4+1}{z+2}$$

$$dx = \left(2 + \frac{1}{z+2}\right)dv$$

Integrating both sides we have

or $x + c = 2z + \ln(z+2)$
or $x + c = 2(x-y) + \ln(x-y+2)$

or the required solution is
 $2y - x + c = \ln(x-y+2)$

Example 8

Solve the equation

$$(x - 2y + 1)dx + (4x - 3y - 6)dy = 0$$

Solution

Let
$$\frac{dy}{dx} = \frac{x-2y+1}{4x-3y-6} \quad (1)$$

Here the ratio of the coefficients of x is given by $\frac{1}{4}$ and $\frac{-2}{-3}$ and $\frac{1}{4} \neq \frac{2}{3}$

Therefore we use the transformation

$$\left. \begin{array}{l} x = X + h \\ y = Y + k \end{array} \right\} \text{ where h and k are to be determined from the equation}$$

$$h - 2k + 1 = 0 \quad (2)$$

$$4h - 3k - 6 = 0 \quad (3)$$

Solving the simultaneous equation (2) and (3) we have $h = 3$ and $k = 2$

Hence
$$\begin{array}{ll} x = X + 3 & dx = dX \\ y = Y + 2 & dy = dY \end{array}$$

Using this substitution, equation (1) reduces to the homogeneous equation of order zero as:

$$\frac{dY}{dX} = \frac{(X-3)-2(Y-2)+1}{4(X+3)-3(Y-2)-6} = \frac{X-2Y}{4X-3Y} \quad (4)$$

we use the substitution $Y = vX$ for the first order homogeneous equation

$Y = vX$ gives

$$\begin{aligned}
v + X \frac{dV}{dX} &= \frac{1-2v}{3v-4} \\
\text{or } X \frac{dV}{dX} &= \frac{1-2v}{3v-4} - v \\
\text{or } X \frac{dV}{dX} &= \frac{1-2v-3v^2+4v}{3v-4} \\
\text{or } X \frac{dV}{dX} &= \frac{-3v^2+2v+1}{3v-4} \\
\text{or } \frac{X}{dX} &= \frac{3v^2+2v-1}{-(3v-4)dv} \\
\text{or } -\frac{dX}{X} &= \frac{(3v-4)dv}{3v^2+2v-1}
\end{aligned}$$

Integrating both sides we have

$$\begin{aligned}
X - \ln(AX) &= \frac{(3v-4)dv}{3v^2+2v-1} \\
-\ln(AX) &= \frac{1}{2} \ln(3v^2-2v-1) - \frac{3}{4} \ln\left(\frac{3v-3}{3v+1}\right) \\
&= \ln(3v^2-2v-1)^{\frac{1}{2}} - \ln\left(\frac{3v-3}{3v+1}\right)^{\frac{3}{4}} \\
\ln(AX) &= \ln\left[\left(\frac{3v-3}{3v+1}\right)^{\frac{3}{4}} \div (3v^2-2v-1)^{\frac{1}{2}}\right]
\end{aligned}$$

$$\text{or finally } \ln\left[\frac{(3v+1)^5}{v-1}\right] = \ln \frac{c_1^4}{x^4}$$

$$\text{or } X^4 |(3v+1)^5| = c(v-1)$$

Replacing v by $\frac{y}{x}$ we have

$$|(3Y+X)^5| = C|(Y-X)|$$

Finally replacing X by x - 3 and Y by y - 3, we obtain the solution of the original equation in the form

$$|3(y+2)+(x-3)|^5 = c|(y-2-x+3)|$$

$$\text{or } |x+3y-9|^5 = C(y-x+1)$$

Exercise 2

Solve the following equations

1. $x^3 dx + (y-1)^2 dy = 0$
2. $(2x+3y)dx + (y-x)dy = 0$
3. $xydx + (1+x^2)dy = 0$
4. $y^2 dx - x^2 dy = 0$

Solve the following homogeneous equations.

5. $2xy \frac{dy}{dx} = x^2 + y^2$
6. $x^2 y dx - (x^3 + y^3) dy = 0$
7. $(x^2 - y^2) \frac{dy}{dx} = 2xy$
8. $xy \frac{dy}{dx} = x^2 + y^2$
9. $x \frac{dy}{dx} = x - y$
10. $xy \frac{dy}{dx} = 3x^2 + xy$
11. $(x^2 + y^2) \frac{dy}{dx} = xy$
12. $y \frac{dy}{dx} = 2x + y$

Solve the following equations reducing them to a homogeneous equation:

13. $\frac{dy}{dx} = \frac{x-y+2}{x+y}$
14. $\frac{dy}{dx} = \frac{2x+y-2}{2x+y+1}$
15. $(x+y) \frac{dy}{dx} = x+y-2$
16. $\frac{dy}{dx} = \frac{2x+3y-8}{5x+y-7}$

Summary

You have learnt the following from this lesson:

1. Solving a first order equation by the method of separation of variables.
2. Putting $y = vx$ will transform a homogeneous equation into an equation with separation of variables.
3. We can transform $\frac{dy}{dx} = \frac{ax+by-c}{px+qy-r}$ into a homogeneous equation of first order and solve the equation.

Further Reading:

1. Pure Mathematics
By Backhouse and others
Longman
2. Differential Equations
By Shepley L. Ross
John Wiley and Sons
New yolk. Toronto. Singapore
3. Frank Ayres, JR, Schaum's outline series
Mc Graw-Hill Book Company
New York. Toronto. Sydney.

LESSON 3

First Order Exact Equations and Integrating Factors

In this lesson you will learn to solve a particular type of first order equation called an **exact differential equation**.

3.2 Objectives of the lesson

By the end of this lesson you should be able to

- i). define an exact differential equation.
- ii). learn the condition for an equation $Mdx + Ndy = 0$ to be exact.
- iii). solve an exact differential equation.
- iv). define an integrating factor.
- v). use the integrating factor to transform an equation into exact and then solve the exact equation obtained.

Let $F(x, y)$ be such that

$$\frac{\partial F}{\partial x} = M \quad \text{and} \quad \frac{\partial F}{\partial y} = N \quad (1)$$

3.3 Total differential

We know that

$$dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy \quad (2)$$

$$= Mdx + Ndy \quad (3)$$

From (2) and (3) we have

$$\frac{\partial M}{\partial y} = \frac{\partial^2 F}{\partial y \partial x} \quad (4)$$

$$\text{and} \quad \frac{\partial N}{\partial x} = \frac{\partial^2 F}{\partial x \partial y} \quad (5)$$

If $F(x, y)$ is continuous in a domain D

$$\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial^2 F}{\partial y \partial x} \quad \text{or}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Theorem

If M , N , $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$ are continuous functions of x and y then a necessary and sufficient condition that $Mdx + Ndy = 0$ to be exact is that $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. The solution of such equation is $F(x, y) = \text{constant}$ where $dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy = Mdx + Ndy = 0$.

The following examples illustrates the methods of solving an exact equation.

Example 1

a) Show that the equation

$$(3x^2 + 4xy)dx + (2x^2 + 2y)dy = 0$$

$$\text{or } \frac{dy}{dx} = -\frac{3x^2 + 4xy}{2x^2 + 2y} \text{ is exact}$$

b) Hence solve the equation

Solution

$$\text{Let } (3x^2 + 4xy)dx + (2x^2 + 2y)dy = 0 \quad (1)$$

This is of the form $Mdx + Ndy = 0$

$$\text{where } M = 3x^2 + 4xy, N = 2x^2 + 2y \quad (2)$$

$$\frac{\partial M}{\partial y} = 4x \text{ and } \frac{\partial N}{\partial x} = 4x$$

$$\text{Since } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x},$$

Hence there exists a function $F(x, y)$ such that the equation is exact.

$$dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy = Mdx + Ndy \quad (3)$$

$$\text{From (3) } \frac{\partial F}{\partial x} = M$$

$$\text{Hence } \partial F = M \partial x$$

$$\text{and } F = \int M \partial x$$

$$= \int (3x^2 + 4xy) \partial x \text{ (Note that this is a partial Integration with respect to } x)$$

$$F(x, y) = x^3 + 2x^2y + \phi(y) \quad (4)$$

y is treated as a constant. Hence constant of integration may be $\phi(y)$

$$\text{Again from (3) } \frac{\partial F}{\partial y} = N$$

From (4) and (2) we have

$$\frac{\partial}{\partial y} = x^3 + 2x^2y + \phi(y) = 2x^2 + 2y$$

$$\text{or } 0 + 2x^2 + \phi'(y) = 2x^2 + 2y \quad (5)$$

$$\text{From (5) } \phi'(y) = 2y \quad (6)$$

$$\text{Integrating we have } \phi(y) = y^2 \quad (7)$$

No constant is needed

Substituting (7) in (4) we have $F(x, y) = x^3 + 2x^2y + y^2 = c$ is the required solution.

Example 2

Solve the equation $(8y - x^2y)dy + (x + xy^2)dx = 0$

Solution

$$(x + xy^2)dx + (8y - x^2y)dy = 0 \quad (1)$$

This is of the form $Mdx + Ndy = 0$

$$\frac{\partial M}{\partial y} = -2xy = \frac{\partial N}{\partial x}$$

Hence it is an exact equation and there exists a function $F(x, y)$ such that

$$dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy = Mdx + Ndy = 0 \quad (2)$$

and $F = \text{Constant}$ is the solution.

$$\frac{\partial F}{\partial x} dx = Mdx = (x - xy^2)dx$$

$$\text{Then } \frac{\partial F}{\partial x} (x - xy^2)$$

$$\partial F = (x - xy^2)\partial x$$

$$\begin{aligned} F = F(x, y) &= \int (x - xy^2)\partial x \\ &= \frac{x^2}{2} - \frac{x^2y^2}{2} + \phi(y) \end{aligned} \quad (3)$$

From (2), $\frac{\partial F(x, y)}{\partial y} = N(x, y)$ gives

$$\frac{\partial}{\partial y} = \left[\frac{x^2}{2} - \frac{x^2y^2}{2} + Q(x, y) \right] = 8y - x^2y + \phi'(y) - x^2y + \frac{d\phi}{dy} = 8y - x^2y$$

$$\frac{d\phi}{dy} = 8y$$

$$d\phi = 8ydy$$

$$\phi = \int 8ydy$$

$$= 4y^2 \text{ (No need for constant)}$$

Substituting $\phi = 4y^2$ in (3) we have $F(x, y) = \frac{x^2}{2} - \frac{x^2 y^2}{2} + 4y^2$

The solution is $F(x, y) = \text{constant}$.

$$\frac{x^2}{2} - \frac{x^2 y^2}{2} + 4y^2 = \text{constant}$$

or $x^2 - x^2 y^2 + 8y^2 = c$ is the required solution of the equation.

Since the solution is $F = \text{constant}$ the required solution is, $x^2 - 2x^2 y + y^2 = c$

Example 3

a) Show that the equation

$$3x - (xy - 2)dx + (x^3 + 2y)dy = 0 \text{ is exact.}$$

b) Hence solve the equation

Solution

$$\text{Let } 3x - (xy - 2)dx + (x^3 + 2y)dy = 0 \quad (1)$$

This equation is the form $Mdx + Ndy = 0$

where $M = 3x(xy - 2)$ and $N = (x^3 + 2y)$

$$\frac{\partial M}{\partial y} = 3x^2 \text{ and } \frac{\partial N}{\partial x} = 3x^2$$

Hence the equation (1) is exact and there exists a function F such that

$$dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy = Mdx + Ndy \quad (2)$$

$$\frac{\partial F}{\partial x} = M \text{ gives}$$

$$\frac{\partial F}{\partial x} = 3x^2 y - 6x$$

$$\text{or } \partial F = (3x^2 y - 6x) \partial x$$

$$\text{or } F = \int (3x^2 y - 6x) \partial x$$

$$= x^3 y - 3x^2 + \phi(y) \quad (3)$$

$$\text{From (2) } \frac{\partial F}{\partial y} = N$$

$$\text{Hence } \frac{\partial}{\partial y} [x^3 y - 3x^2 + \phi(y)] = (x^3 + 2y)$$

$$\text{or } x^3 + \phi'(y) = x^3 + 2y$$

$$\text{or } \phi'(y) = 2y$$

$$\text{Integrating we have } \phi(y) = y^2 \quad (4)$$

Substituting (4) in (3) we have

$$F = x^3 y - 3x^2 + y^2 \text{ and the solution is } F = \text{constant}$$

Hence the required solution is

$$x^3 y - 3x^2 + y^2 = c$$

3.4 Integrating Factor

Consider the equation,

$$M(x, y)dx + N(x, y)dy = 0 \quad (1)$$

Suppose (1) is not an exact equation. When each of the terms of (1) is multiplied by a factor $\phi(x, y)$ the equation (1) may become exact. In other words

$\phi(x, y) M(x, y)dx + \phi(x, y)N(x, y)dy = 0$ may become an exact equation.

The factor $\phi(x, y)$ used to make $Mdx + Ndy = 0$ exact is called an Integrating factor.

Finding the Integrating factor for the equation $Mdx + Ndy = 0$ is generally not easier; However there are some standard forms of Integrating factors which we shall see in the next section.

3.5 Some important differentials and their integrating factors

Consider the equation $Mdx + Ndy = 0$ (1)

1) If $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 0$ (1) is an exact equation

If $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \neq 0$ and $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x)$, a function of x alone then $e^{\int f dx}$ is an integrating factor of (1).

2) If $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \neq 0$ and $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{M} = g(y)$, a function of y alone, then $e^{\int -g dy}$ is an integrating factor of (1).

3) $\frac{1}{Mx + Ny}$ is an integrating factor of (1) if $Mx + Ny \neq 0$.

4) $\frac{1}{x^2}$ is an integrating factor for $x dy - y dx$ since $d\left(\frac{y}{x}\right) = \frac{x dy - y dx}{x^2}$

Example 4

Consider the differential equation

$$(4x + 3y^2)dx + 2x dy = 0$$

a) Show that this equation is not exact.

b) Find an integrating factor of the form x^n where n is a positive integer.

c) Using your integrating factor found in (b) solve the equation.

Solution

Let $(4x + 3y^2)dx + 2xydy = 0$. It is of the form

$$Mdx + Ndy = 0 \text{ where}$$

$$M = 4x + 3y^2 \text{ and } N = 2xy$$

$$\frac{\partial M}{\partial y} = 6y \text{ and } \frac{\partial N}{\partial x} = 2y$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ the equation is not exact.

At this stage, you can find the integrating factor.

Let us try whether x^2 is an integrating factor.

Multiply the equation throughout by x^2

$$x^2(4x + 3y^2)dx + x^2 \cdot 2xydy = 0$$

Now $M = 4x^3 + 3x^2y^2$ and $N = 2x^3y$

$$\frac{\partial M}{\partial y} = 6x^2y = \frac{\partial N}{\partial x}$$

There exists a function $F(x,y)$ such that

$$dF = \frac{\partial F}{\partial x}dx + \frac{\partial F}{\partial y}dy = (4x^3 + 3x^2y^2)dx + 2x^3ydy = 0$$

and $F = \text{constant}$ will be the solution.

$$\frac{\partial F}{\partial x}dx = (4x^3 + 3x^2y^2)dx$$

$$\frac{\partial F}{\partial x} = 4x^3 + 3x^2y^2$$

$$\partial F = (4x^3 + 3x^2y^2)dx$$

$$F = \int (4x^3 + 3x^2y^2)dx$$

$$F = x^4 + x^3y^2 + \phi(y) = 0 \quad (1)$$

$$\frac{\partial F}{\partial y} = N = 2x^3y \text{ gives}$$

$$\frac{\partial}{\partial y} [x^4 + x^3y^2 + \phi(y)] = 2x^3y$$

$$2x^3y + \frac{d\phi}{dy} = 2x^3y$$

$$\frac{d\phi}{dy} = 0 \text{ which gives } \phi = \text{constant } c_0$$

From (1), $F = x^4 + x^3y^2 + c_0$. The solution is $F = \text{constant } c_1$

$x^4 + x^3y^2 + c_0 = c_1$. Then if we call $c_1 - c_0 = c$, the required solution is $x^4 + x^3y^2 = c$

Example 5

a) Show that $ydx - xdy = 0$ is not exact.

b) Show that $\frac{1}{x^2}$ is an integrating factor for the equation.

c) Solve the equation $ydx - xdy = 0$ using the integrating factor.

Solution

a) Let $ydx - xdy = 0$

Comparing this with $Mdx + Ndy = 0$

$M = y$ and $N = -x$

$$\frac{\partial M}{\partial y} = 1 \quad \frac{\partial N}{\partial x} = -1$$

Hence the given equation is not exact.

b) Multiplying each term by $\frac{1}{x^2}$

$$\frac{1}{x^2} ydx - \frac{1}{x^2} xdy = 0$$

$$\frac{y}{x^2} dx - \frac{1}{x} dy = 0$$

This is of the form $Mdx + Ndy = 0$ where $M = \frac{y}{x^2}$ and $N = -\frac{1}{x}$

$$\frac{\partial M}{\partial y} = \frac{1}{x^2} = \frac{\partial N}{\partial x}$$

Now $\frac{y}{x^2} dx - \frac{1}{x} dy = 0$ is exact.

c) There Exists a Function $F(x, y)$ such that $dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy$ $Mdx + Ndy = 0$

$$\frac{\partial F}{\partial x} = M = \frac{y}{x^2} \quad \text{and} \quad \frac{\partial F}{\partial y} = N = -\frac{1}{x}$$

$$\frac{\partial F}{\partial x} = \frac{y}{x^2}$$

$$\partial F = \frac{y}{x^2} dx$$

$$F = y \int \frac{y}{x^2} dx$$

$$F = y \frac{x^{-1}}{-1} + \phi(y)$$

$$F = -\frac{y}{x} + \phi(y) \quad (1)$$

$$\frac{\partial F}{\partial y} = N(x, y)$$

$$\frac{\partial}{\partial y} \left[-\frac{y}{x} + \phi(y) \right] = -\frac{1}{x}$$

$$-\frac{1}{x} + \frac{d\phi}{dy} = -\frac{1}{x}$$

$$\frac{d\phi}{dy} = 0$$

$$\phi(x, y) = \text{constant}$$

From (1) $F(x, y) = -\frac{y}{x} + c_0$. The solution is $F(x, y) = c_1$

Then, $-\frac{y}{x} + c_0 = c_1$ or $\frac{y}{x} = c$ or $y = cx$ is the solution.

Exercise 3

1. Show that the equation

$$(3x + 2y)dx + (2x + y)dy = 0$$

is exact.

2. a). Show that the equation

$$2xy + 1)dx + (x^2 + 4y)dy = 0$$

is exact.

- b). Solve the above equation

3. Solve the equation

$$(2xy - 3)dx + (x^2 + 4y)dy$$

given that $y(1) = 2$

4. If the equation

$$(x^2 + 3xy)dx + (Ax^2 + 4y)dy = 0$$

is exact, determine the constant A and solve the resulting exact equation.

5. If $\left(\frac{1}{x^2} + \frac{1}{y^2}\right)dx + \left(\frac{Ax+1}{y^3}\right)dy = 0$ is exact find the value of A and solve the resulting exact equation.

6. Show that the equation

$$(y \sec^2 x + \sec x \tan x)dx + (\tan x + 2y)dy = 0$$

is exact and solve the equation.

7. Verify that the differential equation

$$(6xy + 2y^2 - 5)dx + (3x^2 + 4xy - 6)dy = 0$$

is exact and solve the equation.

8. Show that the equation

$$\left(\frac{2s-1}{t}\right)ds + \left(\frac{s-s^2}{t^2}\right)dt = 0$$

is exact and solve the equation

9. a). Show that the equation

$$2x \ln x \frac{dy}{dx} + y = 0 \text{ is not exact}$$

b). Prove that $\frac{y}{x}$ is an integrating factor for the equation.

c). Solve the above equation in (a)

10. Show that the equation

$$e^y \sin x \frac{dy}{dx} + (1 + e^y) \cos x = 0$$

is exact and hence solve the equation

11. Show that the equation

$$(y^2 - x^2)dy + 2xydx = 0$$

is exact and solve the equation.

12. Show that

$$(y^4 + 2y^2 - x^3 + 5x^2y - 21xy^2) \frac{dy}{dx} + (x^3 - 3x^2y + 5xy^2 - 7y^3) = 0$$

is exact and solve the equation.

Summary

You have learnt the following from this lesson.

1. Definition of exact differential equation.
2. Condition for the equation $M(x,y)dx + N(x,y)dy = 0$ to be exact.
3. Method of solving $M(x,y)dx + N(x,y)dy = 0$ when it is exact.
4. Definition of integrating factors.
5. The use of integrating factors to solve the equation which is not exact.

Further Reading

1. Differential Equations
By Frank Ayres, Ph. D
Schaum's outline series
McGraw Hill Book Company
New York, Toronto, Sydney.
2. Differential Equations
By Dr. D. Sengottaiyan Ph.D
Oxford Publications
Harrow, London. Nairobi.

LESSON 4

First Order Linear equations and Equations reducible to this form (Bernouli's Equations)

4.1 Introduction;

In the previous lesson we have solved the differential equation of the form,
 $f(x) dx + g(y) dy = 0$

In this lesson we shall consider the differential equation of first order and first degree.
The first order linear differential equation is of the form,

$$\frac{dy}{dx} + Py = Q$$

where p and Q are functions of x only.

4.2 Objectives of the lesson.

By the end of this lesson you will be able to;

- define the first order linear equations
- define the first order Bernouli's equation
- determine Integrating factor of any differential equation.
- state the integrating factor for the linear equation

$$\frac{dy}{dx} + Py = Q \quad (1)$$

where P and Q are functions of x only.

- Solve first order linear equations of the form (1).
- Solve Bernouli's equations of the form

$$\frac{dy}{dx} + Py = Qy^n$$

where P and Q are functions of x only.

4.3 The first order Linear equation

$$\frac{dy}{dx} + Py = Q \quad (1)$$

(where P and Q are functions of x only) has the solution

$$ye^{\int P dx} = \int Qe^{\int P dx} dx \quad (2)$$

Proof

Multiply each term of (1) by $e^{\int P dx}$

$$\text{Then } e^{\int P dx} \frac{dy}{dx} + e^{\int P dx} Py = Q e^{\int P dx}$$

$$\text{Now } \frac{d}{dx} \left[y e^{\int P dx} \right] = e^{\int P dx} \frac{dy}{dx} + y e^{\int P dx} \quad (3)$$

$$\text{Using (3) in (2) we have } \frac{d}{dx} \left[y e^{\int P dx} \right] = Q e^{\int P dx}$$

$$\text{Then } d \left[y e^{\int P dx} \right] = Q e^{\int P dx} dx$$

Integrating both sides with respect to x we have

$$y e^{\int P dx} = \int Q e^{\int P dx} dx \quad (4)$$

is the solution. Remember the equation in (4).

Note: Procedure to solve the Linear equation

$$\frac{dy}{dx} + Py = Q$$

Step 1: Find $\int P dx$

Step 2: Find $e^{\int P dx}$

Step 3: Write the solution as $y e^{\int P dx} = \int Q e^{\int P dx} dx$.

The following examples illustrate the method of solving First order linear differential equations.

Example 1

$$\text{Solve } x \frac{dy}{dx} + y = x^2$$

Writing the equations in standard form

$$\frac{dy}{dx} + \frac{1}{x} y = x$$

comparing this with

$$\frac{dy}{dx} + Py = Q$$

$$P = \frac{1}{x} \text{ and } Q = x^2.$$

Step 1: $\int p dx = \int \frac{1}{x} dx = \ln x$

Step 2: $e^{\int p dx} = e^{\ln x} = x$ (Integrating factor)

Step 3: The solution is

$$ye^{\int p dx} = \int Qe^{\int p dx} dx$$

$$yx = \int x^2 \cdot x \cdot dx$$

$$xy = \int x^3 dx$$

$$xy = \frac{x^4}{4} + c$$

Then $y = \frac{x^3}{4} + \frac{c}{x}$ is the required solution

Example 2

Solve $\frac{dy}{dx} + y \cot x = \operatorname{cosec} x$

This is of the form

$$\frac{dy}{dx} + py = Q$$

where $p = \cot x$ and $Q = \operatorname{cosec} x$.

Step 1: $\int p dx = \int \cot x dx = \int \frac{\cos x}{\sin x} dx$
 $= \ln (\sin x)$

Step 2: $e^{\int p dx} = e^{\ln(\sin x)}$
 $= \sin x$

Step 3: Write the solution as

$$ye^{\int p dx} = \int Qe^{\int p dx} dx$$

$$y \sin x = \int \operatorname{cosec} x \cdot \sin x dx$$

$$y \sin x = \int \frac{1}{\sin x} \cdot \sin x dx$$

$$= \int dx$$

$$y \sin x = x + c$$

Then $y = (\operatorname{cosec} x)(x + c)$.

Example 3

Solve $\frac{dy}{dx} + \frac{2x}{1+x^2}y = \frac{1}{(1+x^2)^2}$

This is of the form

$$\frac{dy}{dx} + py = Q$$

where $p = \frac{2x}{1+x^2}$ and $Q = \frac{1}{(1+x^2)^2}$

- i. $\int p dx = \int \frac{2x}{1+x^2} dx = \ln(1+x^2)$
- ii. $e^{\int p dx} = e^{\ln(1+x^2)} = (1+x^2)$
- iii. The solution of the equation is given by:

$$ye^{\int p dx} = \int Qe^{\int p dx} dx$$

$$y(1+x^2) = \int \frac{1}{(1+x^2)^2} (1+x^2) dx$$

$$y(1+x^2) = \int \frac{1}{1+x^2} dx$$

$$y(1+x^2) = \tan^{-1} x + c$$

Hence $y = \frac{1}{1+x^2}(\tan^{-1} x + c)$ is the required solution.

4.4 Bernoulli's Equations

We now consider a special type of equation that can be reduced to a Linear equation by an appropriate transformation.

The equation of the form

$\frac{dy}{dx} + py = Qy^n$ is called a Bernoulli's differential equation.

We observe that if $n = 0$, the Bernoulli's equation is actually a linear equation of first order. If $n \neq 0$ or 1 the equation is called a Bernoulli equation.

The transformation $v = y^{-(n-1)}$ reduces the Bernoulli equation $\frac{dy}{dx} + py = Qy^n$ to a linear

equation in v and x as $\frac{dv}{dx} - (n-1)pv = -(n-1)Q$ where p and Q are functions of x only.

Example 4

Solve $\frac{dv}{dx} - y = 4y^5$

This is Bernoulli equation where $n = 5$. Multiply each term by y^{-5}

$$y^{-5} \frac{dy}{dx} - y \cdot y^{-5} = 4 \quad (1)$$

put $v = y^{-(n-1)} = y^{-4}$ (2)

$$\begin{aligned} \frac{dv}{dx} &= \frac{d}{dx} y^{-4} \\ &= \frac{d}{dy} y^{-4} \frac{dy}{dx} \\ &= -4y^{-5} \frac{dy}{dx} \end{aligned}$$

Therefore $y^{-5} \frac{dy}{dx} = \frac{1}{-4} \frac{dv}{dx}$ (3)

Substitute (2) and (3) in (1).

$$\begin{aligned} \frac{1}{-4} \frac{dv}{dx} - v &= 4 \\ \frac{dv}{dx} + 4v &= -16 \end{aligned}$$

This is of the form

$$\frac{dy}{dx} + py = Q \text{ where } p = 4 \text{ and } Q = -16$$

Example 5

Solve $\frac{dy}{dx} + y = xy^3$

This is Bernoulli equation where $n = 3$ multiply all the three terms by y^{-3} or divide each term by y^3 .

$$y^{-3} \frac{dy}{dx} + y \cdot y^{-3} = x \quad (1)$$

If we put $v = y^{1-n} = y^{1-3} = y^{-2}$ (2)

$$\begin{aligned} \frac{dv}{dx} &= \frac{d}{dx} y^{-2} \\ &= \frac{dy^{-2}}{dy} \cdot \frac{dy}{dx} \\ &= -2y^{-3} \frac{dy}{dx} \end{aligned}$$

Therefore $y^{-3} \frac{dy}{dx} = \frac{1}{-2} \frac{dv}{dx}$ (3)

Substituting (2) and (3) in 1.

$$\frac{1}{-2} \frac{dv}{dx} + v = x$$

$$\text{or } \frac{dv}{dx} - 2v = -2x$$

This is Linear differential equation of the form $\frac{dy}{dx} + py = Q$

$$\text{Therefore } \int p dx = \int -2 dx = -2x$$

Integrating factor is $e^{\int p dx} = e^{-2x}$

$$\text{Hence } ye^{-2x} = \int -2xe^{-2x} dx$$

$$\begin{aligned} ye^{-2x} &= -2 \int xe^{-2x} dx \\ &= -2xe^{-2x} + \frac{e^{-2x}}{4} \end{aligned}$$

$$\text{The solution is } y = -2x + \frac{1}{4}$$

Example 6

$$\text{Solve } \frac{dy}{dx} - y = 4y^5 \quad (1)$$

Solution

This is Bernoulli's equation where $n = 5$ multiply each term of (1) by y^{-5} .

$$\text{Then } y^{-5} \frac{dy}{dx} - y^{-4} = 4 \quad (2)$$

$$\begin{aligned} \text{Substituting } y^{-(n-1)} = v \text{ or } y^{-4} = v \\ \text{or } v = y^{-4} \end{aligned} \quad (3)$$

$$\text{and } \frac{dv}{dx} = \frac{dy^{-4}}{dx}$$

$$\begin{aligned} \text{or } \frac{dv}{dx} &= \frac{dy^{-4}}{dy} \cdot \frac{dy}{dx} \\ &= -4y^{-5} \frac{dy}{dx} \\ \frac{1}{-4} \frac{dv}{dx} &= y^{-5} \frac{dy}{dx} \end{aligned} \quad (4)$$

Substituting (3) and (4) in (2) we have

$$\begin{aligned} \frac{1}{-4} \frac{dv}{dx} - v &= 4 \\ \frac{dv}{dx} + 4v &= -16 \end{aligned} \quad (5)$$

Equation (5) is of the form $\frac{dy}{dx} + py = Q$

$$\text{i. } \int p dx = \int 4 dx = 4x$$

$$\text{ii. } e^{\int p dx} = e^{4x}$$

$$\text{iii. } ve^{4x} = \int -16e^{4x} dx$$

$$\text{or } ve^{4x} = -4e^{4x} + c$$

$$\text{or } v = -4 + ce^{-4x}$$

$$y^{-4} = -4 + ce^{-4x}$$

$$\text{or } \frac{1}{y^4} = -4 + ce^{-4x}$$

Note:

Bernoulli's Equation

$$\frac{dy}{dx} + py = Qy^n \text{ is reduced to}$$

$$\frac{dv}{dx} - (1-n)pv = -(1-n)Q \text{ using the substitution } v = y^{(1-n)}$$

Example 7

Solve the equation $\frac{dy}{dx} + y = xy^3$

Solution

$$\text{Let } \frac{dy}{dx} + y = xy^3 \quad (1)$$

This is of the form $\frac{dy}{dx} + py = Qy^n$ where $p = 1$ and $Q = x$, $n = 3$. Multiply each term of

(1) by y^{-3}

$$y^{-3} \frac{dy}{dx} + yy^3 = x$$

$$\text{or } y^{-3} \frac{dy}{dx} + y^{-2} = x \quad (2)$$

Substitute $v = y^{-2}$

$$\text{and } \frac{dv}{dx} = \frac{d}{dx} y^{-2} = -2y^{-3} \frac{dy}{dx} \text{ or } y^{-3} \frac{dy}{dx} = \frac{1}{-2} \frac{dv}{dx} \quad (3)$$

Substitute (2) and (3) in (2) we have

$$\frac{1}{-2} \frac{dv}{dx} + v = x$$

$$\text{or } \frac{dv}{dx} - 2v = -2x \quad (4)$$

This is of the form $\frac{dy}{dx} + py = Q$ where $p = -2$ and $Q = -2x$.

$$\text{Now } \int p dx = \int -2 dx = -2x$$

$$\text{Integrating factor } e^{\int p dx} = e^{-2x}$$

$$\text{The solution is } e^{-2x} = \int -2xe^{-2x}$$

$$ve^{-2x} = -2 \int xe^{-2x}$$

$$= \frac{1}{2} e^{-2x} (2x + 1) + c$$

$$v = x + \frac{1}{2} + ce^{2x}$$

Since $v = \frac{1}{y^2}$ we have $\frac{1}{y^2} = x + \frac{1}{2} + ce^{2x}$ which is the required solution

Exercise 4

Solve the following Linear Equations

$$1. \frac{dy}{dx} + 2xy = 4x$$

$$2. \frac{dy}{dx} + y \cot x = \operatorname{cosec} x$$

$$3. \frac{dy}{dx} - \frac{2xy}{1+x^2} = 1+x^2$$

$$4. (1+x^2) \frac{dy}{dx} + y = 1$$

$$5. \frac{dy}{dx} + \frac{2x}{1+x^2} y = \frac{1}{(1+x^2)}$$

$$6. (1+x^3) \frac{dy}{dx} + 3x^2 y = \sin^2 x$$

$$7. \frac{dy}{dx} + y \tan x = \cos^3 x$$

$$8. x \left(\frac{dy}{dx} + y \right) = 1 - y$$

$$9. x \frac{dy}{dx} + y = xe^x$$

$$10. (1-x^2) \frac{dy}{dx} - xy = 1$$

$$11. \frac{dy}{dx} - y \cot x = \sin x$$

$$12. x \frac{dy}{dx} = y + x^3 + 3x^2 - 2x$$

$$13. (x-2)\frac{dy}{dx} = y + 2(x-2)^3$$

Summary

You have learnt the following from third lesson.

- i). Definition of first order linear equation.
- ii). Definition of Bernoulli's equation.
- iii). The integrating factor of linear equation.
- iv). Solving linear equation and Bernoulli's equation.
- v). $\frac{dy}{dx} + Py = Qy^n$ is reduced to $\frac{dv}{dx} - (1-n)v = (1-n)Q$

Further Reading

4. Differential Equations

By Shepley L. Ross

John Wiley & sons

New York Toronto. Singapore.

5. Mathematics method for Science Students

By G. Stephenson

Addison Wesley Longman Limited

Edinburgh Gate, Harlow, England, 1973.

6. Differential equations

By Dr. D. Sengottaiyan

Oxford publications

London Nairobi.

LESSON 5

Second Order Homogeneous Linear Equations with constant coefficients.

5.1 Introduction

In the previous lessons you have learnt different methods of solving first order differential equations. In this lesson you will learn the method of solving second order homogeneous linear equations with constant coefficients.

5.2 Objectives of the lesson.

By the end of this lesson you should be able to

- define homogenous and nonhomogeneous differential equations of second order.
- define linear and non linear differential equations.
- determine the characteristic equation
- solve the second order homogeneous linear equations with constant coefficients using its characteristic equations.

5.3 Definition of second order homogenous Linear equation with constant coefficients.

A second order homogeneous differential equation in the dependent variable y and the independent variable x is an equation that can be expressed in the form,

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0 \quad (1)$$

where $a_0 \neq 0$ and a_0, a_1, a_2 are functions of x only or constants. If a_0, a_1, a_2 are constants, equation (1) is called a second order homogenous equation with constant coefficients.

5.4 Homogenous and nonhomogeneous equations

The equation $a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0$ (1) is called a second order homogenous equation

since all the terms, $\frac{d^2 y}{dx^2}$, $\frac{dy}{dx}$ and y are of the same first degree.

If the equation (1) contains any additional functions of x it is usually written on the right side of the equation and the equation is called a **nonhomogeneous equation of second order.**

For example,

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = F(x)$$

$$2 \frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} - 6y = e^x$$

$$\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 7y = \sin x + x^2 \text{ are nonhomogenous second order equations where as}$$

$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 7y = 0$ is a homogeneous equation.

5.5 Linear and non linear equations

An n^{th} order differential equation $a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_n y = F(x)$ is said to be linear if the following conditions are satisfied.

The degree of y , $\frac{dy}{dx}$, $\frac{d^2 y}{dx^2}$, ..., $\frac{d^n y}{dx^n}$ is all unity (one).

The coefficients of y , $\frac{dy}{dx}$, $\frac{d^2 y}{dx^2}$, ..., $\frac{d^n y}{dx^n}$, and F are all functions of x only.

There should not be any term containing product of derivatives such as $\left(\frac{d^3 y}{dx^3}\right) \left(\frac{dy}{dx}\right)$ or

$$y^2 \frac{dy}{dx}.$$

A non linear differential equation is one that does not satisfy the three conditions above. The following examples will illustrate the meaning of linear and non linear differential equations.

Example 1

$5x \frac{d^3 y}{dx^3} + 8xy \frac{d^2 y}{dx^2} - 9 \frac{dy}{dx} + xy = e^x$ is not a linear equation since the second term $\frac{d^2 y}{dx^2}$ has y in the coefficient. This is a non linear equation

Example 2

$3 \frac{d^2 y}{dx^2} - 2 \left(\frac{dy}{dx}\right)^2 + 8y = \ln x$ is not a linear equation since the degree of $\frac{dy}{dx}$ is not unity. It is a non linear equation.

Example 3

$5 \frac{d^2 y}{dx^2} - 8x \frac{dy}{dx} + 2y = xy$ is not a linear equation since the right hand side is not a function of x only. This is a non – linear equation.

5.6 Operator Notation for a Second order linear equation with constant coefficients.

For the differential equations we can use convenient notations, D , D^2 , D^3 ... instead of $\frac{d}{dx}$, $\frac{d^2}{dx^2}$, $\frac{d^3}{dx^3}$... respectively.

Then the linear equation $5 \frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 8y = 0$ is written as

$$5D^2 y - 3Dy + 8y = 0 \text{ or } (5D^2 - 3D + 8)y = 0$$

In general

$$\left(a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y \right) = 0$$

is written as

$(a_0 D^2 + a_1 D + a_2)y = 0$ or $f(D)y = 0$. where $f(D) = a_0 D^2 + a_1 D + a_2$

5.7 Method of solving the homogeneous second order linear Equation with constant coefficients

Consider the equation,

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0 \quad (1)$$

where a_0 , a_1 and a_2 are constants.

In an attempt to find a solution of the equation (1), we would naturally inquire whether there is any familiar type of function f such that its derivatives are constant multiples of the same function f . The answer is “yes” for the exponential function $y = e^{mx}$.

$$\text{Now } y = e^{mx} \quad (2)$$

$$y' = me^{mx} \quad (3)$$

$$y'' = m^2 e^{mx} \quad (4)$$

Substituting (2), (3) and (4) in (1) we have

$$(a_0 m^2 + a_1 m + a_2)e^{mx} = 0 \quad (5)$$

$$\text{since } e^{mx} \neq 0, \text{ we have } a_0 m^2 + a_1 m + a_2 = 0 \quad (6)$$

This equation (6) is called **the auxiliary equation or the characteristic equation** of (1).

If $y = e^{mx}$ is a solution of (1) then the constant m must satisfy (6). We can write the auxiliary equation and solve it for m . Three cases arise, according as the roots of (6) are;

- real and distinct
- real and repeated or
- complex conjugate

5.8 Real and distinct roots of characteristic equations.

Suppose the roots of (6) are two real and distinct numbers m_1 and m_2

Then $y = e^{m_1 x}$

and $y = e^{m_2 x}$ are two independent solutions.

(using the Wronskian Determinant we can show that they are independent).

Example 4

Consider the equation,

$$\frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 6y = 0$$

The auxiliary equation is

$$m^2 - 5m + 6 = 0$$

or $(m-2)(m-3)=0$
Hence $m_1 = 2, m_2 = 3$

The roots are real and distinct. Thus e^{2x} and e^{3x} are independent solutions. The general solution is written as $y = c_1 e^{2x} + c_2 e^{3x}$ if the Wronskian of e^{2x} and e^{3x} are independent.

We can verify that the Wronskian determinant $w = \begin{vmatrix} e^{2x} & e^{3x} \\ 2e^{2x} & 3e^{3x} \end{vmatrix} = 3e^{5x} - 2e^{5x} = e^{5x} \neq 0$.

Hence $y = c_1 e^{2x} + c_2 e^{3x}$ is the general solution.

5.9 Real and Repeated roots of characteristic equations.

Suppose the characteristic equation has repeated roots m and m . Now $y = c_1 e^{mx}$ is one root of the equation since the second solution should be different from $y = c_1 e^{mx}$ we must find the second solution using the substitution, $y = e^{mx} v$ where v is to be found.

$$\text{Let } y = e^{mx} v. \quad (1)$$

$$\frac{dy}{dx} = e^{mx} \frac{dv}{dx} + m e^{mx} v \quad (2)$$

$$\begin{aligned} \frac{d^2 y}{dx^2} &= e^{mx} \frac{d^2 v}{dx^2} + \frac{dv}{dx} \cdot m e^{mx} + m e^{mx} \frac{dv}{dx} + m^2 e^{mx} v \\ &= e^{mx} \frac{d^2 v}{dx^2} + 2m e^{mx} \frac{dv}{dx} = m^2 e^{mx} v \end{aligned} \quad (3)$$

Substituting (1), (2) and (3) in the given equation, we have

$$= e^{mx} \frac{d^2 v}{dx^2} = 0$$

$$\text{Let } w = \frac{dv}{dx} \text{ then}$$

$$e^{mx} \frac{dw}{dx} = 0$$

or $\frac{dw}{dx} = 0$ since $e^{mx} \neq 0$ or $w = 0$ where c is an arbitrary constant. If we choose $c = 1$

$$w = \frac{dv}{dx} = 1 \text{ or } v = x + c_0$$

Hence the second solution is $y = e^{mx} (x + c_0)$ choosing $c_0 = 0$ we have the second solution $y = x e^{mx}$.

Thus the general solution is $y = c_1 e^{mx} + c_2 x e^{mx}$ if Wronskian determinant is $\neq 0$

5.10 Theorem

Consider the second order homogeneous differential equation,

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0 \text{ where } a_0, a_1 \text{ and } a_2 \text{ are constants.}$$

- i. If the auxiliary equation $a_0m^2 + a_1m + a_2 = 0$ has the two roots real and distinct (m_1 and m_2) then the solution of the equation is $y = c_1e^{m_1x} + c_2e^{m_2x}$.
- ii. If the auxiliary equation has repeated real roots m and m then the general solution is $y = c_1e^{mx} + c_2xe^{mx}$.
- iii. If the auxiliary equation has Conjugate roots $m = a \pm ib$ then the solution is $y = e^{ax}(c_1 \cos bx + c_2 \sin bx)$.

Example 5

Solve the equation,

$$\frac{d^2y}{dx^2} + 10\frac{dy}{dx} + 25y = 0$$

Solution

$$\text{Let } \frac{d^2y}{dx^2} + 10\frac{dy}{dx} + 25y = 0 \quad (1)$$

The auxiliary equation $m^2 + 10m + 25 = 0$ or $(m + 5)^2$ has two equal roots $-5, -5$. (repeated).

Hence the general solution of (1) is

$$y = c_1e^{-5x} + c_2xe^{-5x}$$

Note that e^{-5x} and xe^{-5x} are independent from the Wronskian determinant.

Example 6

Solve the equation

$$4\frac{d^2y}{dx^2} - 12\frac{dy}{dx} + 5y = 0$$

Solution

$$\text{Let } 4\frac{d^2y}{dx^2} - 12\frac{dy}{dx} + 5y = 0 \quad (1)$$

The characteristic equation of (1) is $4m^2 - 12m + 5 = 0$. (2)

The roots of the characteristic equation (2) are

$$\begin{aligned} m &= \frac{12 \pm \sqrt{144 - 4(4)(5)}}{2(4)} \\ &= \frac{12 \pm \sqrt{64}}{8} \\ &= \frac{12 \pm 8}{8} \\ m &= 2\frac{1}{2} \text{ and } \frac{1}{2} \end{aligned}$$

Hence the general solution is $y = c_1e^{\frac{5}{2}x} + c_2e^{\frac{1}{2}x}$.

5.11 The characteristic equation has conjugate complex roots.

$$a + ib \text{ and } a - ib$$

consider the equation

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0 \quad (1)$$

The characteristic equation of (1) is

$$a_0 m^2 + a_1 m + a_2 = 0 \quad (2)$$

Suppose the roots of the equation (2) be $m_1 = a + ib$ and $m_2 = a - ib$.

The general solution of (1) is

$$y = c_1 e^{(a+ib)x} + c_2 e^{(a-ib)x}$$

Since they are not repeated.

$$\begin{aligned} \text{or } y &= c_1 e^{ax} \cdot e^{ibx} + c_2 e^{ax} \cdot e^{-ibx} \\ &= e^{ax} (c_1 e^{ibx} + c_2 e^{-ibx}) \\ &= e^{ax} (c_1 (\cos bx + i \sin bx) + c_2 (\cos bx - i \sin bx)) \\ &= e^{ax} [(c_1 + c_2) \cos bx + i(c_1 - c_2) \sin bx] \\ y &= e^{ax} [A \cos bx + B \sin bx] \text{ is the general solution.} \end{aligned}$$

where a is the real part and b is the imaginary part of the roots and A and B are arbitrary constants.

Example 7

Solve the equation,

$$\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 13y = 0 \quad (1)$$

Solution

$$\text{Let } \frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 13y = 0$$

The characteristic equation of (1) is $m^2 - 4m + 13 = 0$

$$\text{Whose roots are } m = \frac{4 \pm \sqrt{16 - 4(1)(13)}}{2} \text{ or } m = \frac{4 \pm 6i}{2} \text{ or } m = 2 \pm 3i$$

Hence the general solution of (1) is $y = e^{2x} (c_1 \cos 3x + c_2 \sin 3x)$.

Example 8

Solve the equation $\frac{d^2 y}{dx^2} + y = 0$ (This is the equation concerning simple Harmonic motion of a particle)

Solution

$$\text{Let } \frac{d^2 y}{dx^2} + y = 0$$

The characteristic equation is

$$m^2 + 1 = 0 \text{ whose roots are } m = \pm i.$$

Hence the general solution of (1) is $y = e^{\alpha x} (A \cos x + B \sin x)$ or $y = A \cos x + B \sin x$.

Example 9

Solve the initial value problem

$$\frac{d^2 y}{dx^2} - 6 \frac{dy}{dx} + 25y = 0, \quad y(0) = -3, \quad y'(0) = -1$$

Solution

The auxiliary equation is

$$m^2 - 6m + 25 = 0 \text{ whose roots are } m = 3 \pm 4i$$

Hence the general solution may be written

$$y = e^{3x} (c_1 \cos 4x + c_2 \sin 4x)$$

using $y = -3$ when $x = 0$, we have $-3 = c_1(1) + c_2(0)$ or $c_1 = -3$

Then $y = e^{3x} (-3 \cos 4x + c_2 \sin 4x)$

and $y' = e^{3x} (4.3 \sin 4x + 4c_2 \sin 4x) + (-3 \cos 4x + c_2 \sin 4x)(3e^{3x})$

using $y' = -1$ when $x = 0$ we have

$$-1 = 3(0 + 4c_2) + (-3 + 0)(3)$$

$$-1 = 4c_2 - 9 \text{ or } c_2 = 2$$

Hence the general solution is $y = e^{3x} (2 \sin 4x - 3 \cos 4x)$.

5.12 Solution of Higher order homogeneous linear Equations with constant coefficients.

All the results used for the equation

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0$$

in the previous sections can be used for higher order homogeneous linear equations with constant coefficients. The method is illustrated in the following examples:

Example 10

Solve the equation,

$$\frac{d^3 y}{dx^3} + 6 \frac{d^2 y}{dx^2} + 11 \frac{dy}{dx} - 6y = 0 \quad (1)$$

Solution

The characteristic equation of (1) is

$$m^3 - 6m^2 + 11m - 6 = 0 \quad (2)$$

using remainder theorem or factor theorem, (2) becomes $(m - 1)(m - 2)(m - 3) = 0$ whose roots are $m = 1, m = 2, m = 3$.

Hence the general solution may be written as

$$y = c_1 e^x + c_2 e^{2x} + c_3 e^{3x}$$

e^x, e^{2x}, e^{3x} is not zero (you can verify this)

Example 11

Solve the equation

$$\frac{d^5 y}{dx^5} - 2 \frac{d^4 y}{dx^4} + \frac{d^3 y}{dx^3} = 0$$

Solution

The auxiliary equation of (1) is

$$m^5 - 2m^4 + m^3 = 0$$

$$\text{or } m^3(m^2 - 2m + 1) = 0$$

$$m^3(m-1)(m-1) = 0$$

whose roots are $m = 0, 0, 0, 1, 1$

Hence the general solution of (1) is

$$y = c_1 e^{0x} + c_2 x e^{0x} + c_3 x^2 e^{0x} + c_4 e^x + c_5 x e^x$$

$$\text{or } y = c_1 + c_2 x + c_3 x^2 + c_4 e^x + c_5 x e^x.$$

Exercise 5

Solve the following second order homogeneous linear Equations (with constant coefficients) (1 – 8)

$$1. \quad \frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 6y = 0$$

$$2. \quad \frac{d^2 y}{dx^2} - 8 \frac{dy}{dx} + 16y = 0$$

$$3. \quad \frac{d^2 y}{dx^2} + 9y = 0$$

$$4. \quad (D^2 - 2D + 10)y = 0$$

$$5. \quad (D^2 + 2D - 15)y = 0$$

$$6. \quad (D^3 + 4D)y = 0$$

$$7. \quad (D^2 - 2D + 5)^2 y = 0$$

$$8. \quad (D^2 + 25)y = 0$$

Solve the initial value problems (9 to 12)

$$9. \quad \frac{d^2 y}{dx^2} - \frac{dy}{dx} - 12y = 0, \quad y(0) = 3, \quad y'(0) = 5$$

$$10. \quad \frac{d^2 y}{dx^2} + 6 \frac{dy}{dx} + 9y = 0, \quad y(0) = 2, \quad y'(0) = -3$$

$$11. \frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 29y = 0, \quad y(0) = 0, \quad y'(0) = 5$$

$$12. \frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 13y = 0, \quad y(0) = 3, \quad y'(0) = -1$$

Summary of the lesson

You have learnt the following from this lesson.

Definitions of

- i. Homogeneous equation
- ii. Nonhomogeneous equation
- iii. Linear equation
- iv. Non linear equation.

The characteristic equation of $a_0 \frac{d^2y}{dx^2} + a_1 \frac{dy}{dx} + a_2y = 0$ is $a_0m^2 + a_1m + a_2 = 0$

Solving the second order homogeneous linear differential equation with constant coefficients of the form $a_0 \frac{d^2y}{dx^2} + a_1 \frac{dy}{dx} + a_2y = 0$ where a_0, a_1, a_2 are constants. When the characteristic equation has

- i. Distinct real roots
- ii. Repeated real roots
- iii. Conjugate complex roots of the form $a \pm ib$

Further Reading:

1. Differential Equations
By Frank Ayres, Ph. D
Schaum's outlines series
Mc Graw Hill Book company
New York, Toronto, Sydney.
2. Differential Equations
By Shepley L. Ross
John Wiley and sons
New York. Toronto. Singapore
3. Ordinary Differential Equations
By Dr. Sengottaiyan Ph. D
Oxford Publications,
Harrow. London. Nairobi.



Microsoft Equation
3.0

LESSON 6

Second Order *Nonhomogeneous* Linear Equations with Constant Coefficients

6.1 Introduction

In the previous lesson, you have learnt the method of solving a second order homogeneous linear equation with constant coefficients. Such equations contain only the terms $\frac{d^2y}{dx^2}$, $\frac{dy}{dx}$ and y each multiplied by a constant. Suppose the equations contain one more term, a function of x . the additional term $f(x)$ usually written on the right side and it is called the nonhomogeneous term. These kinds of equations are called nonhomogeneous equations.

For example

$2\frac{d^2y}{dx^2} + 8\frac{dy}{dx} + 4y = 0$ is a homogeneous equation where as $2\frac{d^2y}{dx^2} + 8\frac{dy}{dx} + 4y = x^3 + e^x$ is a nonhomogeneous equation

6.2 Objectives of the lesson

By the end of this lesson you should be able to:

- define nonhomogeneous equations of second order with constant coefficients.
- determine the complementary function and the particular integral of the second order nonhomogeneous equation
- solve the nonhomogeneous equations

6.3 The general form of a homogeneous linear equation of second order with constant coefficients.

The equations

$$a_0 \frac{d^2y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0 \quad (1)$$

and $a_0 \frac{d^2y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = Q(x)$ where $a_0 \neq 0$ (2)

are called respectively, the homogeneous and the nonhomogeneous equations of the second order with constant coefficients when a_0, a_1, a_2 are constants. The equation (2) can also be written as,

$$(a_0 D^2 + a_1 D + a_2)y = Q(x) \quad (3)$$

where $a_0 \neq 0$.

6.4 Solution of the equation.

The general solution of the equation

$$F(D)y = (a_0D^2 + a_1D + a_2)y = Q(x) \quad (1)$$

is the sum of the complementary function (solution of $F(D)y = 0$) and any particular Integral, (P.I) of (1). Finding $F(D)y = 0$ is seen already in the previous lesson. There are two important methods for finding the particular integral of $F(D)y = Q(x)$

6.5 Finding particular integral (Method 1)

$$\text{Let } F(D)y = Q(x) \quad (1)$$

$$\text{Then } \frac{F(D)y}{F(D)} = \frac{Q(x)}{F(D)}$$

where $F(D)$ is treated as an operator.

$$\text{or } y = \frac{Q(x)}{F(D)} \quad (2)$$

suppose $F(D)$ is factorisable so that

$$F(D) = (D - m_1)(D - m_2)$$

$$\text{then } y = \frac{Q(x)}{F(D)} = \frac{Q}{(D - m_1)(D - m_2)}$$

We can operate first $\frac{1}{(D - m_2)}$ on Q and the result is further operated by $\frac{1}{(D - m_1)}$

$$\text{Let } u = \frac{Q}{D - m_2} \text{ then } (D - m_2)u = Q$$

Or $\frac{du}{dx} - m_2u = Q$ which is a linear equation. The integrating factor of the above

equation is e^{-m_2x} and the solution is

$$ue^{-m_2x} = \int Qe^{-m_2x} dx$$

$$\text{or } u = e^{m_2x} \int Qe^{-m_2x} dx$$

$$\text{Thus } \frac{Q}{D - m_2} = e^{m_2x} \int e^{-m_2x} Q dx$$

This is a linear equation whose solution is now $y = \frac{Q}{(D-m_1)(D-m_2)} = \frac{1}{D-m_1}u$

$$y = e^{+m_1x} \int e^{-m_1x} u dx$$

the method of finding P.I is illustrated in the following examples:

Example 1

Find the particular integral for the equation

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^{2x}$$

Solution

$$\text{Let } \frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^{2x}$$

$$\text{or } (D^2 - 5D + 6)y = e^{2x}$$

$$\text{or } (D-2)(D-3)y = e^{2x}$$

then the PI is given from

$$\begin{aligned} y &= \frac{e^{2x}}{(D-2)(D-3)} \\ &= \frac{1}{D-2} \left(\frac{e^{2x}}{D-3} \right) \end{aligned}$$

$$\text{let } u = \frac{e^{2x}}{D-3}$$

from the formula in 5.5 we have $\frac{Q}{D-m_2} = e^{+m_1x} \int e^{-m_1x} . Q dx$

$$\text{hence } u = e^{+3x} \int e^{-3x} . e^{2x} dx$$

$$\begin{aligned}
 &= e^{3x} \int e^{-x} dx \\
 &= -e^{3x} e^{-x} \\
 &= e^{2x}
 \end{aligned}$$

$$\begin{aligned}
 \text{Now } y &= \frac{1}{D-2} \left[\frac{e^{2x}}{D-3} \right] = \frac{1}{D-2} u \\
 &= \frac{1}{D-2} u \\
 &= e^{+2x} \int e^{-2x} . u dx \\
 &= e^{2x} \int e^{-2x} . e^{2x} dx \\
 &= e^{2x} \int 1 dx \\
 &= e^{2x} . x
 \end{aligned}$$

there is no constant for the particular integral. Hence the P.I is xe^{2x} .

Thus the solution of the given equation is

$$y = c_1 e^{3x} + c_2 e^{2x} + x e^{2x}$$

Example 2

$$\text{Solve } \frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 6y = e^{2x}$$

Solution

i. The complementary function is obtained from

$$(D^2 + 5D + 6)y = 0 \tag{1}$$

auxiliary equation of (1) is

$$m^2 + 5m + 6 = 0$$

or $(m+3)(m+2) = 0$ whose roots are $m = -3, -2$. then the complementary function is

$$y = c_1 e^{-3x} + c_2 e^{-2x}$$

ii. The particular integral is obtained from $\frac{1}{(D+2)(D+3)}e^{2x}$. Using the formula

$$\text{Let } u = \frac{e^{2x}}{D+2} = e^{-2x} \int Q e^{2x} dx = e^{-2x} \int e^{2x} \cdot e^{2x} dx = \frac{e^{4x}}{4} \cdot x$$

Hence the solution is

$$y = c_1 e^{-3x} + c_2 e^{-2x} + \frac{e^{4x}}{4}$$

Example 3

Solve $(D^2 - 3D + 2)y = e^x$

Solution

$$\text{Let } (D^2 - 3D + 2)y = e^x \quad (1)$$

$$(D-1)(D-2)y = e^x$$

the auxiliary equation of $(D-1)(D-2)y = 0$ is $(m-1)(m-2) = 0$ whose roots are $m = 1$, $m = 2$. Hence the complementary function is

$$y = c_1 e^x + c_2 e^{2x} \quad (2)$$

The particular integral is obtained from $(D-1)(D-2)y = e^x$

$$y = \frac{e^x}{(D-1)(D-2)}$$

$$\begin{aligned} \text{Now let } u &= \frac{e^x}{D-2} = e^{2x} \int Q e^{-2x} dx \\ &= e^{2x} \int e^x \cdot e^{-2x} dx \\ &= e^{2x} \int e^{-x} dx \\ &= e^{2x} \frac{e^{-x}}{-1} \\ &= -e^x \end{aligned}$$

$$y = \frac{1}{D-1} \left[\frac{e^x}{(D-2)} \right] = e^x \int e^{-x} (-e^x) dx$$

$$= -x e^x$$

Hence the general solution is

$$y = C.F + P.I$$

$$\text{or } y = c_1 e^{-3x} + c_2 e^{-2x} - x e^x$$

(For a short method see Lesson 7)

Method of finding particular integral of $(D - a)(D - b)y = Q$ where Q is a function of x only.

Example 4

Let $(D - a)(D - b)y = Q$

Solution

$$\text{Then } y = \frac{Q}{(D - a)(D - b)}$$

$$\text{First we find } u = \frac{Q}{D - b} \text{ and then find } y = \frac{u}{D - a}$$

$$\text{From (), } u = e^{+bx} \int e^{-bx} Q dx$$

$$\text{Then we find } y = \frac{u}{D - a} = e^{+ax} \int e^{-ax} u . dx$$

Example 5

$$\text{Find the particular interval of } (D - 1)(D - 2)y = \frac{e^{5x}}{D - 2} .$$

Solution

This is of the form $(D - a)(D - b)y = Q$. The particular integral is

$$y = \frac{Q}{(D - a)(D - b)} .$$

$$\text{First let us find } u = \frac{Q}{D - b} = \frac{e^{5x}}{D - 2}$$

$$u = \frac{Q}{D - b} = \frac{e^{5x}}{D - 2}$$

$$= e^{+2x} \int e^{-2x} . e^{5x} dx$$

$$= e^{2x} \int e^{3x} dx$$

$$\text{or } u = \frac{e^{2x} . e^{3x}}{3} = \frac{e^{5x}}{3}$$

$$\begin{aligned}\text{Now } \frac{Q}{(D-a)(D-b)} &= \frac{1}{D-a} \left[\frac{Q}{D-b} \right] \\ &= \frac{u}{D-a} \\ &= e^{+ax} \int e^{-ax} u dx\end{aligned}$$

$$\begin{aligned}\text{Here } \frac{Q}{(D-a)(D-b)} &= \frac{1}{D-a} \left(\frac{Q}{D-b} \right) \\ &= \frac{1}{D-a} u \\ &= e^{1x} \int e^{-1x} u dx \\ &= e^x \int \frac{e^{-x} \cdot e^{5x}}{3} dx \\ &= e^x \int \frac{e^{4x}}{3} dx \\ &= \frac{e^x \cdot e^{4x}}{4 \times 3} \\ &= \frac{e^{5x}}{12}\end{aligned}$$

Exercise 6

Use $\frac{Q(x)}{D-a} = e^{ax} \int e^{-ax} Q(x) dx$ wherever necessary.

Evaluate the following (problems 1 to 8)

1) $\frac{e^x}{D+1}$

2) $\frac{e^x}{D-1}$

3) $\frac{\sin 3x}{D+2}$

4) $\frac{x+1}{D+1}$

5) $\frac{x^2+1}{D+1}$

6) $\frac{\cos 6x}{D-3}$

7) $\frac{\cot x}{D^2+1}$

8) $\frac{e^{3x}}{(D+3)(D-2)}$

Solve the following equations (problems 9 – 15)

9) $(D^2 - 3D + 2)y = e^{5x}$

10) $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 4y = 3 - 2x$

11) $(D^2 - 6D + 9)y = e^{2x}$

12) $(D^2 - 1)y = \sin x$

13) $(D^2 - 1)y = 4xe^x$

14) $(D^2 + 9)y = x \cos x$

15) $(D^2 + D - 2)y = 2(1 + x - x^2)$

Summary

You have learnt the following from this lesson.

- i). The equation $\frac{d^2y}{dx^2} + a_1\frac{dy}{dx} + a_2y = Q(x)$ where $a_1 \neq 0$ and a_0, a_1, a_2 and $Q(x)$ are functions of x only is called a nonhomogeneous second order linear equation with constant coefficients if $Q(x)$ is not zero.
- ii). The solution of a nonhomogeneous equation will have two parts namely the complementary function y_c and a particular integral y_p .
- iii). The particular integral $\frac{Q(x)}{D - a} = e^{ax} \int e^{-ax} Q(x) dx$

Further Reading:

- 4. Differential Equations
By Frank Ayres, Ph. D
Schaum's outlines series
Mc Graw Hill Book company
New York, Toronto, Sydney.
- 5. Differential Equations
By Shepley L. Ross
John Wiley and sons
New York. Toronto. Singapore
- 6. Ordinary Differential Equations
By Dr. Sengottaiyan Ph. D
Oxford Publications,
Harrow. London. Nairobi.

Lesson 7

Determination of particular integrals using short methods

7.1 Introduction

In the previous lesson, you have found the particular integrals for non-homogeneous linear equations with constant coefficients of the form

$$\frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = Q \quad (1)$$

or $(a_0 D^2 + a_1 D + a_2)y = Q$

or $f(D)y = Q$

where $f(D) = a_0 D^2 + a_1 D + a_2$

we have already seen that the particular integral of (Q) is

$$y = \frac{Q}{f(D)}$$

where Q is particular types of functions such

as (i) e^{ax} , (ii) $\sin(ax + b)$ or $\cos(ax + b)$, (iii) x^n , (iv) $e^{ax}u$ and (v) xu
where u is a function of x

7.2 Objectives of the lesson

By the end of this lesson, you should be able to determine the particular integrals

$$y = \frac{Q}{f(D)} \text{ when } Q \text{ is of the form}$$

- i. e^{ax}
- ii. $\sin(ax + b)$ and $\cos(ax + b)$
- iii. x^n
- iv. $e^{ax}u$ when u is a function of x.
- v. xu when u is a function of x.

7.3 The particular integral of

Theorem 1

$$\frac{e^{ax}}{f(D)} = \frac{e^{ax}}{f(a)} \text{ if } f(a) \neq 0$$

Proof

First we shall show that

i. $f(D)e^{ax} = f(a)e^{ax}$ if $y = e^{ax}$

Let $f(D) = a_0D^2 + a_1D + a_2$

if $y = e^{ax}$, $Dy = \frac{d}{dx}e^{ax} = ae^{ax}$

$$D^2y = \frac{d^2}{dx^2}e^{ax} = a^2e^{ax}$$

$$\begin{aligned} f(D)y &= a_0(a^2e^{ax}) + a_1(ae^{ax}) + a_2e^{ax} \\ &= (a_0a^2 + a_1a + a_2)e^{ax} \end{aligned}$$

$$\begin{aligned} f(D)e^{ax} &= f(a)e^{ax} \\ \text{or } f(D) &= f(a) \end{aligned}$$

ii. Now consider the equation

$$a_0 \frac{d^2y}{dx^2} + a_1 \frac{dy}{dx} + a_2y = e^{ax}$$

or $(a_0D^2 + a_1D + a_2)y = e^{ax}$

or $f(D)y = e^{ax} \quad (2)$

The particular integral of (2) is written as

$$y = \frac{e^{ax}}{f(D)} \quad [\text{Dividing by } f(D)]$$

$$y = \frac{e^{ax}}{f(a)} \quad \text{if } f(a) \neq 0$$

hence the result.

If $f(a) = 0$, we must use a different method (See example 3)

Example 1

Solve $(D^2 - 5D + 6)y = e^{4x}$

Solution

Let $(D^2 - 5D + 6)y = e^{4x}$

The characteristic equation is

$$m^2 - 5m + 6 = 0$$

Or $(m - 3)(m - 2) = 0$

Then $m_1 = 3$ and $m_2 = 2$

Hence the complementary function is

$$y = c_1 e^{3x} + c_2 e^{2x}$$

A particular integral of the equation is $y = \frac{e^{4x}}{f(D)} = \frac{e^{4x}}{D^2 - 5D + 6} = \frac{e^{4x}}{f(4)}$

Then $y = \frac{e^{4x}}{16 - 20 + 6} = \frac{e^{4x}}{2}$

Hence the general solution of the equation is $y = c_1 e^{3x} + c_2 e^{2x} + \frac{e^{4x}}{2}$

Example 2

Solve $(D + 2)(D + 3)(D - 4)y = e^{5x}$

Solution

Let $(D + 2)(D + 3)(D - 4)y = e^{5x}$

(1)

The characteristic equation of (1) is $(m + 2)(m + 3)(m - 4) = 0$

Then $m_1 = -2$, $m_2 = -3$, $m_3 = 4$. The complementary function is

$$y = c_1 e^{-2x} + c_2 e^{-3x} + c_3 e^{4x}$$

A particular integral is given by $y = \frac{e^{4x}}{f(D)} = \frac{e^{5x}}{(D + 2)(D + 3)(D - 4)}$

Using the short method

$$\begin{aligned} y &= \frac{e^{4x}}{f(5)} = \frac{e^{5x}}{(5 + 2)(5 + 3)(5 - 4)} \\ &= \frac{e^{5x}}{7 \cdot 8 \cdot 1} \\ &= \frac{e^{5x}}{56} \end{aligned}$$

Hence the general solution is

$$y = c_1 e^{-2x} + c_2 e^{-3x} + c_3 e^{4x} + \frac{e^{5x}}{56}$$

Example 3

Solve $(D^2 - 5D + 6)y = e^{3x}$

Solution

Let $(D^2 - 5D + 6)y = e^{3x}$

Or $(D - 3)(D - 2)y = e^{3x}$

The characteristic equation is $(D - 3)(D - 2) = 0$ whose roots are $m_1 = 3, m_2 = 2$.

Hence the complementary function is

$$y = c_1 e^{3x} + c_2 e^{2x}$$

the particular integral is $y = \frac{e^{3x}}{(D-3)(D-2)} = \frac{e^{3x}}{0}$

[f(a) = 0 since the nonhomogeneous term e^{3x} is already in the complementary function $c_1 e^{3x}$] Hence the short method does not apply. However we may write

$$\begin{aligned} y &= \left(\frac{1}{D-3} \right) \frac{1}{D-2} e^{3x} \\ &= \frac{1}{D-3} \frac{e^{3x}}{3-2} \\ &= \frac{1}{D-3} e^{3x} \\ &= e^{3x} \int e^{-3x} \cdot e^{3x} dx \end{aligned}$$

$$\begin{aligned} \text{(Note that: } \frac{1}{D-\alpha} Q &= e^{\alpha x} \int e^{-\alpha x} Q dx) \\ &= e^{3x} \int dx \\ &= x e^{3x} \end{aligned}$$

Hence the general solution of the equation is

$$y = c_1 e^{3x} + c_2 e^{2x} + c_3 x e^{3x}$$

Example 4

Solve $(D - 2)^2 y = 5e^{3x} + 4e^{2x}$

Solution

The characteristic equation is $(m - 2)^2 = 0$. Hence $m_1 = 2$ and $m_2 = 2$ (repeated)

Hence the complementary function is $y = c_1 e^{2x} + c_2 x e^{2x}$

A particular integral is $y = \frac{5e^{3x} + 4e^{2x}}{(D-2)^2}$

One particular integral corresponding to $5e^{3x}$ is given by

$$\begin{aligned}
 y_1 &= \frac{5e^{3x}}{(D-2)^2} = \frac{5e^{3x}}{(3-2)^2} \\
 &= \frac{5e^{3x}}{1^2} \\
 &= 5e^{3x}
 \end{aligned}$$

The particular integral corresponding to $4e^{2x}$ is given by

$$y_2 = \frac{4e^{2x}}{(D-2)^2} = \frac{4e^{2x}}{0}$$

Hence the short method doesn't apply however

$$\begin{aligned}
 y_2 &= \frac{4e^{2x}}{(D-2)(D-2)} \\
 &= \frac{1}{D-2} \left[\frac{1}{D-2} 4e^{2x} \right] \\
 &= \frac{1}{D-2} \left[e^{2x} \int e^{-2x} \cdot 4e^{2x} dx \right] \\
 &= \frac{1}{D-2} \left[e^{2x} \cdot 4 \int dx \right] \\
 &= \frac{1}{D-2} [4xe^{2x}] \\
 &= \frac{4}{D-2} [xe^{2x}] \\
 &= 4 \cdot e^{2x} \int e^{-2x} \cdot xe^{2x} dx \\
 &= 4e^{2x} \int x dx \\
 &= 4e^{2x} \cdot \frac{x^2}{2} \\
 &= 2x^2 e^{2x}
 \end{aligned}$$

Hence the general solution is $y_2 = c_1 e^{2x} + c_2 x e^{2x} + 5e^{3x} + 2x^2 e^{2x}$

Formula

$$\text{If } \frac{e^{ax}}{f(D)} = \frac{e^{ax}}{(D-a)^r} = \frac{x^r e^{ax}}{r!}$$

$$\text{Hence } \frac{4e^{2x}}{(D-2)^2} = \frac{4x^2}{2!} e^{2x}$$

Example 5

Solve the equation $(D - 1)(D + 3)(D - 5)y = \cosh x$

Solution

$$\text{Let } (D - 1)(D + 3)(D - 5)y = \frac{e^x + e^{-x}}{2}$$

The complementary function is $y = c_1 e^x + c_2 e^{-3x} + c_3 e^{5x}$

A particular integral corresponding to $\frac{e^x}{2}$ is given by

$$\begin{aligned} y &= \frac{e^x}{2(D-1)(D+3)(D-5)} \\ &= \frac{1}{2(D-1)} \left[\frac{e^x}{(1+3)(1-5)} \right] \\ &= \frac{1}{2(D-1)} \left(\frac{e^x}{-16} \right) \\ &= \frac{1}{-32} e^x \int e^{-x} \cdot e^x dx \\ &= -\frac{1}{32} x e^x \end{aligned}$$

The particular integral corresponding to $\frac{e^{-x}}{2}$ is given by

$$\begin{aligned} y &= \frac{e^{-x}}{2(D-1)(D+3)(D-5)} \\ &= \frac{e^{-x}}{2(-1-1)(-1+3)(-1-5)} \\ &= \frac{e^{-x}}{48} \end{aligned}$$

Hence the general solution of the equation is

$$y = c_1 e^x + c_2 e^{-3x} + c_3 e^{5x} + \frac{e^{-x}}{48} - \frac{1}{32} x e^x$$

7.4 Non homogeneous term is of the form $\sin(ax + b)$

Theorem 2

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = \sin(ax + b)$$

$$\begin{aligned} \text{or} \quad & (a_0 D^2 + a_1 D + a_2)y = \sin(ax + b) \\ \text{or} \quad & f(D)y = \sin(ax + b) \end{aligned}$$

$$\begin{aligned} \text{is} \quad & \frac{\sin(ax + b)}{f(D)} = \frac{\sin(ax + b)}{f(-a^2)} \quad \text{if } f(-a^2) \neq 0 \\ & \text{(or replace } D^2 \text{ by } -a^2) \end{aligned}$$

Proof

Suppose $y = \sin(ax + b)$

$$D^2 y = \frac{d^2 y}{dx^2} = -a^2 \sin(ax + b)$$

Then $f(D^2)y = f(-a^2)\sin(ax + b)$

$$\text{Hence } \frac{1}{f(D^2)}\sin(ax + b) = \frac{1}{f(-a^2)}\sin(ax + b) \quad \text{if } f(-a^2) \neq 0$$

Note that D^2 is replaced by $-a^2$.

In the same way,

$$\frac{1}{f(D^2)}\cos(ax + b) = \frac{\cos(ax + b)}{f(-a^2)}$$

$$\text{if } f(-a^2) \neq 0$$

Example 6

Solve $(D^2 + 9)y = \sin 5x$

The characteristic equation is $m^2 + 9 = 0$ whose roots are $m_1 = 3i$, $m_2 = -3i$

The complementary function is

$$Y_c = c_1 \cos 3x + c_2 \sin 3x.$$

A particular integral is

$$\begin{aligned} y_p &= \frac{\sin 5x}{f(D^2)} = \frac{\sin 5x}{D^2 + 9} \\ &= \frac{\sin 5x}{-(5)^2 + 9} \\ &= \frac{\sin 5x}{-16} \end{aligned}$$

$$\text{Hence the general solution is } y = c_1 \cos 3x + c_2 \sin 3x - \frac{1}{16} \sin 5x$$

Example 7

Solve the equation $(D^2 + 3D - 10)y = \sin 2x$

Solution

Let $(D^2 + 3D - 10)y = \sin 2x$

The characteristic equation is $(m^2 + 3D - 10) = 0$ or $(m - 2)(m + 5) = 0$, Hence $m_1 = 2$ and $m_2 = -5$

The complementary function is $y = c_1 e^{2x} + c_2 e^{-5x}$

The particular integral corresponding to $\sin 2x$ is given by

$$y = \frac{\sin 2x}{D^2 + 3D - 10}. \text{ This is of the form } y = \frac{\sin ax}{f(D)^2}$$

Here, only D^2 can be replaced by $-2^2 = -4$. Then

$$\begin{aligned} y &= \frac{\sin 2x}{-4 + 3D - 10} = \frac{\sin 2x}{3D - 14} \\ &= \frac{1}{3} \frac{\sin 2x}{\left(D - \frac{14}{3}\right)} \\ &= \frac{1}{3} \frac{\left(D + \frac{14}{3}\right) \sin 2x}{\left(D - \frac{14}{3}\right) \left(D + \frac{14}{3}\right)} \\ &= \frac{1}{3} \frac{\left(D - \frac{14}{3}\right) \sin 2x}{D^2 - \left(\frac{14}{3}\right)^2} \\ &= \frac{1}{3} \frac{\left(D - \frac{14}{3}\right) \sin 2x}{-4 - \frac{196}{9}} \\ &= \frac{9}{3(-36 - 196)} \left[\frac{d}{dx} \sin 2x + \frac{14}{3} \sin 2x \right] \\ &= \frac{3}{-232} \left(2 \cos 2x + \frac{14}{3} \sin 2x \right) \\ &= \frac{3}{-116} \cos 2x - \frac{7}{-116} \sin 2x \end{aligned}$$

7.5 Non-homogeneous term is of the form x^n

Theorem 3

The particular integral of

$$= a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = x^n$$

or $(a_0 D^2 + a_1 D + a_2)y = x^n$

or $f(D)y = x^n$

is now

$$\begin{aligned}
 y &= \frac{x^n}{f(D)} = [f(D)]^{-1} x^n \\
 &= [a_2 + a_1 D + a_0 D^2]^{-1} x^n \\
 &= a_2^{-1} \left[1 + \frac{a_1 D}{a_2} + \frac{a_0 D^2}{a_2} \right]^{-1} x^n \\
 &= a_2^{-1} \left[1 + \left(\frac{a_1 D}{a_2} + \frac{a_0 D^2}{a_2} \right) \right]^{-1} x^n
 \end{aligned}$$

Using Binomial theorem: $(1 + x)^{-1} = 1 - x + 2x^2 - \dots$

$1 + \left(\frac{a_1 D}{a_2} + \frac{a_0 D^2}{a_2} + \dots \right)$ is expanded up to the term D^n and operate on x^n

Example 8

Solve $(2D^2 + 2D + 3)y = x^2$

Solution

Let $(D^2 + D + 3)y = x^2$

The characteristic equation is $2m^2 + 2m + 3 = 0$ whose roots are

$$\begin{aligned}
 m &= -2 \pm \frac{\sqrt{4 - 4(2)(3)}}{2(2)} \\
 &= -1 \pm \frac{\sqrt{1 - 16}}{2} \quad \text{or} \quad m_1 = -1 + \frac{i\sqrt{5}}{2} \quad \text{or} \quad m_2 = -1 - \frac{i\sqrt{5}}{2}
 \end{aligned}$$

Hence the complementary function is

$$y = e^{-\frac{1}{2}x} \left(c_1 \cos \frac{\sqrt{5}}{2} x + c_2 \sin \frac{\sqrt{5}}{2} x \right)$$

and a particular integral is

$$y = \frac{1}{2D^2 + 2D + 3} x^2$$

Since the power of x is 2, we need up to the term D^2 only.

$$\text{Then } y = \frac{x^2}{3 \left(1 + \frac{2}{3} D + \frac{2D^2}{3} \right)}$$

$$\text{Or } y = \frac{1}{3} \left(1 + \frac{2D}{3} + \frac{2D^2}{3} \right)^{-1} x^2$$

$$\begin{aligned}
&= \frac{1}{3} \left[1 - \left(\frac{2D}{3} + \frac{2D^2}{3} \right) + \left(\frac{\frac{2D}{3} + \frac{2D^2}{3}}{2!} \right) - \dots \right] x^2 \\
&= \frac{1}{3} \left[1 - \frac{2D}{3} - \frac{2D^2}{3} + \frac{\frac{2D^2}{3}}{2!} + \dots \right] x^2 \\
&= \frac{1}{3} \left[1 - \frac{2D}{3} - \frac{2D^2}{3} + \frac{4D^2}{18} + \dots \right] x^2 \\
&= \frac{1}{3} \left[x^2 - \frac{2}{3} \cdot \frac{dx^2}{dx} - \frac{2}{27} \frac{dx^2}{dx} \right] \\
&= \frac{1}{3} \left[x^2 - \frac{2}{3} \cdot 2x - \frac{2}{27} (2) \right] \\
&= \frac{1}{3} x^2 - \frac{4}{9} x - \frac{4}{27}
\end{aligned}$$

the general solution is

$$y = e^{-\frac{1}{2}x} \left(c_1 \cos \frac{\sqrt{5}}{2} + c_2 \sin \frac{\sqrt{5}}{2} + \frac{1}{3} x^2 - \frac{4}{9} x - \frac{4}{27} \right)$$

7.6 Nonhomogeneous term is of the form $e^{ax}u$ where u is a function of x .

Theorem 4

The particular integral of

$$(a_0 D^2 + a_1 D + a_2)y = e^{ax}u$$

$$\text{or } f(D)y = e^{ax}u$$

$$\text{is given by } y = \frac{e^{ax}}{f(D)} = e^{ax} \cdot \left[\frac{u}{f(D+a)} \right]$$

Proof

Suppose $y = e^{ax}u$

$$Dy = a e^{ax}u + e^{ax} \frac{du}{dx}$$

$$= e^{ax}(au + Du)$$

$$= e^{ax}(D+a)u$$

$$D^2 y = a e^{ax}(D+a)u + e^{ax}D(D+a)u$$

$$= e^{ax}(D+a)^2 u$$

$$(D)e^{ax}u = e^{ax}(D+a)^2 u$$

Let $v = f(D+a)u$ then $u = \frac{v}{f(D+a)}$

Hence $f(D)e^{ax} \cdot \frac{v}{f(D+a)} = e^{ax}v$

Then $\frac{1}{f(D)}e^{ax}v = \frac{1}{f(D)} \left[f(D)e^{ax} \cdot \frac{1}{f(D+a)}v \right]$

$$= \frac{e^{ax}}{f(D+a)}$$

Example 9

Solve $(D-4)y = e^{3x}x^2$

Solution

We use the formula $\frac{1}{f(D)}e^{ax}u = e^{ax} \frac{1}{f(D+a)}u$ for finding a particular integral. The complementary function is $y = c_1e^{2x} + c_2e^{-2x}$ (you work out)

A particular integral is given by

$$y_1 = \frac{x^2 e^{3x}}{D^2 - 4} = e^{3x} \left[\frac{x^2}{(D+3)^2 - 4} \right]$$

$$= e^{3x} \left[\frac{x^2}{D^2 + 6D + 5} \right]$$

$$= e^{3x} \left[\frac{1}{5 \left(1 + \frac{6D}{5} + \frac{D^2}{5} \right)} \right] x^2$$

$$= \frac{e^{3x}}{5} \left[1 + \frac{6D}{5} + \frac{D^2}{5} \right]^{-1} x^2$$

$$\begin{aligned}
&= \frac{e^{3x}}{5} \left[1 - \frac{6D}{5} + \frac{31}{25} D^2 \right] x^2 \\
&= \frac{e^{3x}}{5} \left[x^2 - \frac{6}{5} \cdot 2x + \frac{31}{25} \cdot 2 \right] \\
&= e^{3x} \left[\frac{x^2}{5} - \frac{12x}{25} + \frac{62}{125} \right]
\end{aligned}$$

The general solution is $y = c_1 e^{2x} + c_2 e^{-2x} + e^{3x} \left(\frac{x^2}{5} - \frac{12x}{25} + \frac{62}{125} \right)$

7.7 Nonhomogeneous term is of the form xu where u is a function of x .

Theorem 5

The particular integral of $(a_0 D^2 + a_1 D + a_2)y = xu$

Or $f(D)y = xu$ (where u is a function of x) then the particular integral of $f(D)y = xu$ is

$$y = \frac{xu}{f(D)} = x \frac{1}{f(D)} u - \frac{f'(D)}{[f(D)]^2} u$$

Consider $(a_0 D^2 + a_1 D + a_2)y = xu$ or $f(D)y = xu$

Then the particular integral is $y = \frac{xu}{f(D)}$

Let

$$y = xu$$

$$y' = xu' + u$$

$$\begin{aligned}
y'' &= xu'' + u' + u' \\
&= xu'' + 2u'
\end{aligned}$$

$$f(D)xu = xf(D)u + f'(D)u$$

let $v = f(D)u$, then $u = \frac{1}{f(D)}v$

Hence $f(D).x. \frac{1}{f(D)}v = xf(D). \frac{1}{f(D)}v + f'(D). \frac{1}{f(D)}v$

$$\text{or } xv = f(D).x. \frac{1}{f(D)}v - f'(D). \frac{1}{f(D)}v$$

$$\begin{aligned}
\text{or } \frac{xu}{f(D)} &= x. \frac{1}{f(D)}u - \frac{1}{f(D)}f'(D). \frac{1}{f(D)}u \\
&= x \frac{1}{f(D)}u - \frac{f'(D)}{[f(D)]^2}u
\end{aligned}$$

Theorem 5

The particular integral of $(a_0 D^2 + a_1 D + a_2)y = xu$ or $f(D)y = xu$ (where u is a

function of x) is $y = \frac{xu}{f(D)} = x \cdot \frac{1}{f(D)}u - \frac{f'(D)}{[f(D)]^2}u$

Example 10

Solve $(D^2 - 1)y = xe^{3x}$.

Solution

Let $(D^2 - 1)y = x \cdot e^{3x}$ (1)

The characteristic equation is $m^2 - 1 = 0$ whose roots are $m_1 = 1$, $m_2 = -1$. Hence the complementary function is

$$y = c_1 e^x + c_2 e^{-x}.$$

A particular integral is given by

$$y_1 = \frac{x e^{3x}}{(D^2 - 1)} \quad (2)$$

We use the formula:

The particular integral of

$$y_1 = \frac{xv}{f(D)} \text{ is}$$

$$y_1 = x \left[\frac{v}{f(D)} \right] - \frac{f'(D)}{[f(D)]^2} v$$

then from (2) we have

$$y_1 = x \left[\frac{e^{3x}}{(D^2 - 1)} \right] - \frac{2D \cdot e^{3x}}{(D^2 - 1)^2}$$

$$= x \cdot \frac{e^{3x}}{8} - 2D \left[\frac{e^{3x}}{64} \right]$$

$$= x \frac{e^{3x}}{8} - \frac{2}{64} (3e^{3x})$$

Hence the general solution is $y_1 = c_1 e^x + c_2 e^{-x} + \frac{x e^{3x}}{8} - \frac{3}{32} e^{3x}$

Exercise 7

Solve the following equations:

1. $(D^2 + D + 1)y = e^{2x}$
2. $(D^2 - 1)y = e^x$
3. $(D^4 - 1)y = \sin 2x$
4. $(D^2 + 4D + 4)y = 2 \sin 2x$ (you may use: $2 \sin 2x = e^{2ix} - e^{-2ix}$)
5. $(D^2 + 4)y = \cos 2x + \cos 4x$

6. $(D^2 - 1)y = x^2$
7. $\frac{d^2 y}{dx^2} - 2\frac{dy}{dx} + 3y = \cos x + x^2$
8. $\frac{d^2 y}{dx^2} + y = e^{-x} + \cos x + x^3$
9. $(D^2 - 2D + 1)y = x^2 e^{3x}$
10. $\frac{d^2 y}{dx^2} + 2y = x^2 e^{3x} + e^x \cos 2x$
11. $\frac{d^2 y}{dx^2} - y = x^2 \cos x$
12. $\frac{d^2 y}{dx^2} - 2\frac{dy}{dx} + y = x \sin x$

Summary

You have learnt the following from this lesson.

Particular integrals for the following:

- (i) $y = \frac{1}{f(D)} e^{ax}$ when $f(a) \neq 0$ and $f(a) = 0$
- (ii) a. $y = \frac{1}{f(D^2)} \sin(ax + b) = \frac{1}{f(-a^2)} \sin(ax + b)$ when $f(-a^2) \neq 0$
 b. $y = \frac{1}{f(D^2)} \cos(ax + b) = \frac{1}{f(-a^2)} \cos(ax + b)$ when $f(-a^2) \neq 0$
- (iii) $y = \frac{1}{f(D)} x^n$ by expanding $[f(D)]^{-1}$ up to D^n and operate
- (iv) a. $y = \frac{1}{f(D)} e^{ax} u = e^{ax} \left[\frac{1}{f(D+a)} u \right]$
 b. $y = \frac{1}{f(D)} xu = x \left[\frac{1}{f(D)} u \right] - \frac{f'(D)}{[f(D)]^2} u$ when u is a function of x .

Further Reading

1. Differential Equations
 By Frank Ayres J.R Ph.D
 Schaum's outline series
 McGraw – Hill Book Company
 New York. Toronto – Sydney
2. Differential equations
 By Dr. D. Sengottaiyan Ph.D
 Oxford publications
 Harrow – London-Nairobi.

LESSON 8

Second Order Equations with Variable Coefficients

8.1 Introduction

in the previous lessons you have learnt various methods of solving second order differential equations with constant coefficients. In this lesson you will learn different methods of solving the second order equations with variable coefficients.

8.2 Objectives of this lesson

By the end of this lesson you should be able to

- i). Define the general form of a second order equation with variable coefficients.
- ii). Solve the second order equation with variable coefficients:
 - a) By changing the independent variable
 - b) By removal of the first derivative or by reduction of the equation to normal form.

8.3 Definition of second order Linear equations with variable coefficients.

An equation of the form

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = f(x), a_0 \neq 0$$

$$\text{or } \frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R$$

(where $\frac{a_1}{a_0} = P$, $\frac{a_2}{a_0} = Q$, and $\frac{f(x)}{a_0} = R$ are functions of x only) is called a second order linear differential equation with variable coefficients.

8.4 Solving the differential equation (Method I)

Transformation of equations by changing the independent variable

Theorem

$$\text{Let } \frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R \quad (1)$$

Where P, Q, R are functions of x be in standard form (coefficients of y'' is unity).

$$\text{Choosing } Q = \left(\frac{dz}{dx} \right)^2 \quad (2)$$

The equation is transformed into linear equation with constant coefficients of the form

$$\frac{d^2 y}{dx^2} + P_1 \frac{dy}{dx} + R_1$$

$$\text{where } P_1 = \frac{\frac{d^2 z}{dx^2} + P \frac{dz}{dx}}{\left(\frac{dz}{dx}\right)^2}, Q_1 = \frac{Q}{\left(\frac{dz}{dx}\right)^2} \text{ and } R_1 = \frac{R}{\left(\frac{dz}{dx}\right)^2} \quad (3)$$

Proof

$$\text{Let } \frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R \quad (1)$$

Where P, Q, R are functions of x. Let the independent variable x be changed from x to z where $z = f(x)$.

$$\text{Then } \frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} \quad (4)$$

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{d}{dx} \left(\frac{dy}{dx} \right) \\ &= \frac{d}{dx} \left[\frac{dy}{dz} \cdot \frac{dz}{dx} \right] \\ &= \frac{d}{dx} \left[\frac{dy}{dz} \right] \cdot \frac{dz}{dx} + \frac{dy}{dz} \cdot \frac{d^2 z}{dx^2} \\ &= \frac{d}{dz} \left(\frac{dy}{dz} \right) \cdot \frac{dz}{dx} \cdot \frac{dz}{dx} + \frac{dy}{dz} \cdot \frac{d^2 z}{dx^2} \\ &= \frac{d^2 y}{dz^2} \left(\frac{dz}{dx} \right)^2 + \frac{dy}{dz} \cdot \frac{d^2 z}{dx^2} \end{aligned} \quad (5)$$

Substituting (4) and (5) in (1) we have

$$\left(\frac{d^2 y}{dz^2} \right) \left(\frac{dz}{dx} \right)^2 + \left(\frac{d^2 z}{dx^2} + P \frac{dz}{dx} \right) \frac{dy}{dx} + Qy = R \quad (6)$$

Dividing (6) by $\left(\frac{dz}{dx}\right)^2$ we have

$$\begin{aligned} \frac{d^2 y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y &= R_1 \\ \text{where } P_1 &= \frac{\frac{d^2 z}{dx^2} + P \frac{dz}{dx}}{\left(\frac{dz}{dx}\right)^2}, Q_1 = \frac{Q}{\left(\frac{dz}{dx}\right)^2} \text{ and } R_1 = \frac{R}{\left(\frac{dz}{dx}\right)^2} \end{aligned} \quad (7)$$

Here P_1 , Q_1 and R_1 are function of x but they can be converted to functions of z. Suppose

$$\text{we take } Q = k f(x) = \left(\frac{dz}{dx}\right)^2 \text{ then } \frac{dz}{dx} = \sqrt{k f(x)}$$

or $z = \int (\sqrt{k f(x)}) dx$ is the Transformation.

The method of solving a second order equation with variable coefficients is illustrated in the following examples.

Example 1

Solve the equation $x \frac{d^2 y}{dx^2} - \frac{dy}{dx} + 4x^3 y = x^5$

Solution

Let $xy'' - y' + 4x^3 y = x^5$.

The standard form of (1) is $\frac{d^2 y}{dx^2} - \frac{1}{x} \frac{dy}{dx} + 4x^2 y = x^4$

Here $P = -\frac{1}{x}$, $Q = 4x^2$ and $R = x^4$.

Let $\left(\frac{dz}{dx}\right)^2 = Q$ or $\frac{dz}{dx} = \sqrt{4x^2}$

Then $dz = 2x dx$ and the transformation is $z = x^2$ (we need a relation between s and x and we don't need constant).

Now using this transformation the equation becomes, $\frac{d^2 y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = R_1$

Where $P_1 = \frac{\frac{d^2 z}{dx^2} + P \frac{dz}{dx}}{\left(\frac{dz}{dx}\right)^2} = \frac{2 + \left(-\frac{1}{x}\right) \cdot 2x}{4x^2} = 0$

$$Q_1 = \frac{Q}{\left(\frac{dz}{dx}\right)^2} = \frac{4x^2}{4x^2} = 1$$

$$R_1 = \frac{R}{\left(\frac{dz}{dx}\right)^2} = \frac{x^4}{4x^2} = \frac{x^2}{4} \text{ or } \frac{z}{4}$$

Then the equation to $\frac{d^2 y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = R$ is connected to $\frac{d^2 y}{dz^2} + 0 + y = \frac{z}{4}$ or

$$(D^2 + 1)y = \frac{z}{4}.$$

The complementary function

The characteristic equation is $m^2 + 1 = 0$ whose roots are $m_1 = i$, $m_2 = -i$. The complementary function is $y = A \cos z + B \sin z$

The particular integral is

$$\begin{aligned} y &= \frac{z}{4(D^2 + 1)} \\ y &= \frac{1}{D^2 + 1} \left(\frac{z}{4} \right) \\ &= \frac{1}{4} (1 + D^2)^{-1} z \\ &= \frac{1}{4} (1 - \dots\dots) z] \\ &= \frac{z}{4} \end{aligned}$$

Hence the general solution is

$$y = c_1 \cos z + c_2 \sin z + \frac{z}{4}$$

using $z = x^2$ we have,

$$y = c_1 \cos x^2 + c_2 \sin x^2 + \frac{x^2}{4} \text{ is general solution.}$$

8.5 Solving the Differentiation equation (Method II)

Theorem

Let the second order equation be

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R \quad (1)$$

can be reduced to the form

$$\frac{d^2 v}{dx^2} + sv = T \quad (2)$$

$$\text{where } S = Q - \frac{1}{4} P^2 - \frac{1}{2} \frac{dP}{dx} \quad (3)$$

$$\text{and } T = \frac{R}{u}.$$

Proof

Let the complete solution of (1) be $y = uv$ where u and v are functions of x . Differentiating $y = uv$ twice we have

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx} \quad (6)$$

$$\frac{d^2 y}{dx^2} = \frac{d^2 u}{dx^2} v + 2 \frac{du}{dx} \frac{dv}{dx} + u \frac{d^2 v}{dx^2} \quad (7)$$

Substituting $\frac{d^2y}{dx^2}$ and $\frac{du}{dx}$ in (1) we have

$$u \frac{d^2v}{dx^2} + \left(Pu + 2 \frac{du}{dx} \right) \frac{dv}{dx} + v \left(\frac{d^2u}{dx^2} + P \frac{du}{dx} + Qu \right) = R$$

$$\text{or } \frac{d^2v}{dx^2} + \left(P + \frac{2}{u} \frac{du}{dx} \right) \frac{dv}{dx} + \frac{1}{u} \left(\frac{d^2u}{dx^2} + P \frac{du}{dx} + Qu \right) v = \frac{R}{u}$$

to remove first derivative $\frac{dv}{dx}$ from (8) we take,

$$P + \frac{2}{u} \frac{du}{dx} = 0$$

$$\text{or } \frac{du}{u} = -\frac{1}{2} P dx \quad (9)$$

$$\text{or } \ln u = -\frac{1}{2} \int P dx$$

$$\text{or } u = e^{-\frac{1}{2} \int P dx}$$

Hence the required substitution is $y = u v$

Where $u = e^{-\frac{1}{2} \int P dx}$

More over the coefficient of v in (8) is

$$\frac{1}{u} \left(\frac{d^2u}{dx^2} + P \frac{du}{dx} + Qu \right) = Q - \frac{1}{4} P^2 - \frac{1}{2} \left(\frac{dp}{dx} \right) \quad (10)$$

(Since $\frac{du}{dx} = -\frac{1}{2} Pu$ from 9 and $\frac{d^2u}{dx^2} = -\frac{1}{2} P \frac{du}{dx} - \frac{1}{2} \frac{dp}{dx} u$)

Thus the equation (1) is transformed into

$$\frac{d^2v}{dx^2} + Sv = T$$

where $S = Q - \frac{1}{4} P^2 - \frac{1}{2} \frac{dp}{dx}$ and $T = \frac{R}{u}$.

Example 2

Solve $\frac{d^2y}{dx^2} - \frac{2}{x} \frac{dy}{dx} + \left(4 + \frac{2}{x^2} \right) y = 0$

Here $P = -\frac{2}{x}$, $Q = 4 + \frac{2}{x^2}$, $R = 0$

The substitution is now $-\int \frac{1}{2} P dx = -\frac{1}{2} \int -\frac{2}{x} dx$

$$= \int \frac{1}{x} dx$$

$$= \ln x$$

$$u = e^{-\int \frac{1}{2} P dx} = e^{\ln x} = x$$

$$\begin{aligned} S &= Q - \frac{p^2}{4} - \frac{1}{2} \frac{dp}{dx} \\ &= 4 + \frac{2}{x^2} - \frac{1}{4} \left(\frac{4}{x^2} \right) - \frac{1}{2} \frac{d}{dx} \left(\frac{-2}{x} \right) \\ &= 4 + \frac{2}{x^2} - \frac{1}{x^2} - \frac{1}{x^2} \\ &= 4 \end{aligned}$$

$$T = \frac{R}{u} = 0$$

$$\frac{d^2 v}{dx^2} + SV = T \text{ gives } \frac{d^2 v}{dx^2} + 4v = 0$$

The auxiliary equation is $m^2 + 4 = 0$, $m = \pm 2i$

$$v = A \cos 2x + B \sin 2x$$

The solution is

$$y = uv$$

$$y = x (A \cos 2x + B \sin 2x)$$

Example

$$\text{Solve } \frac{d^2 y}{dx^2} - 4x \frac{dy}{dx} + (4x^2 - 3)y = e^{x^2}$$

$$\text{Here } P = -4x, Q = 4x^2 - 3, R = e^{x^2}$$

$$\text{Then } -\int \frac{1}{2} P dx = \frac{-1}{2} \int -4x dx = 2 \frac{x^2}{2} = x^2$$

$$u = e^{-\int \frac{1}{2} P dx} = e^{x^2}$$

$$\begin{aligned} S &= Q - \frac{P^2}{4} - \frac{1}{2} \frac{dP}{dx} \\ &= 4x^2 - 3 - \frac{16x^2}{4} - \frac{1}{2} \frac{d}{dx} (-4x) \\ &= -1 \end{aligned}$$

$$R = \frac{R}{u} = \frac{e^{x^2}}{e^{x^2}} = 1$$

$$\frac{d^2 v}{dx^2} + Tv = S \text{ becomes}$$

$$\frac{d^2v}{dx^2} - v = 1$$

The auxiliary equation is

$$m^2 - 1 = 0, m = 1, -1$$

The complementary function is

$$v = c_1 e^x + c_2 e^{-x}$$

$$\text{P.I: } y = \frac{1}{D^2 - 1} = \frac{1e^{0x}}{D^2 - 1} = \frac{1}{0 - 1} = -1$$

$$v = CF + PI = c_1 e^x + c_2 e^{-x} - 1$$

The complete solution is $y = uv$

$$y = e^{x^2} (c_1 e^x + c_2 e^{-x} - 1)$$

Exercise

1) Solve $\frac{d^2y}{dx^2} - \frac{2}{x} \frac{dy}{dx} + \left(25 + \frac{2}{x^2}\right)y = 0$

2) Solve $\frac{d^2y}{dx^2} + \frac{2}{x} \frac{dy}{dx} - 16y = 0$

3) Solve $y'' + 4xy' + 4x^2y = 0$

4) Solve $y'' + 2xy' + (x^2 + 5)y = xe^{-\frac{1}{2}x^2}$

5) Reduce the differential equation

$$y'' - 4xy' + (4x^2 - 1)y = -3xe^{x^2} \sin 2x \text{ to normal form and hence find its solution.}$$

6) Reduce the equation

$$\frac{d^2y}{dx^2} - 4x \frac{dy}{dx} + 4x^2y = xe^{x^2} \text{ to normal form and hence solve the equation.}$$

7) Solve $y'' - 2 \tan xy' + 5y = 0$

8) Solve $\frac{d^2y}{dx^2} - 4x \frac{dy}{dx} + (4x^2 - 3)y = e^{x^2}$

9) Solve $y'' - \frac{2}{x}y' + 9y = 0$

10) Solve $y'' - \frac{2}{x}y' - 25y = 0$

Summary

You have learnt the following from this lesson

1. The general form of the second order linear equation with variable coefficients is given by $\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = R$
Where P, Q, R are functions of x.
2. The equation $y'' + Py' + Qy = R$ can be transformed into an equation of constant coefficients by choosing $Q = \left(\frac{dz}{dx}\right)^2$ and solved.
3. The equation $y'' + Py' + Qy = R$ can be transformed into an equation $\frac{d^2v}{dx^2} + Sv = I$ in which the first derivative is not present, using the transformation: $y = uv$ where $u = e^{-\frac{1}{2}\int Pdx}$

Further Reading

1. Differential Equations
By Frank Ayres
Schaum's outline series
McGraw – Hill Book Company
New York, Toronto, Sydney.
2. Elementary Differential Equations
By Dr. D. Sengottaiyan
Oxford Publications
Harrow. London. Nairobi.

LESSON 9

Linear Equations of Order 9

9.1 Introduction

You have already solved second order linear homogeneous and nonhomogeneous equations. Different methods of finding the particular integrals have also been studied. All those methods are applicable for solving n^{th} order equations or higher order equations. In this lesson we shall solve the equations of third, fourth and higher order linear equations.

9.2 Objectives of the lesson

By the end of this lesson you will be able to

- i). define an n^{th} order equation in general
- ii). recognize Cauchy-Euler Equation of order n .
- iii). recognize Legendre's Linear equations of order n .
- iv). apply Cauchy's and Legendre's Linear equation to solve Differential Equations.

9.3 Definition of a linear equation of order n

A linear differential equation of order n has the form

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = Q \quad (1)$$

where $a_0 \neq 0$ and $a_0, a_1, a_2, \dots, a_n$ and Q are functions of x only or constants.

If $Q = 0$, then the equation (1) is called homogeneous to indicate that all of the terms are of the same first degree in y and its derivatives.

Example 1

- i). $2x \frac{d^4 y}{dx^4} - 5x^3 \frac{dy}{dx} + 9xy = e^x$ is a nonhomogeneous linear equation of order 4 with variable coefficients.
- ii). $5x \frac{d^3 y}{dx^3} - 8x^3 \frac{d^2 y}{dx^2} + 10 \frac{dy}{dx} + 8y = 0$ is an example of **homogeneous** linear differential equation of order three.

9.4 Solution of n^{th} order linear equation.

The method of solving an n^{th} order linear equation with constant or variable coefficients is exactly the same as solving the second order equations.

The solution consists of a complementary function y_c and a particular integral y_p .

An n^{th} order equation will have n independent solutions $y = y_1, y = y_2, \dots, y_n$ and the general solution is $y = c_1 y_1 + c_2 y_2 + \dots + c_n y_n + \text{any particular integral}$. The method of solving the higher order equation is illustrated in the following examples.

Example 2

Solve the equation

$$\frac{d^3 y}{dx^3} - \frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = e^{3x}$$

Solution

Writing the equation in D notation we have

$$(D^3 - D - 4D + 4)y = e^{3x} \quad (1)$$

The auxiliary equation of (1) is $m^3 - m^2 - 4m + 4 = 0$

or $(m + 2)(m - 1)(m - 2) = 0$ whose roots are $m_1 = -2$, $m_2 = 1$ and $m_3 = 2$

Then the complementary function is

$$y_c = c_1 e^{-2x} + c_2 e^x + c_3 e^{2x} \quad (2)$$

The particular integral of (1) is given by

$$\begin{aligned} y_p &= \frac{e^{3x}}{f(D)} = \frac{e^{3x}}{D^3 - D^2 - 4D + 4} \\ &= \frac{e^{3x}}{3^3 - 3^2 - 4(3) + 4} \quad (\text{see lesson 7}) \\ &= \frac{e^{3x}}{10} \end{aligned}$$

Then the general solution of the given equation is

$$y = c_1 e^{-2x} + c_2 e^x + c_3 e^{2x} + \frac{e^{3x}}{10}$$

Example 3

Solve the equation

$$\frac{d^4 y}{dx^4} + 2 \frac{d^3 y}{dx^3} + \frac{d^2 y}{dx^2} = \sin 3x$$

Solution

The auxiliary equation is $m^4 + 2m^3 + m^2 = 0$ or $m^2(m + 1)^2 = 0$

Hence $m = 0, 0, -1, -1$.

Then the complementary function is $y_c = c_1 + c_2 x + c_3 e^{-x} + c_4 x e^{-x}$

The particular integral is

$$\begin{aligned}
y_c &= \frac{\sin 3x}{D^4 + 2D^3 + D^2} \\
&= \frac{\sin 3x}{(D^2)^2 + 2D^2 \cdot D + D^2} \\
&= \frac{\sin 3x}{(-9)^2 + 2(-9)D + (-9)} \\
&= \frac{\sin 3x}{81 - 18D - 9} = \frac{\sin 3x}{(-18D + 72)} \\
&= \frac{\sin 3x}{-18(D - 4)} \\
&= \frac{(D + 4)\sin 3x}{-18(D - 4)(D + 4)} \\
&= \frac{(D + 4)\sin 3x}{-18(D^2 - 16)} \\
&= \frac{(D + 4)\sin 3x}{-18(-9 - 16)} \\
&= \frac{1}{450} \left(\frac{d}{dx} \sin 3x + 4 \sin 3x \right) \\
&= \frac{1}{450} (3 \cos 3x + 4 \sin 3x)
\end{aligned}$$

Hence the general solution of the given equation is

$$y = c_1 + c_2 x + c_3 e^{-x} + c_4 x e^{-x} + \frac{1}{450} (3 \cos 3x + 4 \sin 3x)$$

10.5 Cauchy – Euler Equation

An equation of the form

$$x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} x \frac{dy}{dx} + a_n y = f(x)$$

where $a_1, a_2 \dots a_{n-1}, a_n$ are functions of x or constants is usually called Cauchy's Equation or Cauchy – Euler equation of order n .

Note that each term of Cauchy's equation is a constant times $x^r \frac{d^r y}{dx^r}$

10.6 Solution of Cauchy – Euler Equation

By a change of the independent variable x Cauchy's equation can be transformed into an equation with constant coefficients which can be solved by familiar methods:

If we set $x = e^z$, the $\frac{dx}{dz} = e^z$ or $\frac{dz}{dx} = e^{-z}$ on writing $D = \frac{d}{dz}$ we get

$$\begin{aligned}
 y' &= \frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = e^{-z} Dy \\
 y'' &= \frac{dy'}{dx} = \frac{dy'}{dz} \cdot \frac{dz}{dx} = e^{-z} Dy' \\
 &= e^{-2z} (D^2 - D)y \\
 &= e^{-2z} D(D-1)y
 \end{aligned}$$

In a similar way

$$\frac{d^n y}{dx^n} = e^{-nz} D(D-1)(D-2)\dots(D-n+1)y$$

Substituting for $y', y'', \dots y^{(n)}$

Cauchy's equation becomes,

$$[D(D-1)(D-2)\dots(D-n+1) + a_1 D(D-1)\dots(D-n+2) + \dots + a_{n-1} D + a_n]y = f(e^z) \quad (4)$$

If a solution of (4) is denoted by $y = Q(z)$ then the solution of the original equation is

$y = Q(\ln x)$ since $x = e^z$ gives $z = \ln x$.

The method of solving Cauchy – Euler equation is illustrated in the following examples.

Since the nonhomogeneous term is ze^z the particular integral is of the form $z^4 e^z$. Note that the integral of e^z is same after integration of n times and the integral of z is z^4 after 3 times integration.

Let us assume $y = a z^4 e^z$ as a particular integral. Since the particular integral is a solution of $(D^3 - 3D^2 + 3D - 1)y = ze^z$

We substitute $y = az^4 e^z$ in the above equation to get, $a = \frac{1}{24}$.

Hence the general solution of (1) is $y = (c_1 + c_2 z + c_3 z^2)e^z + \frac{1}{24} z^4 e^z$

Note: If Cauchy's equation is homogeneous (if the right member $f(x) = 0$ then the substitution $y = x^m$ will yield a general solution easily.

Example 4

Solve the equation,

$$x^2 y'' + 2xy' = 0$$

Solution

$$\text{Let } x^2 y'' + 2xy' = 0 \quad (1)$$

$$\text{Put } y = x^m \quad (2)$$

Example 5

Find the general solution of

$$x^3 \frac{d^3 y}{dx^3} + x \frac{dy}{dx} - y = x \ln x$$

Solution

$$x^3 y''' + xy' - y = x \ln x \quad (1)$$

This is Cauchy's equation of order 3 and we put $x = e^z$ or $z = \ln x$. Then the equation becomes

$$D(D-1)(D-2)y + (D-1)y = e^z \cdot z$$

$$\text{or } (D-1)[D(D-2)+1]y = ze^z$$

$$\text{or } (D-1)(D-1)(D-1)y = ze^z$$

The roots of the auxiliary equation $(m-1)(m-1)(m-1) = 0$ are $m_1 = m_2 = m_3 = 1$.

Hence the solution of the associated homogeneous equation is, $y = c_1 e^z + c_2 z e^z + c_3 z^2 e^z$

$$\text{or } y = (c_1 + c_2 z + c_3 z^2) e^z$$

To find a particular integral

$$\text{Put } y = x^m$$

$$\text{Then } y' = mx^{m-1} \quad (3)$$

$$y'' = m(m-1)x^{m-2} \quad (4)$$

Substitute (3) and (4) in the given equation to get

$$m(m-1)x^m + 2mx^m = 0$$

$$\text{or } m(m-1) + 2m = 0$$

$$\text{or } m^2 - m + 2m = 0$$

$$\text{or } m(m-1) = 0, \text{ then } m = 0 \text{ or } m = -1$$

Then $y = x^m$ gives either $y = x^0 = 1$ or $y = x^{-1}$. 1 and x^{-1} are independent solution.

Then $y = c_1 + c_2 x^{-1}$ is the general solution of the given equation.

Legendre linear equation of order n

An equation of the form

$$(ax+b)^n \frac{d^n y}{dx^n} + a_1(ax+b)^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1}(ax+b) \frac{dy}{dx} + a_n y = f(x)$$

where $a_1, a_2, \dots, a_n$ and $f(x)$ are function of x only or constants is called the Legendre's linear equation.

Cauchy-Euler equation is a particular case of Legendre's linear equation when we put $ax+b$ as just x .

to solve Cauchy's equation we put $x = e^z$ and $\frac{x^r dy^r}{dx^r} = D(D-1)(D-2)\dots(D-r+1)y$

In the same way when we put $(ax+b) = e^z$ then

$$(ax+b)^r \frac{dy^r}{dx^r} = a^r D(D-1)(D-2)\dots(D-r+1)y$$

The following example illustrates the method of solving Legendre's linear differential equations.

Example 6

Solve the equation

$$(2x+1)^2 \frac{d^2 y}{dx^2} - 2(2x+1)y' - 12y = 6x \quad (1)$$

Putting $2x+1 = e^z$

The equation is reduced to

$$2^2 D(D-1)y - 2 \cdot 2Dy - 12y = ze^z - z$$

$$\text{or } [4D(D-1) - 4D - 12]y = 3e^z - 3$$

$$\text{or } (4D(D-1) - 4D - 12)y = 3e^z - 3$$

$$\text{or } (D^2 - 2D - 3)y = \frac{3}{4}e^z - \frac{3}{4}$$

$$\text{or } (D-3)(D+1)y = \frac{3}{4}e^z - \frac{3}{4}$$

The complimentary function is

$$y = c_1 e^{3z} + c_2 e^{-z}$$

The particular integral corresponding to $\frac{3}{4}e^z$ is given by:

$$\frac{\frac{3}{4}e^z}{(D-3)(D+1)} = \frac{3e^z}{4(1-3)(1+1)}$$

Example 4

Solve the equation.

$$(3x+2)^2 \frac{d^2 y}{dx^2} + 3(3x+2) \frac{dy}{dx} - 36y = 3x^2 + 4x + 1$$

Solution

$$\text{Let } (3x+2)^2 D^2 + 3(3x+2)D - 36y = 3x^2 + 4x + 1$$

Using the transformation $3x+2 = e^z$ the equation (1) reduces to

$$[3^2(D)(D-1) + 3 \cdot 3D - 36]y = 3x^2 + 4x + 1 \text{ where } D = \frac{d}{dz}$$

$$\text{or } 9(D^2 - 4)y = 3x^2 + 4x + 1 = \frac{9x^2 + 12x + 3}{3}$$

$$\begin{aligned} \text{or } (D^2 - 4)y &= \frac{9x^2 + 12x + 3}{27} \\ &= \frac{9x^2 + 12x + 3}{27} = (3x+2)^2 - 1 \\ &= \frac{e^{2z} - 1}{27} \end{aligned}$$

The Auxiliary equation is $m^2 - 4 = 0$ whose roots are $m_1 = 2, m_2 = -2$

The complementary function is

$$y_c = c_1 e^{2z} + c_2 e^{-2z}$$

The particular integral is

$$\begin{aligned} y_p &= \frac{e^{2z} - 1}{D^2 - 4} \\ &= \frac{e^{2z}}{D^2 - 4} - \frac{1e^0}{D^2 - 4} \\ &= \frac{e^{2z}}{(D-2)(D+2)} - \frac{1}{0-4} \end{aligned}$$

$$\begin{aligned} 27y_p &= \frac{e^{2z}}{4(D-2)} \\ &= \frac{1}{4} e^{2z} \int e^{-2z} \cdot e^{2z} dz \quad \text{Note that } \frac{1}{D-\alpha} Q = e^\alpha \int e^{-\alpha} Q_{dx} \\ &= \frac{1}{4} e^{2z} \cdot \int dz \\ &= \frac{ze^{2z}}{4} + \frac{1}{4} \end{aligned}$$

$$y_p = \frac{1}{108} (ze^{2z} + 1)$$



Microsoft Equation
3.0

LESSON 10

Reduction of nth order Equations to a First order system of n equations and solve the system.

10.1 Introduction

You have already studied different methods of solving nth order differential equations. In this LESSON you will know a new method of solving an nth order equation.

First we reduce an nth order equation into a First order system of n linear (simultaneous) equations using new variables and then we solve the n linear equations using the matrix method. We revise a section of matrix algebra to find the Eigen values and the Eigen vectors of a square matrix and use them to solve the n linear simultaneous equations. In this LESSON we also solve nth order equations by reducing the order of the equations.

10.2 Objectives of the LESSON

- by the end of this LESSON you should be able to: -
- reduce an nth order equation to a first order system of n equations.
- write the first order system of equations in matrix form.
- determine the Eigen values and the Eigen vectors of the matrix obtained.
- determine the solution of the nth order homogeneous and non homogeneous equation using the Eigen values and the Eigen vectors of the matrix.

10.3 Reduction of nth order equations to a first order system

We introduce the new variables:

$$\begin{aligned}y' &= u \\u' &= v (= y'') \\v' &= w (= y''') \\w' &= z (= y^{(4)}) \\&\dots\dots\dots\end{aligned}$$

and reduce the nth order equation to a system of n linear equations. The following examples will illustrate the method of reducing the equations to a first order system.

Example 1

Reduce the fourth order equation

$$\frac{d^4 y}{dx^4} - 5 \frac{d^3 y}{dx^3} + 7 \frac{d^2 y}{dx^2} + 9 \frac{dy}{dx} - 6y = e^x \quad (A)$$

$$\text{or } y^{(4)} = 5y''' - 7y'' + 9y' + 6y + e^x \quad (B)$$

to a system of four linear equations.

Solution

We introduce the new variables

$$\begin{aligned}
y' &= \frac{dy}{dx} = u \\
u' &= \frac{d^2y}{dx^2} = v \\
v' &= \frac{d^3y}{dx^3} = w \\
w' &= \frac{d^4y}{dx^4}
\end{aligned} \tag{2}$$

Then the fourth order equation (1) becomes, a first order system of four equations as shown below:

$$y' = u \tag{1}$$

$$u' = v \tag{2}$$

$$v' = w \tag{3}$$

$$w' = 5w - 7v - 9u + 6y + e^x \tag{4}$$

Thus we have reduced the fourth order equation (A) or (B) to a first order system of four equations (1) to (4).

Example 2

Reduce the third order equation

$$\frac{d^3y}{dx^3} + 8\frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 7y = 2x^3 \tag{1}$$

to a first order system of 3 equations.

Solution

Using the new variables

$$y' = u$$

$$u' = v$$

$$\text{and } v' = y'''$$

$$\frac{d^3y}{dx^3} + 8\frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 7y = 2x^3 \quad \text{becomes the system:}$$

$$y' = u$$

$$u' = v$$

$$v' = -8v + 10u + 7y + 2x^3 \quad \text{from equation (1)}$$

Example 3

Reduce the equation

$$y'''' = -y \quad \text{into a system of four linear equations.}$$

Solution

Let

$$y' = u$$

$$u' = v$$

$$v' = w$$

$$w' = \dots$$

Then $y''' = -y$ is reduced to the following four linear equations.

$$y' = u \quad (1)$$

$$u' = v \quad (2)$$

$$v' = w \quad (3)$$

$$w' = -y \quad (4)$$

10.4 System of equations in matrix form

We shall study some three important theorems (without proof) before we consider the solution of a system of equations. Consider the system of equations $X' = AX$ where A is

a square matrix and $X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$. The following theorems give a relationship connecting

the Eigen values and the Eigen vectors of the matrix A and the solutions of the system.

A system of n linear equations are written conveniently in a matrix form as

$X' = AX$ if the equations are homogeneous or $X' = AX + B$ if the equations are non homogeneous. Here A and B are matrices.

Consider the two linear equations

$$\frac{dx}{dt} = 4y + 3x \quad (1)$$

$$\text{and } \frac{dy}{dx} = -2x - 3y \quad (2)$$

The system of equations (1) and (2) can be written in a matrix form.

$$\begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} = \begin{pmatrix} 3 & 4 \\ -2 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \quad (3)$$

Note that the order of left side is x and then y. Hence each row of the matrix should be in the same order x and y. The column vector multiplying the matrix should also be of the same order x and y.

More over (3) can be written as $\begin{pmatrix} x' \\ y' \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix}$ or $\boxed{X' = AX}$

Here $X = \begin{pmatrix} x \\ y \end{pmatrix}$, $A = \begin{pmatrix} 3 & 4 \\ -2 & -3 \end{pmatrix}$ and $X' = \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}'$

Note

Generally when we write $X' = AX$,

$$X' = \begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \\ \vdots \\ x'_n \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix}'$$

A is a square matrix of order $n \times n$ and $X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix}$

Example 4

Consider the first order system of three equations:

$$y' = u \quad (1)$$

$$u' = v \quad (2)$$

$$v' = -8v + 9u + 7y \quad (3)$$

These three equations can be written in matrix form as,

$$\begin{pmatrix} y' \\ u' \\ v' \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 7 & 9 & -8 \end{pmatrix} \begin{pmatrix} y \\ u \\ v \end{pmatrix} \quad \text{or} \quad X' = AX$$

$$\text{where } X' = \begin{pmatrix} y' \\ u' \\ v' \end{pmatrix}, A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 7 & 9 & -8 \end{pmatrix}, X = \begin{pmatrix} y \\ u \\ v \end{pmatrix}$$

How do you write the first row of the matrix A?

$y' = u$ is written as $y' = 0y + 1u + 0v$

Each row of the matrix should be written in the same order y, u, and v. Their coefficients should be written in each row in the order y, u and v.

The second row is $u' = v$ is written as $u' = 0y + 0u + 1v$. The third row $v' = -8v + 9u + 7y$. Writing this third row as $v' = 7y + 9u - 8v$, the coefficient of the third row of the matrix is 7, 9, and -8.

The following examples will illustrate the method of writing a system of equations into a matrix form.

Example 5

Consider the system of equations,

$$X'_1 = X_1 - X_2 - 4X_3 \quad (1)$$

$$X'_2 = -2X_2 + 3X_3 \quad (2)$$

$$X'_3 = X_3 + 8X_2 \quad (3)$$

Write down the three equations in matrix form.

Solution

Right hand side of the three equations should be in the same order as the left side namely X_1, X_2, X_3

Then,

$$X'_1 = X_1 - X_2 - 4X_3 \quad (4)$$

$$X'_2 = 0 - 2X_2 + 3X_3 \quad (5)$$

$$X'_3 = 0 + 8X_2 + X_3 \quad (6)$$

The system of three linear equations (4), (5) and (6) is written in matrix form as

$$X' = AX$$

$$\text{where } X' = \begin{pmatrix} X'_1 \\ X'_2 \\ X'_3 \end{pmatrix}, A = \begin{pmatrix} 1 & -1 & -4 \\ 0 & -2 & 3 \\ 0 & 8 & 1 \end{pmatrix} \text{ and } X = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}$$

Example 6

Consider the equation

$$\frac{d^4 y}{dx^4} - 3 \frac{d^3 y}{dx^3} + 2 \frac{dy}{dx} - 8y = 0$$

- Reduce the equation to a first order system of four equations.
- Write down the system in matrix form.

Solution

Let

$$\begin{aligned} y' &= u \\ u' &= v \\ v' &= w \text{ and} \\ w' &= y''' \end{aligned}$$

Then $\frac{d^4 y}{dx^4} - 3 \frac{d^3 y}{dx^3} + 5 \frac{d^2 y}{dx^2} + 8y = 0$ becomes

$$y' = u \quad (1)$$

$$u' = v \quad (2)$$

$$v' = w \quad (3)$$

$$w' = 3w + 5v - 2u - 8y \quad (4)$$

or

$$y' = 0y + 1u + 0v + 0w \quad (5)$$

$$u' = 0y + 0u + 1v + 0w \quad (6)$$

$$v' = 0y + 0u + 0v + 1w \quad (7)$$

$$w' = -8y - 2u - 5v + 3w \quad (8)$$

The system of equations (5), (6), (7) and (8) can be written as

$$X' = AX$$

Where $X' = \begin{pmatrix} y' \\ u' \\ v' \\ w' \end{pmatrix}$, $A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -8 & -2 & -5 & 3 \end{pmatrix}$ and $X = \begin{pmatrix} y \\ u \\ v \\ w \end{pmatrix}$

10.5 Solution of First order system of equations using matrix form.

It is assumed that the reader is familiar with some knowledge of matrix. However we shall explain a particular section of matrix namely determination of Eigen values and Eigen vectors of a square matrix. They are required to find the solution of the first order system of equations.

10.6 Characteristic equation and the Eigen value of a square matrix.

Let A be a square matrix of order n x n and I be the unit matrix of the same order. The **determinant** $|A - mI| = 0$ (1)
where m is a constant is called the characteristic equation of A

In equation (1) m is unknown and m is called the Eigen value of the matrix A.
For example consider the matrix

$$A = \begin{pmatrix} 3 & 4 \\ -2 & -3 \end{pmatrix}$$

$$|A - mI| = \left| \begin{pmatrix} 3 & 4 \\ -2 & -3 \end{pmatrix} - m \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right| = \begin{vmatrix} 3-m & 4 \\ -2 & -3-m \end{vmatrix} = 0$$

The determinant $|A - mI| = 0$ or $\begin{vmatrix} 3-m & 4 \\ -2 & -3-m \end{vmatrix} = 0$ (2)

is called the characteristic equation of the matrix A and m is called the Eigen values of A.

Expanding the determinant in (2) we have $m^2 - 1 = 0$ or $m_1 = 1, m_2 = -1$.

Thus the Eigen values of the matrix A are $m_1 = 1$ and $m_2 = -1$

10.7 Eigen vector of a matrix

Let $\begin{pmatrix} a \\ b \end{pmatrix}$ be a column vector and $\begin{vmatrix} 3-m & 4 \\ -2 & -3-m \end{vmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0$, for one Eigen value (say $m_1 = 1$). Then the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is called the Eigen vector of A corresponding to one value of m.

In this example corresponding to the Eigen value $m_1 = 1$

$$\begin{vmatrix} 3-1 & 4 \\ -2 & -3-1 \end{vmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{ or } \begin{vmatrix} 2 & 4 \\ -2 & -4 \end{vmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{ gives } 2a + 4b = 0 \text{ and } -2a - 4b = 0.$$

Both the equations are the same as $a + 2b = 0$. we can choose any value for a and find the corresponding value for b; here if $a = 2, b = -1$. Hence the vector $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ or $k \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ where $k \neq 0$ is an Eigen vector. Similarly, corresponding to the Eigen value $m_2 = -1$ the Eigen vector $\begin{pmatrix} a_1 \\ b_1 \end{pmatrix}$ is given from $\begin{vmatrix} 3-(-1) & 4 \\ -2 & -3-(-1) \end{vmatrix} \begin{pmatrix} a_1 \\ b_1 \end{pmatrix} = 0$

$$\text{or } \begin{vmatrix} 4 & 4 \\ -2 & -2 \end{vmatrix} \begin{pmatrix} a_1 \\ b_1 \end{pmatrix} = 0 \text{ or}$$

$$\left. \begin{array}{l} 4a_1 + 4b_1 = 0 \\ -2a_1 - 2b_1 = 0 \end{array} \right\} \text{ or } a_1 + b_1 = 0$$

If $a = 1, b = -1$

Thus the second Eigen vector corresponding to $m_2 = -1$ is $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

Conclusion

If $A = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ is a square matrix, then

$$\text{i). } \begin{vmatrix} p-m & q \\ r & s-m \end{vmatrix} = 0 \text{ or } (p-m)(s-m) - qr = 0 \quad (1)$$

(1) is called the characteristic equation of A.

ii). Solving the equation (1) we get the values of m and these values are called the Eigen values of A.

iii). Let m_1 is one Eigen value and $\begin{pmatrix} a \\ b \end{pmatrix}$ is the Eigen vector corresponding to m_1 . Then

$$\begin{vmatrix} p-m_1 & q \\ r & s-m_1 \end{vmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0 \text{ gives the Eigen vector } \begin{pmatrix} a \\ b \end{pmatrix}$$

Example 7

Let $A = \begin{pmatrix} 12 & -15 \\ 4 & -4 \end{pmatrix}$ be a square matrix

- Write down the characteristic equation of A.
- Calculate the Eigen values of A.
- Determine the Eigen vectors of A.
- Check using the Wronskian whether the two Eigen vectors are independent.

Solution

$$\text{i. Let } A = \begin{pmatrix} 12 & -15 \\ 4 & -4 \end{pmatrix}$$

The characteristic equation is

$$\begin{vmatrix} 12-m & -15 \\ 4 & -4-m \end{vmatrix} = 0$$

$$\text{or } (12-m)(-4-m) + 60 = 0$$

$$\text{or } m^2 - 8m + 12 = 0$$

$$\text{or } (m - 2)(m - 6) = 0$$

ii. $m_1 = 2$ and $m_2 = 6$ are the Eigen values of A.

iii. Let the Eigen vector corresponding to $m_1 = 2$ be $\begin{pmatrix} a \\ b \end{pmatrix}$

$$\begin{pmatrix} a \\ b \end{pmatrix} \text{ is obtained from } \begin{pmatrix} 12-2 & -15 \\ 4 & -4-2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0$$

$$\text{or } \begin{pmatrix} 10 & -15 \\ 4 & -6 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0$$

$$\text{or } \begin{cases} 10a - 15b = 0 \\ 4a - 6b = 0 \end{cases} \text{ or both the equation are reduced to } 2a - 3b = 0 \text{ or } 2a = 3b.$$

If we choose $b = 2$, $a = 3$.

Hence the Eigen vector corresponding to $m_1 = 2$ is given by $\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$.

Let the Eigen vector corresponding to $m_2 = 6$ be $\begin{pmatrix} a_1 \\ b_1 \end{pmatrix}$. It is obtained from

$$\begin{pmatrix} 12-6 & -15 \\ 4 & -4-6 \end{pmatrix} \begin{pmatrix} a_1 \\ b_1 \end{pmatrix} = 0$$

$$\text{or } \begin{cases} 6a_1 - 15b_1 = 0 \\ 4a_1 - 10b_1 = 0 \end{cases} \text{ or } 2a_1 - 5b_1 = 0 \text{ or } 2a_1 = 5b_1$$

if $b = 2$ then $a = 5$

Hence the Eigen vector corresponding to $m_2 = 6$ is given by $\begin{pmatrix} a_1 \\ b_1 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$.

iv. If the two Eigen vectors $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 5 \\ 2 \end{pmatrix}$ are independent then the Wronskian of

$$\begin{pmatrix} 3 & 5 \\ 2 & 2 \end{pmatrix} \text{ or } W = \begin{vmatrix} 3 & 5 \\ 2 & 2 \end{vmatrix} \text{ should not be zero.}$$

$$W = (6 - 10) = -4 \text{ which is not zero.}$$

Note: If the Eigen vectors are distinct (not repeated) then they are always independent.

10.8 Solution of the First order system of Equations.

In order to solve the First order System of equations we use the fundamental Theorem which states the relationship between the solution of the First order system of equations $X' = AX$ and the Eigen values and the Eigen vectors of the matrix A.

Theorem

Let the first order system of equations be represented in matrix form as $X' = AX$. where A is a square matrix of order $n \times n$. If the Eigen values of A are $m_1, m_2, \dots, m_n$ (all distinct) and their corresponding Eigen vectors are respectively

$$\begin{pmatrix} a_1 \\ a_2 \\ \cdot \\ \cdot \\ a_n \end{pmatrix}, \begin{pmatrix} b_1 \\ b_2 \\ \cdot \\ \cdot \\ b_n \end{pmatrix} \dots \dots \dots \begin{pmatrix} l_1 \\ l_2 \\ \cdot \\ \cdot \\ l_n \end{pmatrix}$$

then the general solution of the equation $X' = AX$ is

$$X = C_1 \begin{pmatrix} a_1 \\ a_2 \\ \cdot \\ \cdot \\ a_n \end{pmatrix} e^{m_1 t} + C_2 \begin{pmatrix} b_1 \\ b_2 \\ \cdot \\ \cdot \\ b_n \end{pmatrix} e^{m_2 t} + \dots C_n \begin{pmatrix} l_1 \\ l_2 \\ \cdot \\ \cdot \\ l_n \end{pmatrix} e^{m_n t}$$

Theorem 1: Eigen values are distinct and real

If m_1, m_2 and $m_3, \dots$ (all distinct) are the Eigen values and v_1, v_2 , and v_3 are the corresponding Eigen vectors respectively then the general solution of $X' = AX$ is given by

$$X = c_1 v_1 e^{m_1 t} + c_2 v_2 e^{m_2 t} + c_3 v_3 e^{m_3 t}$$

Theorem 2 Eigen values real and repeated

If the Eigen values m_1 and m_2 are repeated so that $m_1 = m_2 = m$, then one solution of the system $X' = AX$ is given by

$$X_1 = v_1 e^{mt}$$

and the second solution is

$$X_2 = v_1 t e^{mt} + \begin{pmatrix} a \\ b \end{pmatrix} e^{mt}$$

where the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is to be determined by substituting X_2 in the equation $X'_2 = AX_2$.

Then the general solution is

$$X = c_1 v_1 e^{m_1 t} + c_2 v_1 t e^{m_1 t} + \begin{pmatrix} a \\ b \end{pmatrix} e^{m_1 t}$$

Theorem 3 The solution of the nonhomogeneous systems.

If $X' = AX + B$ is a nonhomogeneous system of equations where A is a constant matrix and B is a vector function of t , we must find a particular solution X_p of the system and add it to the general solution X_c of the associated homogeneous system $X' = AX$

If $X_c = c_1 v_1 e^{m_1 t} + c_2 v_2 e^{m_2 t}$ then
 $X_p = g_1 v_1 e^{m_1 t} + g_2 v_2 e^{m_2 t}$ where g_1 and g_2 are to be determined using the vector function B .

Example 8

Consider the first order system

$$\frac{dx}{dt} = 3x + 4y \quad (1)$$

$$\frac{dy}{dt} = -2x - 3y \quad (2)$$

These two equations of the first order system is first written in matrix form as

$$X' = AX \quad (3)$$

where $X' = \begin{pmatrix} x \\ y \end{pmatrix}'$, $A = \begin{pmatrix} 3 & 4 \\ -2 & -3 \end{pmatrix}$ and $X = \begin{pmatrix} x \\ y \end{pmatrix}$.

The Eigen values of A are $m_1 = 1$, $m_2 = -1$ (see example 2.7)

The Eigen vectors corresponding to Eigen values $m_1 = 1$ and $m_2 = -1$ are $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ respectively.

Since they are not repeated they are independent.

Then the general solution of the system (1) and (2) is given by

$$X = C_1 \begin{pmatrix} 2 \\ -1 \end{pmatrix} e^t + C_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t} = \begin{pmatrix} x \\ y \end{pmatrix}$$

Hence $x = 2C_1 e^t + C_2 e^{-t}$
 $y = -C_1 e^t - C_2 e^{-t}$

Example 9

Consider the system $X' = AX$ where $A = \begin{pmatrix} 4 & -1 \\ -4 & 4 \end{pmatrix}$

Determine the following

- i). The characteristic equations A
- ii). The Eigen values of A.
- iii). The Eigen vector corresponding to each Eigen value.
- iv). Show that the two solutions are independent using the Wronskian of the two solutions.
- v). Write down the general solution of the equation $X' = AX$.

Solution

i). The characteristic equation of A is $\begin{vmatrix} 4-m & -1 \\ -4 & 4-m \end{vmatrix} = 0$ or $m^2 - 8m + 12 = 0$ or $(m-2)(m-6) = 0$

ii). The Eigen values of A are $m_1 = 2$ and $m_2 = 6$.

iii). If the Eigen vector of A corresponding to $m_1 = 2$ is $\begin{pmatrix} a \\ b \end{pmatrix}$, then

$$\begin{pmatrix} 2 & -1 \\ -4 & 2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0$$

or $2a - b = 0$

If we choose $a = 1, b = 2$

Then $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is one Eigen vector when $m = 2$. Similarly, when $m_2 = 6$ and the Eigen vector is $\begin{pmatrix} a \\ b \end{pmatrix}$ then $\begin{pmatrix} -2 & -1 \\ -4 & -2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0$ gives $-2a - b = 0$. We choose $a = 1, b = -2$.

Then $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$ is the second Eigen vector corresponding to the Eigen values $m_1 = 2$

and $m_2 = 6$ are $x_1 = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t}$ and $x_2 = c_2 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{6t}$

The Wronskian of x_1 and x_2 are $w = w(x_1, x_2) = \begin{vmatrix} 1e^{2t} & 1e^{6t} \\ 2e^{2t} & -2e^{6t} \end{vmatrix}$

for all values of t.

$$\text{then } w = \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix} \text{ for } t = 0$$

$$= -4 \neq 0$$

Hence x_1 and x_2 are independent and the general solution of $X' = AX$ is

$$X = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{6t}$$

Example 10

Consider the system of equations

$$x' = x - y - z \quad (1)$$

$$y' = 3y + 3z \quad (2)$$

$$z' = 3y + z \quad (3)$$

- Write the system of equations in matrix form $X' = AX$.
- What is the characteristic form of equation and the Eigen value of the matrix
- Calculate the Eigen vectors of the matrix?
- Write down the three solutions of the system
- Show that the three solutions are independent.
- Write down the general solution of the equation

Solution.

- a) The matrix form of the system is given by: $X' = AX$ where $X' = \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}$,

$$A = \begin{pmatrix} 1 & -1 & -1 \\ 0 & 3 & 3 \\ 0 & 3 & 1 \end{pmatrix} \text{ and } X = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad (1)$$

- b) The characteristic equation of A is given by $\begin{vmatrix} 1-m & -1 & -1 \\ 0 & 1-m & 3 \\ 0 & 3 & 1-m \end{vmatrix} = 0$ or

$$(1-m)(m-4)(m+2) = 0$$

The Eigen values of A are $m = 1$, $m = 4$, $m = -2$

- c) For $m_1 = 1$, the Eigen vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ are given by $\begin{pmatrix} 0 & -1 & -1 \\ 0 & 0 & 3 \\ 0 & 3 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 0$

or $-a - b = 0$, $3c = 0$, $3b = 0$. Hence $b = 0$, $c = 0$, a is arbitrary,

Eigen vector corresponding to m_1 is $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

d) Hence $X_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^t$ is one solution.

The Eigen vector corresponding to $m_2 = 4$ is given by $\begin{pmatrix} -3 & -1 & -1 \\ 0 & -3 & 3 \\ 0 & 3 & -3 \end{pmatrix} \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix} = 0$

$$3a + b + c = 0$$

$$-3b + 3c = 0$$

$$\text{or } b = c$$

Choosing $b = c = -3$, $a = 2$ from which we obtain $x_2 = \begin{pmatrix} 2 \\ -3 \\ -3 \end{pmatrix} e^{4t}$ as second solution. Now

considering $m_3 = -2$, we have $\begin{pmatrix} 3 & -1 & -1 \\ 0 & 3 & 3 \\ 0 & 3 & 3 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 0$

$$\text{Or } 3a - b - c = 0, 3b + 3c = 0$$

Choosing $b = 1$, $c = -1$, $a = 0$, we have $x_3 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} e^{-2t}$ as the third solution.

e) Now the Wronskian of the three solutions when $t = 0$ is given

by $\begin{vmatrix} 1 & 2 & 0 \\ 0 & -3 & 1 \\ 0 & -3 & -1 \end{vmatrix} = 6 \neq 0$. Then the three solutions are independent.

$$\text{Hence } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^t + c_2 \begin{pmatrix} 2 \\ -3 \\ -3 \end{pmatrix} e^{4t} + c_3 \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} e^{-2t}$$

Example 11:

Consider the first order system of equations,

$$\frac{dx}{dt} = 3x + 4y$$

$$\frac{dy}{dt} = -2x - 3y$$

- Write down the equations in matrix form.
- Write down the characteristic equation of the matrix
- Determine the Eigen values of the matrix
- Find the Eigen vectors corresponding to each of the two Eigen values.
- Write the general solution of the system.
- Verify your general solution is correct.

Solution

Let $\frac{dx}{dt} = 3x + 4y$ (1)

$$\frac{dy}{dt} = -2x - 3y$$
 (2)

- a) The system of equations (1) and (2) is written in the matrix form as $X' = AX$

where $X' = \begin{pmatrix} x \\ y \end{pmatrix}'$, $A = \begin{pmatrix} 3 & 4 \\ -2 & -3 \end{pmatrix}$, $X = \begin{pmatrix} x \\ y \end{pmatrix}$

b) The characteristic equation of A is $\begin{vmatrix} 3-m & 4 \\ -2 & -3-m \end{vmatrix} = 0$ (4)

where m is the Eigen value.

- c) To obtain the Eigen values we expand the determinant (4) and solve for m.

$$(3-m)(-3-m) - 3 = 0$$

$$m^2 - 1 = 0 \text{ or } (m-1)(m+1) = 0$$

The Eigen values are $m_1 = 1$ and $m_2 = -1$.

- d) Eigen vector $\begin{pmatrix} x \\ y \end{pmatrix}$ corresponding to $m_1 = 1$ is given from the characteristic

equation (4) for $m = 1$

$$\begin{pmatrix} 3-1 & 4 \\ -2 & -3-1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\text{or } \begin{pmatrix} 2 & 4 \\ -2 & -4 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$2a + 4b = 0$$

$$\text{and } -2a - 4b = 0$$

Both the equations give $a + 2b = 0$. If we take any value for a , say $a = 2$, $b = -1$.

Hence the Eigen vector corresponding to $m = 1$ is $k \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ where k is any constant. We can take conveniently by the Eigen vector corresponding to $m_1 = 1$ as $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$. Then $X_1 = \begin{pmatrix} 2 \\ -1 \end{pmatrix} e^t$ when $m_1 = 1$. Similarly the Eigen vector

corresponding to $m_2 = -1$ is obtained from $\begin{pmatrix} 4 & 4 \\ -2 & -2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$$\text{Or } \begin{aligned} 4a + 4b &= 0 \\ -2a - 2b &= 0 \end{aligned}$$

both the equations give $a + b = 0$. Taking $a = 1$, $b = -1$. Hence the Eigen vector

corresponding to $m_2 = -1$ is $\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ or $x_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

e) The two Eigen vectors are $x_1 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $x_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ when $m_2 = -1$. Hence the

general solution of the system (1) and (2) is given by $x = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^t + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t}$

where c_1 and c_2 are arbitrary constants.

f) If $x = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^t + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t}$ is the solution $\begin{pmatrix} 1 \\ 2 \end{pmatrix} e^t$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t}$ should be

independent. In other words the Wronskian of $\begin{pmatrix} 1 \\ 2 \end{pmatrix} e^t$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t}$ should not be zero for all t or when $t = 0$

$$w[x_1(0), x_2(0)] = \begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix} = -3. \quad \text{Because } w \neq 0, \quad \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^t \quad \text{and} \quad \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t} \quad \text{are}$$

independent and the general solution of the system is $x(t) = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^t + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t}$

Example 11.

Consider the system of equations

$$\left. \begin{aligned} x_1' &= x_1 - x_2 - x_3 \\ x_2' &= x_2 + 3x_3 \\ x_3' &= 3x_2 + x_3 \end{aligned} \right\} \quad (1)$$

i). Write down the first order system (1) into matrix form.

- ii). Determine the characteristic equation of the matrix.
- iii). Find the Eigen values of the matrix.
- iv). Calculate the Eigen vectors of the matrix.
- v). Write down the general solution of the system and verify the general solution using the Wronskian.

Solution.

i). The matrix form of (1) is $\mathbf{X}' = \mathbf{A}\mathbf{X}$

where

$$\mathbf{X}' = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}', \mathbf{A} = \begin{pmatrix} 1 & -1 & -1 \\ 0 & 1 & 3 \\ 0 & 3 & 1 \end{pmatrix}, \mathbf{X} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

ii). The characteristic equation of A is given by

$$\begin{vmatrix} 1-m & -1 & -1 \\ 0 & 1-m & 3 \\ 0 & 3 & 1-m \end{vmatrix} = 0 \quad (2)$$

expanding the third order determinant we get $(1-m)(m-4)(m+2) = 0$

iii). The Eigen values are $m_1 = 1, m_2 = 4, m_3 = -2$.

iv). The Eigen vectors corresponding to the Eigen value $m_1 = 1$ is

$$\begin{pmatrix} 1-1 & -1 & -1 \\ 0 & 1-1 & 3 \\ 0 & 3 & 1-1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \quad (3)$$

$$\text{or} \quad \begin{pmatrix} 0 & -y & -z \\ 0 & 0 & 3z \\ 0 & 3y & 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \text{or} \quad -x - z = 0 \quad \text{or} \quad x = z$$

$$3z = 0 \quad \text{or} \quad z = 0$$

$$3y = 0 \quad \text{or} \quad y = 0$$

$$\text{Then } 0x - y - z = 0, \quad 0x = 0$$

x is arbitrary.

Let $x = 1$. Then Eigen vector corresponding to $m_1 = 1$ is given by

$$x_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^t \quad (4)$$

Eigen vector corresponding to $m_2 = 4$ is given by

$$\begin{pmatrix} 1-4 & -1 & -1 \\ 0 & 1-4 & 3 \\ 0 & 3 & 1-4 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \quad (5)$$

$$\begin{aligned} -3x - y - z &= 0 \\ -3x_2 + 3y_3 &= 0 \text{ or } y = z \\ -3y + 3z &= 0 \text{ or } y = z \\ -3x - y - z &= 0 \text{ or } -3x - 2y = 0 \\ \text{Taking } x &= 2, y = 3. \end{aligned}$$

The Eigen vector corresponding to $m_2 = 4$ is given by $\begin{pmatrix} 2 \\ -3 \\ -3 \end{pmatrix}$

$$\text{Hence } x_2 = \begin{pmatrix} 2 \\ -3 \\ -3 \end{pmatrix} e^{4t} \quad (6)$$

Eigen vector corresponding to $m_3 = -2$ is given by

$$\begin{pmatrix} 3 & -1 & -1 \\ 0 & 3 & 3 \\ 0 & 3 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \quad (7)$$

$$3x - y - z = 0, \quad 3y + 3z = 0,$$

using $y = -z$, $3x + z - z = 0$, hence $x = 0$

$$\text{Taking } y = 1, z = -1. \text{ Hence } x_3 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} e^{-2t} \quad (8)$$

The general solution could be $x = c_1x_1 + c_2x_2 + c_3x_3$

$$\text{Or } x = c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^{4t} + c_2 \begin{pmatrix} 2 \\ -3 \\ -3 \end{pmatrix} e^{4t} + c_3 \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} e^{-2t} \quad (9)$$

provided that x_1, x_2 and x_3 are independent. Wronskian of x_1, x_2, x_3 when $t = 0$ is given by

$$\begin{vmatrix} 1 & 2 & 0 \\ 0 & -3 & 1 \\ 0 & -3 & -1 \end{vmatrix} = 6 \text{ which is not zero.}$$

Since the Wronskian $w \neq 0$ the solution are linearly independent and hence the equation (9) is the general solution of the system of equation (1)

10.9 Real and Repeated Eigen values of the matrix.

Example

Consider the system

$$X' = AX \quad (1)$$

$$\text{where } A = \begin{pmatrix} 0 & 1 \\ -4 & 4 \end{pmatrix}, \quad X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

The Eigen value $m = 2, 2$, repeated. The Eigen vector $\begin{pmatrix} p \\ q \end{pmatrix}$ is obtained from

$$\begin{pmatrix} -2 & 1 \\ -4 & 2 \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix} = 0 \quad \text{or} \quad -2p + q = 0, \quad -4p + 2q = 0$$

Choosing $p = 1$ and $q = 2$

Then the Eigen vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is repeated. Hence we obtain the first solution:

$$X_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t} \quad (2)$$

By the main Theorem, the second solution is of the form,

$$X_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} t e^{2t} + \begin{pmatrix} a \\ b \end{pmatrix} e^{2t} \quad (3)$$

where $\begin{pmatrix} a \\ b \end{pmatrix}$ is to be found.

Since X_2 must satisfy the system $X' = AX$ we have $X_2' = AX_2$. Differentiating (3) we have

$$X_2' = \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} 2t e^{2t} + \begin{pmatrix} a \\ b \end{pmatrix} 2e^{2t} \quad (4)$$

$$AX_2' = \begin{pmatrix} 0 & 1 \\ -4 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t} + \begin{pmatrix} 0 & 1 \\ -4 & 4 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} e^{2t} \quad (5)$$

Equating (4) and (5) and canceling the term involving $t e^{2t}$ we have

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} + 2 \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -4 & 4 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 2a \\ 2b \end{pmatrix} = \begin{pmatrix} 0 + b \\ -4a + 4b \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} b \\ -4a + 4b \end{pmatrix} - \begin{pmatrix} 2a \\ 2b \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -2a & + & b \\ -4a & + & 2b \end{pmatrix}$$

Then $-2a + b = 1$ and $-4a + 2b = 2$. We can choose $a = 0$, $b = 1$

Therefore $\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and from (3)

$x_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} te^{2t} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{2t}$ is the second solution.

The general solution of the problem is

$$\begin{aligned} X &= c_1 x_1 + c_2 x_2 \\ &= c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} te^{2t} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{2t} \end{aligned}$$

10.10 Eigen values are complex conjugates

When the Eigen values are complex numbers the procedure is exactly the same. Only the final answer is separated into real and imaginary parts using Euler's notation:

$$e^{i\theta} = \cos \theta + i \sin \theta \quad \text{and}$$

$$e^{-i\theta} = \cos \theta - i \sin \theta$$

The following example illustrates this method.

Example 12

Consider the system

$$X' = AX \tag{1}$$

$$\text{where } A = \begin{pmatrix} 2 & -5 \\ 2 & -4 \end{pmatrix}, X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, X' = \begin{pmatrix} x_1' \\ x_2' \end{pmatrix}$$

Solution

$$\text{Let } A = \begin{pmatrix} 2 & -5 \\ 2 & -4 \end{pmatrix}$$

The characteristic equation is

$$\begin{pmatrix} 2-m & -5 \\ 2 & -4-m \end{pmatrix} = 0 \quad \text{or} \quad m^2 + 2m + 2 = 0 \quad (2)$$

Solving $m^2 + 2m + 2 = 0$ we have $m_1 = -1 + i$ and $m_2 = -1 - i$

The Eigen vector $\begin{pmatrix} a \\ b \end{pmatrix}$ corresponding to $m_1 = -1 + i$ is given by

$$\begin{pmatrix} 3-i & -5 \\ 2 & -3-i \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0$$

$(3-i)a - 5b = 0$ or $b = \frac{(3-i)a}{5}$. Choosing $a = 5$, $b = 3 - i$. Thus the Eigen

vector is $\begin{pmatrix} 5 \\ 3-i \end{pmatrix}$ (3)

Hence the first solution is $x_1 = \begin{pmatrix} 5 \\ 3-i \end{pmatrix} e^{(-1+i)t}$ (4)

Corresponding to the second Eigen value $m_2 = -1 - i$ the Eigen vector is $\begin{pmatrix} 5 \\ 3+i \end{pmatrix}$ (work

out yourself) Thus the second solution is $x_2 = \begin{pmatrix} 5 \\ 3+i \end{pmatrix} e^{(-1-i)t}$.

When $t = 0$, $W = \begin{vmatrix} 5 & 5 \\ 3 & 3+i \end{vmatrix} = 10i \neq 0$. Hence x_1 and x_2 are independent

The general solution is

$$x = c_1 \begin{pmatrix} 5 \\ 3-i \end{pmatrix} e^{(-1+i)t} + c_2 \begin{pmatrix} 5 \\ 3+i \end{pmatrix} e^{(-1-i)t} \quad (5)$$

When $t = 0$, $W = \begin{vmatrix} 5 & 5 \\ 3 & 3+i \end{vmatrix} = 10i \neq 0$ Hence x_1 and x_2 are independent

Now let us use Euler's notation namely $e^{i\theta} = \cos \theta + i \sin \theta$, $e^{-i\theta} = \cos \theta - i \sin \theta$

$$\begin{aligned}
e^{(-1-i)} &= e^{-t} \cdot e^{it} \\
&= e^{-t} (\cos t + i \sin t) \\
&= e^{-t} \cdot e^{-it} \\
&= e^{-t} (\cos t - i \sin t)
\end{aligned} \tag{6}$$

Using (6) and (7) in the general solution (5) we have

$$\begin{aligned}
X &= c_1 \begin{pmatrix} 5 \\ 3-i \end{pmatrix} e^{-t} \cdot e^{it} + c_2 \begin{pmatrix} 5 \\ 3+i \end{pmatrix} e^{-t} \cdot e^{-it} \\
&= e^{-t} \left[c_1 \begin{pmatrix} 5 \\ 3-i \end{pmatrix} e^{it} + c_2 \begin{pmatrix} 5 \\ 3+i \end{pmatrix} e^{-it} \right] \\
&= e^{-t} \left[c_1 \begin{pmatrix} 5 \\ 3-i \end{pmatrix} (\cos t + i \sin t) + c_2 \begin{pmatrix} 5 \\ 3+i \end{pmatrix} (\cos t - i \sin t) \right] \\
&= e^{-t} \left[(c_1 + c_2) \begin{pmatrix} 5 \cos t \\ 3 \cos t + \sin t \end{pmatrix} + i(c_1 - c_2) \begin{pmatrix} 5 \sin t \\ -\cos t + 3 \sin t \end{pmatrix} \right] \\
X &= e^{-t} \left[\left\{ b_1 \begin{pmatrix} 3 \\ 5 \end{pmatrix} \cos t - \begin{pmatrix} 0 \\ -1 \end{pmatrix} \sin t \right\} + b_2 \begin{pmatrix} 0 \\ -1 \end{pmatrix} \cos t + \begin{pmatrix} 3 \\ 5 \end{pmatrix} \sin t \right]
\end{aligned}$$

the two solutions are independent since the Wronskian at $t = 0$ is -5 hence () is the general solution.

2.11 Solution of the First Order Nonhomogeneous System of Equation

The first order non-homogeneous system of equation is of the form

$$X' = AX + B \tag{1}$$

where A is a constant $n \times n$ matrix. B is a vector function of t. the solution of the system consists of two parts namely the complementary function (CF) and particular integral (PI). The complementary function X_c is obtained by solving the associated homogeneous system.

$$X' = AX \tag{2}$$

we need to find the particular integral X_p and add it to the general solution of the system

$X' = AX$. Thus the general solution of the homogeneous system is

$$X = X_c + X_p \tag{3}$$

The following example will illustrate the method of solving a non-homogeneous first order system.

Example 14:

Consider the system

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e^t \\ 1 \end{pmatrix} \quad (4)$$

we write this system in matrix form as

$$\mathbf{X}' = \mathbf{A}\mathbf{X} + \mathbf{B} \quad (5)$$

where $A = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} e^t \\ 1 \end{pmatrix}$, $x' = \begin{pmatrix} x \\ y \end{pmatrix}'$

- i). The characteristic equation of A is $\begin{vmatrix} 0-m & 1 \\ -2 & 3-m \end{vmatrix} = (m-1)(m-2) = 0$
- ii). The Eigen values of A are $m_1 = 1$, and $m_2 = 2$
- iii). The Eigen vectors are $\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ (you can find out)
- iv). The general solution of $\mathbf{X}' = \mathbf{A}\mathbf{X}$ is given by

$$x = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + c_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t} \quad (6)$$

Hence the c.f is given by $\begin{pmatrix} x \\ y \end{pmatrix}_c = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + c_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t} \quad (7)$

The particular integral PI is given by a similar expression

$$\begin{pmatrix} x \\ y \end{pmatrix} = g_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + g_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t} \quad (8)$$

where g_1 and g_2 are functions of t depending on the vector B.

$$\text{since } \begin{pmatrix} x \\ y \end{pmatrix}_p = g_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + g_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t}$$

$$\begin{pmatrix} x \\ y \end{pmatrix}_p^i = g_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t g_1^1 + 2g_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t} g_2^1$$

The particular integral should also satisfy the equation $X' = AX + B$.

Using $X = \begin{pmatrix} x \\ y \end{pmatrix}_p$ in $X' = AX + B$ we have

$$\begin{aligned} & g_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + g_1^1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + 2g_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t} + g_2^1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t} \\ &= \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix} \left[\begin{pmatrix} 1 \\ 1 \end{pmatrix} g_1 e^t + \begin{pmatrix} 1 \\ 2 \end{pmatrix} g_2 e^{2t} \right] + \begin{pmatrix} e^t \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} g_1 e^t + \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} g_2 e^{2t} + \begin{pmatrix} e^t \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 \\ 1 \end{pmatrix} g_1 e^t + 2g_2 e^{2t} + \begin{pmatrix} e^t \\ 1 \end{pmatrix} \end{aligned}$$

canceling the common term in both sides we have

$$g_1^1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + g_2^1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t} = \begin{pmatrix} e^t \\ 1 \end{pmatrix}$$

$$\text{or, in matrix form } \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} g_1^1 & e^t \\ g_2^1 & e^{2t} \end{pmatrix} = \begin{pmatrix} e^t \\ 1 \end{pmatrix}$$

we can solve the simultaneous equation (6) using Crammer's Rule.

$$\text{Thus } g_1^1 e^t = \frac{\begin{vmatrix} e^t & 1 \\ 1 & 2 \end{vmatrix}}{\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix}} = 2e^t - 1$$

$$g_1^1 = 2 - e^{-t}$$

$$g_2^1 e^{2t} = \frac{\begin{vmatrix} 1 & e^t \\ 1 & 1 \end{vmatrix}}{1} = 1 - e^t$$

$$\text{or } g_2^1 = e^{-2t} - e^{-t}$$

Integrating g_1^1 and g_2^1 we have $g_1 = 2t + e^{-t}$ and $g_2 = e^{-t} - \frac{1}{2} e^{-2t}$

Hence the general solution of $X' = AX + B$ is

$$X' = C_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + C_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t} + (2t + e^{-t}) \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + \left(e^{-t} - \frac{1}{2} e^{-2t} \right) \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t}$$

$$\text{or } X' = C_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + C_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} (2te^{-t} + 1) + \begin{pmatrix} 1 \\ 2 \end{pmatrix} \left(e^{-t} - \frac{1}{2} \right)$$

Exercise 10

Write down the given equations (1 to 4) by a system of first order equations.

1. $\frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 8y = x + 3$
2. $y'' + 4y' + 5y = e^x$
3. $y''' + 2y'' + 3y' + 4y = f(x)$
4. $y''' - 2y = 0$

In exercise 5 through 7, write the given system of differential equations as a matrix equation.

$$5. \quad \frac{dx}{dt} = 2x + 3y$$

$$\frac{dy}{dt} = x - y$$

$$6. \quad \frac{dx}{dt} = 2x - y + z + t$$

$$\frac{dy}{dt} = x + 2y - z + 1$$

$$\frac{dz}{dt} = 2x - y + z = e^{2t}$$

$$7. \quad \frac{dx}{dt} = 2x - y + e^t$$

$$\frac{dy}{dt} = x + y + t$$

$$8. \quad \text{Consider the system } X' = AX \text{ where } A = \begin{pmatrix} 4 & -1 \\ -4 & 4 \end{pmatrix}, \quad X' = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

- a) Find the characteristic equation of the matrix A.
- b) Find the Eigen values and the Eigen vectors corresponding to each Eigen values.
- c) Find the general solution of the system of equation after verifying that the two solutions are independent.

9. Consider the system $X' = AX$ where $A = \begin{vmatrix} 1 & -1 & -1 \\ 0 & 1 & 3 \\ 0 & 3 & 1 \end{vmatrix}$

- a) Show that the Eigen values of A are 1, 4 and -2.
- b) Prove that the Eigen vectors of A are $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 2 \\ -3 \\ -3 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$
- c) Write down the three solutions of the equation $X' = AX$.
- d) Show that the Wronskian of $\begin{pmatrix} 1 & 2 & 0 \\ 0 & -3 & 1 \\ 0 & -3 & -1 \end{pmatrix} \neq 0$ and hence write down the general solution of $X' = AX$.

10. Solve the system of equations $X' = AX$ where $A = \begin{pmatrix} 8 & -3 \\ 16 & -8 \end{pmatrix}$ and $X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$.

11. Find the general solution of the equation, $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}' = \begin{pmatrix} 12 & -15 \\ 4 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ using matrix algebra.

12. Solve the system of equations $X' = AX$ where $A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & -1 & 3 \\ 0 & 0 & 2 \end{pmatrix}$ X

13. Solve the following First order non-homogeneous equations.

$$X' = AX + B \text{ where } A = \begin{pmatrix} 2 & 1 \\ -4 & 2 \end{pmatrix}, B = \begin{pmatrix} 3 & e^{2t} \\ t & e^{2t} \end{pmatrix}$$

14. If $\begin{pmatrix} x \\ y \end{pmatrix}' = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 0 \\ 3 \end{pmatrix} e^t$, find the particular integral of the equation.

Summary of the lesson

You have learnt the following from this LESSON.

- i. Reducing a nth order equation to a system of n linear equations.
- ii. Writing the n linear equations in matrix form as
 - a. $X' = AX$ for homogeneous equations
 - b. $X' = AX + B$ for nonhomogeneous equations
where $A = n \times n$ matrix

B = Corresponding matrix to the nonhomogeneous part of the equations

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{pmatrix}$$

$$X' = \begin{pmatrix} x_1' \\ x_2' \\ \cdot \\ \cdot \\ x_n' \end{pmatrix}$$

- c. Determination of Eigen values and Eigen vectors of the matrix A
- d. The important theorems connecting the relations between the Eigen vectors of the matrix and the solution of the system of equations
- e. Solving homogeneous and nonhomogeneous equations of the system of equations when the Eigen values are:
 - i. Real and distinct
 - ii. Real and repeated
 - iii. Complex conjugate

Further Reading

1. Differential Equations
By Frank Ayres JR. Ph.D
Schaum's outline series
McGraw - Hill Book Company
New York, Toronto, Sydney
2. Differential Equations
By Dr. D Sengottaiyan
Oxford Publications
London – Nairobi.

LESSON 11

Singular Points of Differential Equations and Power Series Solutions near Singular points.

11.1 Introduction

In this lesson we are going to learn about the singular points of a differential equation and nature of power series solutions near different types of singular points.

11.2 Objectives of the lesson

By the end of this lesson you should be able to:

- define ordinary points and the singular points of a differential equation.
- classify the singular points into Regular Singular points and Irregular Singular points of the differential equation.
- write the nature of power series solution which may exist at an Ordinary point, Regular Singular points and Irregular singular points of the given differential equations.

11.3 Power series solutions

We have learnt that many types of differential equations have solutions that can be expressed as finite linear combinations of known elementary functions. However, many equations may not have solutions that can be found in the usual methods we have studied so far. In such case we must seek other means of expression for the solutions of such equations. One such method of obtaining solutions is infinite series form. The solution so obtained is called as Power Series Solution of the given differential equation.

Definition of power series solutions.

All differential equations either first order or higher order need not possess solutions. Under certain conditions some equations may have solutions which can be expressed as an infinite series of the form.

$$y = C_0 + C_1 (x - x_0) + C_2 (x - x_0)^2 + \dots + C_n (x - x_0)^n + \dots \text{ to } \infty \text{ or}$$
$$y = \sum_{n=0}^{\infty} C_n (x - x_0)^n \quad (5.1)$$

where $C_0, C_1, C_2, \dots$ are constants

Solutions of the form (5.1) are called **Power Series Solutions** of the differential equations.

In order to confirm whether an equation does have solution of a power series or not we must know certain basic definitions that will be seen now

11.4 Definition of Analytic Function at a point

A function $f(x)$ is said to be analytic at x_0 if its Taylor series about x_0

$\sum_{n=0}^{\infty} \frac{f^n(x_0)}{n!} (x - x_0)^n$ exists and converges to $f(x)$ for all x in some open interval including x_0 .

Sometimes we say that any function that can be differentiable at the point x_0 is called an **analytic function** at x_0

Alternatively we can say that $f(x)$ is analytic at the point x_0 if the function does not become infinite when $x = x_0$.

11.5 Definition: Normalized form of a second order equation.

Consider a second order equation

$$a_0(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_2(x) y = 0 \quad (1)$$

where $a_0(x) \neq 0$

when each term of the equation is divided by $a_0(x)$ we have

$$\frac{d^2 y}{dx^2} + \frac{a_1(x)}{a_0(x)} \frac{dy}{dx} + \frac{a_2(x)}{a_0(x)} y = 0$$

This equation is represented as $\frac{d^2 y}{dx^2} + p_1(x) \frac{dy}{dx} + p_2(x) y = 0$ (2)

where $p_1(x) = \frac{a_1(x)}{a_0(x)}$, $p_2(x) = \frac{a_2(x)}{a_0(x)}$

Equation (2) is called **the normalized form** of a second order equation. In the normalized form, the coefficient of $\frac{d^2 y}{dx^2}$ is unity.

Example 1

Write down the equation $(x^2 - 1) \frac{d^2 y}{dx^2} + 3x \frac{dy}{dx} + xy = 0$ into the normalized form.

Solution

Let equation $(x^2 - 1) \frac{d^2 y}{dx^2} + 3x \frac{dy}{dx} + xy = 0$

Dividing throughout by $(x^2 - 1)$ the equation becomes

$$\frac{d^2 y}{dx^2} + \frac{3x}{x^2 - 1} \frac{dy}{dx} + \frac{xy}{x^2 - 1} y = 0 \quad (5.3)$$

Equation (5.3) is said to be in the normalized form

$$\frac{d^2y}{dx^2} + p_1(x)\frac{dy}{dx} + p_2(x)y = 0 \quad \text{where } p_1(x) = \frac{3x}{x^2-1}, p_2(x) = \frac{xy}{x^2-1}.$$

11.6 Definition of an ordinary point and singular point of the differential equation

The point x_0 is called an ordinary point of the equation (5.3) if both of the functions p_1 and p_2 are analytic at x_0 .

If either p_1 or p_2 or both of these functions is not analytic at x_0 then x_0 is called a **singular point** of the normalized equation (5.3).

Example 2

Determine the singular points of the equation

$$(x^2 - 9)\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + (x+3)y = 0 \quad (1)$$

Solution

The normalized form of the equation (1) is

$$\frac{d^2y}{dx^2} + \frac{3x}{(x+3)(x-3)}\frac{dy}{dx} + \frac{(x+3)y}{(x+3)(x-3)} = 0 \quad (2)$$

$$\text{Here } p_1(x) = \frac{3x}{(x+3)(x-3)}, p_2(x) = \frac{1}{(x-3)}y$$

$p_1(x)$ becomes infinite (not analytic) at $x = -3$ and $x = 3$.

Hence $x = -3$ and $x = 3$ are singular points for the equation (1) or (2) p_2 becomes infinite at $x = 3$. Hence $x = 3$ and $x = -3$ are singular points for the equation (1) or (2).

Example 3

Determine if the equation

$$\frac{d^2y}{dx^2} + (x-2)\frac{dy}{dx} + (x^2+1)y = 0 \text{ has any singular points.}$$

Solution

The equation is already in the normalized form. $P_1 = (x-2)$ and $p_2 = (x^2+1)$. Both of them are polynomials and hence they are analytic. Then the equation has no singular points.

11.7 Definition of Regular Singular points and Irregular Singular points.

Consider the second order equation

$$\frac{d^2 y}{dx^2} + p_1 \frac{dy}{dx} + p_2 y = 0 \quad (1)$$

Suppose equation (1) has a singular point at x_0 .

If both $(x - x_0) p_1$ and $(x - x_0)^2 p_2$ are analytic then the singular point x_0 is said to be a **regular singular point**.

Example 4

Locate and classify the singular point of the equation (1)

Solution

Dividing each term of the equation (1) by $x^2 - 8x$ or $x(x - 8)$

$$\frac{d^2 y}{dx^2} + \frac{x+2}{x(x-8)} \frac{dy}{dx} + \frac{1}{x(x-8)} y = 0$$

$$\text{Here } p_1 = \frac{x+2}{x(x-8)} \text{ and } p_2 = \frac{1}{x(x-8)}$$

p_1 is not analytic since p_1 becomes infinite when $x = 0$ and $x = 8$

p_2 also becomes infinite when $x = 0$ hence $x = 0$ and $x = 8$ are singular points

Nature of the singular point at $x = x_0 = 0$

$$\text{Here } xp_1(x) = \frac{x(x+2)}{x(x-8)} = \frac{(x+2)}{(x-8)} \text{ is analytic at } x = 0$$

$$x^2 p_2(x) = \frac{x}{x(x-8)} = \frac{x}{x-8} \text{ is analytic at } x = 0$$

Hence $x = 0$ is a regular singular point.

Nature of the singular point at $x = 8$

$$\begin{aligned} (x-8)p_1(x) &= \frac{(x-8)(x+2)}{x(x-8)} \\ &= \frac{(x+2)}{x} \text{ which is analytic at } x = 8 \end{aligned}$$

$$\begin{aligned} (x-8)^2 p_2(x) &= \frac{(x-8)^2}{x(x-8)} \\ &= \frac{x-8}{x} \text{ which is analytic at } x = 8 \end{aligned}$$

Hence $x = 8$ is also a regular singular point.

Example 5

Locate and classify the singular point of the equation

$$(x^4 - 2x^3 + x^2) \frac{d^2 y}{dx^2} + 2(x-1) \frac{dy}{dx} + x^2 y = 0$$

Solution

The normalized form of the equation is

$$\frac{d^2 y}{dx^2} + \frac{2(x-1)}{x^2(x^2 - 2x + 1)} \frac{dy}{dx} + \frac{x^2 y}{x^2(x^2 - 2x + 1)} = 0$$

$$\text{or } \frac{d^2 y}{dx^2} + \frac{2(x-1)}{x^2(x-1)^2} \frac{dy}{dx} + \frac{y}{(x-1)^2} = 0$$

$$\text{or } \frac{d^2 y}{dx^2} + \frac{2}{x^2(x-1)} \frac{dy}{dx} + \frac{y}{(x-1)^2} = 0$$

then $p_1 = \frac{x+2}{x^2(x-1)}$ and $p_2 = \frac{1}{(x-1)^2}$. Hence $x = 0$ and $x = 1$ are singular points.

Nature of the singular point at $x = 0$

$$\begin{aligned} (x-0)p_1(x) &= \frac{2x}{x^2(x-1)} \\ &= \frac{2}{x(x-1)} \text{ which is not analytic at } x = 0 \end{aligned}$$

$$(x-0)^2 p_2(x) = \frac{x^2 y}{(x-1)^2} \text{ is analytic at } x = 0.$$

Hence $x = 0$ is an irregular singular point since $x^2 p_2(x)$ is not analytic

Nature of the singular point at $x = 1$

$$\begin{aligned} (x-1)p_1(x) &= \frac{(x-1) \times 2}{x^2(x-1)} \\ &= \frac{2}{x^2} \text{ which is analytic at } x = 1 \end{aligned}$$

$$\begin{aligned} (x-1)^2 p_2(x) &= (x-1)^2 \times \frac{1}{(x-1)^2} \\ &= 1 \text{ which is analytic at } x = 1 \end{aligned}$$

Hence $x = 1$ is a regular singular point.

11.8 Existence Theorem and the nature of Power Series solutions

Consider the equation (normalized form)

$$\frac{d^2 y}{dx^2} + p_1 \frac{dy}{dx} + p_2 y = 0 \quad (1)$$

- I. If the point x_0 is an ordinary point of the equation (1), it has two nontrivial solutions of the form**

$$y = \sum_{n=0}^{\infty} C_n (x - x_0)^n$$

- II. If the point x_0 is a regular singular point of the equation (1) it has at least one nontrivial solution of the form.**

$$y = |x - x_0|^r \sum_{n=0}^{\infty} C_n (x - x_0)^n$$

or $y = \sum_{n=0}^{\infty} C_n (x - x_0)^{n+r}$ where r is a constant (real or complex). This solution is valid in some deleted interval $0 < |x - x_0| < R$ where $R > 0$ about x_0 .

- III. If the point x_0 is an irregular singular point, we are not assured about the existence of solution of the equation. It may or may not have the power series solution near the irregular point at x_0 .**

Hence before we assume a power series solution at x_0 for the normalized equation we must examine the nature of the point x_0 and proceed.

Example 6

What kind of power series solution exists for the equation

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2) y = 0$$

where n is a non-negative constant. (This equation is called Bessel's equation).

Solution

The normalized form of the equation is

$$\frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} + \frac{(x^2 - n^2)}{x^2} y = 0$$

Now $x = 0$ is a singular point of the equation

Here $p_1(x) = \frac{1}{x}$ and $p_2(x) = \frac{x^2 - n^2}{x^2} = 1 - \frac{n^2}{x^2}$

Nature of the singular point at $x = 0$

$x p_1(x) = x \times \frac{1}{x} = 1$ which is analytic at $x = 0$

$x^2 p_2(x) = \frac{x^2(x^2 - n^2)}{x^2} = x^2 - n^2$ which is also analytic at $x = 0$

hence $x = 0$ is a regular singular point. The Bessel's equation has at least one nontrivial solution of the form.

$$y = \sum_{n=0}^{\infty} C_n (x - x_0)^{n+r}$$

where r is a constant real (positive or negative or fraction) or complex.

Exercise 11

Express the following equations in the normalized form

1. $(x-1) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + \frac{1}{x} y = 0$
2. $(x^2 - 1) \frac{d^2 y}{dx^2} + 3x \frac{dy}{dx} + xy = 0$
3. $(x^3 - 1) \frac{d^2 y}{dx^2} + \sin x \frac{dy}{dx} + 8y = 0$
4. $\sin x \frac{d^2 y}{dx^2} + \cos x \frac{dy}{dx} + \frac{1}{x} y = 0$

Locate and Classify the singular points if any in the following equations: State the nature of power series solution near each of the singular points.

5. $2x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + (x-5)y = 0$
6. $x^2(x-1)^2 \frac{d^2 y}{dx^2} + 2(x-2) \frac{dy}{dx} + (x+1)y = 0$
7. $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - \left(x^2 + \frac{5}{4}\right)y = 0$

$$8. \quad x^2 \frac{d^2 y}{dx^2} + (x^2 - 3x) \frac{dy}{dx} + 3y = 0$$

$$9. \quad (x^2 - 3x) \frac{d^2 y}{dx^2} + (x + 2) \frac{dy}{dx} + y = 0$$

$$10. \quad (x^4 - 2x^3 + x^2) \frac{d^2 y}{dx^2} + 2(x - 1) \frac{dy}{dx} + x^2 y = 0$$

$$11. \quad x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - p^2) y = 0$$

where p is a constant (Bessel's Equation)

12. Consider the Legendre's equation

$$(1 - x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + p(p + 1)y = 0$$

Show that $x = 0$ is an ordinary point for the Legendre's equation.

13. Define

- i. an ordinary point
- ii. singular point
- iii. regular singular point.
- iv. an irregular singular point of a normalized equation of the form.

$$\frac{d^2 y}{dx^2} + p_1 \frac{dy}{dx} + p_2 y = 0$$

Summary

You have learnt the following from this lesson.

- i). normalized form of a second order linear equation
- ii). an ordinary point and a singular point of a second order linear equation
- iii). a regular singular point and an irregular singular point for a second order linear equation.
- iv). The form of power series near an ordinary point and near a Regular Singular point.
- v). The power series solution may or may not exist near an Irregular Singular point.

Further Reading

1. Differential Equations

By Frank Ayres Ph. D

Schaum's Outline series

McGraw – Hill Book Company

New York. Toronto. Singapore.

2. Differential Equations
By Dr. D. Sengottaiyan
Oxford Publications
London Nairobi.

LESSON 12

Power series solutions of First and Second order Equations

12.1 Introduction

You have learnt from the previous lesson that power series solutions need not exist for all the differential equations and the nature of power series is different at ordinary points and singular points of the equation.

In this lesson we shall find the power series solutions for the First and Second order equations if the conditions for the existence of power series solutions are satisfied.

12.2 Objectives of the lesson

By the end of this lesson you should be able to:

- obtain the power series solutions of first and second order equations with variable coefficients when the conditions for the existence of power series solution are satisfied.

12.3 Power series solution of the first order equation.

The solution of the first order equation of the form $\frac{dy}{dx} = f(x, y)$ exists and is

unique if and only if the following three conditions are satisfied.

- i). $f(x, y)$ is continuous and single valued in a domain D of the x, y – plane.
- ii). $f(x, y)$ is bounded or $|f(x, y)| \leq M$ where M is the definite constant. In other words $f(x, y)$ does not tend to infinity in the domain D .
- iii). $\frac{\partial}{\partial y} f(x, y)$ exists and continuous at all points in the domain D .

If $f(x, y)$ satisfies all the three conditions the solution of the equation exists.

Example 1

- a). Obtain the power series solution of the same equation, $\frac{dy}{dx} = y$ in ascending powers of x .
- b). Solve the equation $\frac{dy}{dx} = y$ using the method of separation of variables.
- c). Verify that the two solutions are equivalent.

Solution

- a). Let $\frac{dy}{dx} = y$ This is of the form $\frac{dy}{dx} = f(x, y)$ (1)

- i). Now $f(x, y)$ is a polynomial and hence continuous and single valued in the whole of $x y$ plane as the domain.
- ii). Also it is bounded, since it is a polynomial.
- iii). $\frac{\partial}{\partial y} f(x, y) = \frac{\partial}{\partial y} y = 1$ which exists. Then $\frac{dy}{dx} = y$ has a power series solution of the form.
- $$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots \quad (2)$$

$$\text{Let } y = \sum_{n=0}^{\infty} C_n x^n$$

$$y^1 = \sum_{n=0}^{\infty} n C_n x^{n-1}$$

Substitution of y and y^1 in the given equation

$$\sum_{n=0}^{\infty} n C_n x^{n-1} = \sum_{n=0}^{\infty} C_n x^2$$

$$\text{or } c_1 x^0 + 2c_2 x + 3c_3 x^2 + \dots = c_0 x^0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots \quad (3)$$

Equating coefficient of x^r on both sides of (3)

$$(r+1) c_{r+1} = c_r \dots \quad (4)$$

Equation (4) is called the recurrence relation between the coefficients. From (4)

Let

$$r = 0, c_1 = c_0$$

$$r = 1, 2c_2 = c_1 \text{ or } c_2 = \frac{c_1}{2} = \frac{c_0}{2}$$

$$r = 2, 3c_3 = c_2 \text{ or } c_3 = \frac{c_2}{3} = \frac{c_0}{6}$$

$$r = 3, 4c_4 = c_3 \text{ or } c_4 = \frac{c_3}{4} = \frac{c_0}{24}$$

$$\text{Now } y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots$$

$$\text{Then } y = c_0 + c_0 x + \frac{c_0}{2} x^2 + \frac{c_0}{6} x^3 + \frac{c_0}{24} x^4 + \dots$$

$$y = c_0 \left[1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots \right]$$

$$\text{or } y = c \left[1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \right]$$

- b). Let us solve the same equation using method of separation of variables.

$$\text{Let } \frac{dy}{dx} = y$$

$$\text{Then } \frac{dy}{y} = dx$$

$$\int \frac{dy}{y} = \int dx$$

$$\ln Ay = x \text{ or } Ay = e^x$$

$$y = c e^x$$

$$y = c \left[1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \right]$$

- c). Hence the two solutions are equivalent Generally power series solution is obtained only when it is difficult or impossible to solve the given equation and find elementary functions as solution.

12.4 Power series solution of second order Equations

Method of obtaining the solution for the equation

$$a_0(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_2(x) y = 0 \quad (1)$$

Before we assume a power series solution for the equation (1) we should check for the singular points of the equation and confirm that the equation has a power series solution such as

$$y = \sum_{n=0}^{\infty} C_n x^n \text{ or } y = \sum_{n=0}^{\infty} C_n x^{n+r} \quad (2)$$

If the power series solution exists we assume the solution in any of the form in (2)

Next we find y' and y'' from (2) and substitute in the given equation (1).

Since the resulting equation is identically equal to zero we equate the coefficient of x^r to zero and establish the recurrence relation between the coefficients from which we find all the constants in term of c_0

Hence the expression for the solution y is obtained.

The method of solving the equation is illustrated in the following examples.

12.5 Power series solution at an ordinary point of the second order equation.

Example 2

a). Show that $x = 0$ is an ordinary point for the equation

$$(x^2 - 1) \frac{d^2 y}{dx^2} + 3x \frac{dy}{dx} + xy = 0$$

b). Find the power series solution for the equation given in (a)

Solution

The normalized form of the equation $(x^2 - 1) \frac{d^2 y}{dx^2} + 3x \frac{dy}{dx} + xy = 0$ (1)

$$\text{is } \frac{d^2 y}{dx^2} + \frac{3x}{x^2 - 1} \frac{dy}{dx} + \frac{x}{x^2 - 1} y = 0$$

the singular points of the equation are obtained by solving $x^2 - 1 = 0$ or $(x + 1)(x - 1) = 0$ and $x = -1$ and $x = 1$ are the only singular points of the equation and all other points are ordinary points. Then $x = 0$ is an ordinary point for the equation (1)

From the existence theorem given in (10.8) two linearly independent power series solutions at $x = 0$ exist.

They are in the form of

$$y = \sum_{n=0}^{\infty} c_n x^n. \quad c_0 \neq 0 \quad (2)$$

6). Let the power series solution of (1) be

$$y = \sum_{n=0}^{\infty} c_n x^n.$$
$$y' = \sum_{n=0}^{\infty} n c_n x^{n-1} \quad (3)$$

$$y'' = \sum_{n=0}^{\infty} n(n-1) c_n x^{n-2} \quad (4)$$

Substituting the series (2), (3) and (4) into the equation

$$(x^2 - 1)y'' + 3xy' + xy = 0, \quad \text{we have}$$

$$(x^2 - 1) \sum_{n=0}^{\infty} n(n-1)c_n x^{n-2} + 3x \sum_{n=0}^{\infty} n c_n x^{n-1} + x \sum_{n=0}^{\infty} c_n x^n = 0$$

$$x^2 \sum_{n=0}^{\infty} n(n-1)c_n x^{n-2} - 1 \sum_{n=0}^{\infty} n(n-1)c_n x^{n-2} + 3x \sum_{n=0}^{\infty} n c_n x^{n-1} + x \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\text{or } \sum_{n=0}^{\infty} n(n-1)c_n x^n - \sum_{n=0}^{\infty} n(n-1)c_n x^{n-2} + \sum_{n=0}^{\infty} 3n c_n x^n + \sum_{n=0}^{\infty} c_n x^{n-1} = 0$$

$$\text{or } \sum_{n=0}^{\infty} \{n(n-1)c_n + 3n c_n\} x^n + \sum_{n=0}^{\infty} c_n x^{n-1} - \sum_{n=0}^{\infty} n(n-1)c_n x^{n-2} = 0$$

$$, \quad \text{or } \sum_{n=0}^{\infty} n(n-2)c_n x^n + \sum_{n=0}^{\infty} c_n x^{n+1} - \sum_{n=0}^{\infty} n(n-1)c_n x^{n-2} = 0 \quad (5)$$

In (5) there are three infinite series

The coefficient of x^r

In the first series is $r(r+2)c_r$ when $n = r$,

In the second series is c_{r-1} when $n = r - 1$

In the third series is $-(r+2)(r+1)c_{r+2}$ when $n = r + 2$

Thus the coefficient of x^r in the left side of (5) is

$$r(r+2)c_r + c_{r-1} - (r+2)(r+1)c_{r+2} = 0 \quad (6)$$

Since the equation is identically zero.

From the equation (6)

$$c_{r-2} = \frac{r(r+2)c_r + c_{r-1}}{(r+1)(r+2)} \quad (7)$$

equation (7) is the recurrence relation between the coefficients.

$$\text{Let } r = 0 \text{ in (7)} \quad c_2 = \frac{0 + c_{-1}}{2} = 0$$

Let $r = 1$ in (7) $c_3 = \frac{3c_1 + c_0}{6} =$

Let $r = 2$ in (7) $c_4 = \frac{8c_2 + c_1}{12} = \frac{c_1}{12}$ since $c_2 = 0$.

Let $r = 3$ in (7) $c_5 = \frac{15c_3 + c_2}{20} = \frac{15(3c_1 + c_0)}{120} = \frac{3c_1}{8} + \frac{c_0}{8}$

Substituting the values of $C_0, C_1, C_2, C_3, C_4, C_5$ in (1) we have

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + \dots$$

$$y = c_0 + c_1x + 0 + \left(\frac{1}{2}c_1 + \frac{c_0}{6}\right)x^3 + \frac{c_1}{12}x^4 + \left(\frac{3c_1}{8} + \frac{c_0}{8}\right)x^5 + \dots$$

$$y = c_0 \left(1 + \frac{1}{6}x^3 + \frac{1}{8}x^5 + \dots\right) + c_1 \left(x + \frac{1}{2}x^3 + \frac{1}{12}x^4 + \frac{3}{8}x^5 + \dots\right) \quad (8)$$

equation (8) gives the general power series solution of the given equation. The solution has two independent constants C_0 and C_1

12.6 Power series solution at a regular singular point.

Example 3

- a. Show that the equation

$$2x^2y'' - xy' + (1 - x^2)y = 0 \quad (1)$$

has a regular singular point at $x = 0$

- b. What will be the nature of the power series solution of the equation (1)
c. Solve the equation (1) in ascending powers of x .

Solution

a. Let $2x^2y'' - xy' + (1 - x^2)y = 0$

Then $y'' - \frac{1}{2x}y' + \frac{1 - x^2}{2x^2}y = 0$

Hence $x = 0$ is a singular point.

This is of the form $y'' + p_1y' + p_2y = 0$

where $p_1 = -\frac{1}{2x}$ and $p_2 = \frac{1-x^2}{2x^2}$

$$xp_1(x) = x\left(\frac{-1}{2x}\right) = \frac{-1}{2} \text{ which is analytic}$$

$$xp_2(x) = x\left(\frac{1-x^2}{2x^2}\right) = \frac{1-x^2}{2x} \text{ which is also analytic}$$

Hence $x = 0$ is a regular singular point.

b. By existence the theorem there exists at least one non-trivial solution of the form

$$y = \sum_{n=0}^{\infty} c_n x^{n+r} \text{ where } c_0 \neq 0 \text{ and } r \text{ is a non-negative constant to be}$$

found.

c. To solve the equation

$$\text{Let } y = \sum_{n=0}^{\infty} c_n x^{n+r} \quad (2)$$

$$\text{Then } y' = \sum_{n=0}^{\infty} (n+r)c_n x^{n+r-1} \quad (3)$$

$$\text{and } y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1)c_n x^{n+r-2} \quad (4)$$

substituting these values of y , y' and y'' in $2x^2 y'' - x y' + (1-x^2)y = 0$, we have

$$2x^2 \sum_{n=0}^{\infty} (n+r)(n+r-1)c_n x^{n+r-2} - x \sum_{n=0}^{\infty} (n+r)c_n x^{n+r-1} + (1-x^2) \sum_{n=0}^{\infty} c_n x^{n+r} = 0$$

or

$$2 \sum_{n=0}^{\infty} (n+r)(n+r-1)x^{n+r} - \sum_{n=0}^{\infty} c_n (n+r)x^{n+r} + \sum_{n=0}^{\infty} c_n x^{n+r} - \sum_{n=0}^{\infty} c_n x^{n+r+2} = 0$$

Combining the first three series together

$$\sum_{n=0}^{\infty} [2(n+r)(n+r-1) - (n+r) + 1]c_n x^{n+r} = \sum_{n=0}^{\infty} c_n x^{n+r+2}$$

or

$$\sum_0^{\infty} [(2n + 2r)(n + r - 1) - (n + r - 1)]c_n x^{n+r} = \sum_0^{\infty} c_n x^{n+r+2}$$

$$\text{or } \sum_0^{\infty} (n + r - 1)(2n + 2r - 1)c_n x^{n+r} = \sum_0^{\infty} c_n x^{n+r+2} \quad (5)$$

This is an identity and the coefficient of x^r (the smallest power of x) on both sides should be equal.

In the left side the coefficient of x^r is obtained when $n = 0$ and it is $(r-1)(2r-1)$. In the right side the coefficient of x^r is zero since the lowest power of x is x^{r+2} .

$$\text{Hence } (r-1)(2r-1) = 0$$

This equation is called the **indical equation** in r .

From the indicial equation we get $r = 1$ or $r = \frac{1}{2}$.

To obtain the recurrence relation

Equating the coefficient of x^{n+r} on both sides of (5)

$$(n + r - 1)(2n + 2r - 1)c_n = c_{n-2}$$

$$\text{Hence } c_n = \frac{c_{n-2}}{(n + r - 1)(2n + 2r - 1)}$$

This is the recurrence relation between the constants $c_0, c_1, c_2, \dots$

$$\text{Let } n = 1, \text{ in (b) } c_1 = \frac{c_{-1}}{(n + r - 1)(2n + 2r - 1)} = 0 \text{ since } c_{-1} = 0$$

$$\text{Let } n = 2, c_2 = \frac{c_0}{(r+1)(2r+3)}$$

$$\text{Let } n = 3, c_3 = \frac{c_1}{(2+r)(5+2r)} = 0 \text{ since } c_1 = 0$$

$$\text{Let } n = 4, c_4 = \frac{c_2}{(r+3)(2r+5)} = \frac{c_0}{(r+1)(r+3)(2r+3)(2r+5)}$$

$$\text{Let } n = 5, c_5 = \frac{c_3}{(r+4)(9+2r)} = 0$$

Substitution of these values in

$$y = c_0 x^r + c_1 x^{r+1} + c_2 x^{r+2} + c_3 x^{r+3} + c_4 x^{r+4} + \dots$$

$$\begin{aligned}
&= x^r (c_0 + 0x + \frac{c_0 x^2}{(r+1)(2r+3)} + 0x^3 + \frac{c_0 x^4}{(r+1)(r+3)(2r+3)(2r+5)}) \\
&= c_0 x^r [1 + \frac{x^2}{(r+1)(2r+3)} + \frac{x^4}{(r+1)(r+3)(2r+3)(2r+5)} + \dots]
\end{aligned}$$

Putting $r = 1$ and $r = \frac{1}{2}$

$$y = Ax[1 + \frac{x^2}{2.4} + \frac{x^4}{2.4.5.9} + \dots]$$

$$\text{and } y = Bx^{\frac{1}{2}}[1 + \frac{x^2}{2.3} + \frac{x^4}{2.3.4.7} + \dots]$$

Hence the general solution is

$$y = Ax[1 + \frac{x^2}{2.5} + \frac{x^4}{2.4.5.9} + \dots] + y = Bx^{\frac{1}{2}}[1 + \frac{x^2}{2.3} + \frac{x^4}{2.3.4.7} + \dots]$$

where A and B are arbitrary constants.

12.7 Power series solutions of nonhomogeneous second order equations

Consider the nonhomogeneous second order equation

$$a_0(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_2(x)y = f(x)$$

we shall test the corresponding homogeneous part

$$a_0(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_2(x)y = 0$$

for singular points and apply the existence theorem for the power series solution. The method of solving the non-homogeneous equation is similar to the homogeneous equations. The method of finding power series solutions of (1) is illustrated in the following example.

Example 4

Solve the equation $y'' - y = x$ at $x = 0$. (1)

Solution

It is verified that $x = 0$ is an ordinary point of the equation. Let the power series solution of (1) be

$$y = \sum_0^{\infty} c_n x^n \quad (2)$$

Then

$$y' = \sum_0^{\infty} n c_n x^{n-1} \quad (3)$$

$$y'' = \sum_0^{\infty} n(n-1) c_n x^{n-2} \quad (4)$$

Substituting the values of y , y' and y'' in (1)

$$\sum_0^{\infty} n(n-1) c_n x^{n-2} - \sum_0^{\infty} c_n x^n = x$$

coefficient of x^r in the first series is obtained when $n^{n-2} = x^r$ or when $n = r + 2$. Then $c_{r+2}(r+2)(r+1)$ is the coefficient of x^r on the second series the term containing x^r is obtained using $x^n = x^r$ or when $n=r$. Then c_r is the coefficient of x^r . Thus the coefficient of x^r in the left side is $c_{r+2}(r+2)(r+1) - c_r$.

The coefficient of x^r on the right side is zero except when $r = 1$.

$$\text{When } r = 1, 6c_3 - c_1 = 1, c_3 = \frac{c_1 + 1}{6}$$

$$\text{Then } c_{r+2} = \frac{c_r}{(r+1)(r+2)} \text{ except when } r = 1.$$

$$\text{If } r = 0, c_2 = \frac{c_0}{2}$$

$$\text{If } r = 1, c_3 = \frac{c_1 + 1}{6}$$

$$\text{If } r = 2, c_4 = \frac{c_2}{12} = \frac{c_0}{24}$$

$$\text{If } r = 3, c_5 = \frac{c_3}{20} = \frac{c_1 + 1}{120}$$

If $r = 4, c_6 = \frac{c_4}{30} = \frac{c_0}{720}$

Substituting the values of $c_1, c_2, c_3, c_4, c_5 \dots c_6$ in (1)

$$y = c_0 + c_1 x + \frac{c_0}{2} x^2 + \frac{c_{1+1}}{3} x^3 + \frac{c_0}{24} x^4 + \frac{c_{1+1}}{120} x^5 + \frac{c_0}{720} x^6 + \dots$$

$$y = c_0 \left(1 + c_1 x + \frac{x^2}{2} + \frac{x^4}{4} + \frac{x^6}{720} + \dots \right)$$

$$+ c_1 \left(x + \frac{x^3}{6} + \frac{x^5}{120} + \dots \right)$$

Thus the general solution of the nonhomogeneous equation is obtained.

Exercise 12

1. Define the following:
 - a). Analytic function.

- b). Ordinary point and singular point of the equation $\frac{d^2 y}{dx^2} + p_1 \frac{dy}{dx} + p_2 y = 0$

- c). A regular singular point and an irregular singular point.
2. State three conditions necessary for the existence of the solution of the equation

$$\frac{dy}{dx} = f(x, y)$$

3. State the theorem for the existence of power series solution for the second order equation at

- i). An ordinary point of the equation.

- ii). A regular singular point of the equation.

- iii). An irregular singular point of the equation.

4. Solve $(1 - xy) y' - y = 0$ in power of x.

5. Solve $\frac{dy}{dx} = 2x^2 + 3y$ in power of x. Locate and classify the singular points of each of the differential equations.

6. $(x^2 - 3x) \frac{d^2 y}{dx^2} + (x + 2) \frac{dy}{dx} + y = 0$

7. $(x^4 - 2x^2 + x^2) \frac{d^2 y}{dx^2} + 2(x-1)(x+10) \frac{dy}{dx} + x^2 y = 0$
8. $3x^2 y^{11} - x y^1 + (x-6) y = 0$

Find power series solutions in powers of x of each of the differential equations in problems 9 to 12.

9. $\frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$
10. $\frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (2x^2 + 1)y = 0$
11. $\frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (3x + 2)y = 0$
12. $(x^3 + 1) \frac{d^2 y}{dx^2} + x^2 \frac{dy}{dx} + xy = 0$

Find the power series solutions of each of the initial value problems in 13 and 14

13. $\frac{d^2 y}{dx^2} - x \frac{dy}{dx} - y = 0 \quad y(0) = 1, y^1(0) = 0$
14. $(x^2 + 1) y^{11} + x y^1 + 2xy = 0 \quad y(0) = 2, y^1(0) = 3$

Find power series solutions in powers of (x - 1) of each of the differential equations in problems 15 and 16

15. $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$
16. $x \frac{d^2 y}{dx^2} + \frac{dy}{dx} + 2y = 0 \quad y(1) = 2, y^1(1) = 4$
17. Show that for Legendre's differential equation

$(1 - x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0$, $x = 0$ is an ordinary point where n is a non negative integer.

18. Find the power series solution of the equation. $3x \frac{d^2 y}{dx^2} - (x-2) \frac{dy}{dx} - 2y = 0$
near $x = 0$.

Summary

You have learnt the following from this lesson

Nature of power series solution near an ordinary point, a regular singular point and an irregular singular point.

Solving the first and the second order equation with constant or variable coefficients in a power series.

Further Reading

Differential Equations

By Shepley L. Ross

John Wiley and sons

New York, Toronto, Singapore

Advanced Differential Equations

By Dr. D. Sengottaiyan

Oxford Publications,

Harrow, London, Nairobi.

Lesson 13

Application of Differential Equations

13.1 Introduction

Differential Equations originate from the mathematical formulation of great variety of problems in science and engineering. In this lesson we consider problems that give rise to some of the types of **first and second order ordinary differential equations** and the **system of equations** studied in the previous lessons. First we formulate the problems mathematically and obtain a differential equation and then we solve the equation and interpret the solutions.

13.2 Objective of the lesson

By the end of this lesson you will be able to:

- formulate the differential equations from the given physical statements.
- solve the first and second order differential equations and system of equations formed.

13.3 Application of Differential Equation for Rate of Growth and Decay.

In many problems the rate at which a quantity changes is a known function of the amount present at the time and we must find the quantity at any particular time after the initial observation.

If x denotes the amount of the quantity present at time t , then $\frac{dx}{dt}$ denotes the rate at which the quantity changes. Suppose the amount of quantity grows or decays at a rate proportion to the amount present, we form the differential equation,

$$\frac{dx}{dt} = +kx \text{ if the amount of the quantity grows}$$
$$= -kx \text{ if the amount of the quantity decays.}$$

The following examples illustrate these types of problems.

Example 1

The rate at which radio active nuclei decay is proportional to the number of such nuclei that are present in a given sample. Half of the original number of radio – active nuclei **have undergone** disintegration in a period of 1000 years (**half- life**).

- a) What percentage of the original radioactive nuclei will remain after 3000 years?
- b) In how many years will only one–tenth of the original number remain?

Solution

Formulation of the problem

Let x be the amount (number) of radioactive nuclei present after t years. Then $\frac{dx}{dt}$ represents the rate at which the nuclei decay.

Since the nuclei decay at a rate proportional to the amount present we have

$$\frac{dx}{dt} = -kx \quad (1)$$

where k is positive (since x is decreasing we have kx is negative)

Using separation of variables,

$$\frac{dx}{x} = -kdt$$

Integrating both sides we have

$$\int \frac{dx}{x} = -k \int dt$$

$$\ln x + \ln c = -kt$$

$$\ln(cx) = -kt$$

$$cx = e^{-kt}$$

$$\text{or } x = Ae^{-kt} \quad (2)$$

Let x_0 be the amount present initially, (when $t = 0$)

It is given that $x = x_0$ when $t = 0$

Then $x_0 = A$. Hence (2) becomes $x = x_0 e^{-kt}$

using the initial condition that

$$x = \frac{x_0}{2} \text{ when } t = 1000 \text{ we have}$$

$$\frac{x_0}{2} = x_0 e^{-1000k}$$

$$\text{or } (e^{-k})^{1000} = 0.5$$

$$\text{or } e^{-k} = (0.5)^{\frac{1}{1000}} \quad (3)$$

substitution of (3) in (2) we have

$$x = x_0 (0.5)^{\frac{t}{1000}} \quad (4)$$

Equation (4) gives the amount of radioactive nuclei that are present at any particular time t in terms of the original number of nuclei present, that is x_0 , when $t = 0$.

a) To find the percentage of x_0 that remains after 3000 years.

we let $t = 3000$ in the equation (4)

$$\text{Then } x = x_0 (0.5)^{\frac{3000}{1000}} = x_0 (0.5)^3$$

$$= \frac{1}{8} x_0$$

Then after 3000 years $= \frac{1}{8} x_0$ will remain

or $= \frac{1}{8} x_0 \times 100\%$ will remain.

or 12.5 percent of the original amount will be present after 3000 years.

b) To find the number of years t when there is only 10 percent of x_0 will remain.

we let $x = \frac{10}{100} x_0$ in the solution (4) and find t .

$$\text{Thus } \frac{x_0}{10} = x_0 (0.5)^{\frac{t}{1000}}$$

$$\text{or } (0.5)^{\frac{t}{1000}} = \frac{1}{10}$$

$$\text{or } \frac{t}{1000} \ln\left(\frac{1}{2}\right) = \ln\left(\frac{1}{10}\right)$$

$$\text{or } t = \frac{1000 \ln\left(\frac{1}{10}\right)}{\ln\left(\frac{1}{2}\right)} = \frac{1000 \ln 10}{\ln 2} = 3330 \text{ years (Approximately)}$$

Hence after 3330 years the amount of radioactive nuclei will be 10% of the original amount.

Example 2

The population of a city increases at a rate proportional to the number of inhabitants at any time. If the population of the city is 30000 in 1990 and 35000 in 2000 what will be the population of the city in 2010?

Solution

Let x be the number of individuals in the city at time t . If we assume that the rate of change of the population is proportional to the number of individuals in the city at any time t we get the differential equation

$$\frac{dx}{dt} = kx \quad (1)$$

where k is a positive constant since the population is increasing

We have the initial condition that $t = t_0$ when $x = x_0$

From (1) we have

$$\int \frac{dx}{x} = \int k dt$$

$$\text{or } \ln x + \ln c = kt$$

$$\text{or } \ln(cx) = kt$$

$$\text{Then } cx = e^{kt}$$

$$\text{or } x = Ae^{kt} \quad (2)$$

Applying the initial condition that $x = x_0$ when $t = t_0$ we have

$$x_0 = Ae^{kt_0}$$

$$\text{Then } A = x_0 e^{-kt_0} \quad (3)$$

substitution of (3) in (2) gives

$$x = x_0 e^{-kt_0} \cdot e^{kt}$$

$$x = x_0 e^{k(t-t_0)} \quad (4)$$

The equation (4) is the required solution of the equation (1) satisfying the given conditions.

Example 3

The population x of a certain city satisfies the law $\frac{dx}{dt} = \frac{1}{100} x$

where time t is measured in years. Given that the population of this city is 200 000 in 1980,

- Find the equation connecting the population x and time t . (> 1980).
- Determine in what year does the population double.

Solution

- Let x be the population at time t after 1980. The differential equation governing the population x and time t in years is given by

$$\frac{dx}{dt} = \frac{1}{100} x \quad (1)$$

using variables separable method,

$$\frac{dx}{x} = \frac{dt}{100}$$

Integrating both sides

$$\ln x + \ln a = \frac{t}{100}$$

$$\ln ax = \frac{t}{100}$$

$$ax = e^{\frac{t}{100}}$$

$$x = A e^{0.01t}$$

When $t = 0$, $x = 200\,000$

Therefore $200\,000 = A$

$$\text{Therefore } x = 200\,000 e^{0.01t} \quad (2)$$

- It is required to find t when $x = 400\,000$
from the equation (2) we get

$$400\,000 = 200\,000 e^{0.01t}$$

$$e^{0.01t} = 2$$

$$0.01t = \ln 2$$

$$t = \frac{\ln 2}{0.01} = 69.3 \text{ years}$$

In 2049.3 the population will become double.

Example 4

The rate at which radioactive nuclei decay is proportional to the number of nuclei that are present in a given sample. Half of the original number of radioactive nuclei have undergone disintegration in a period of 50 years.

- a) What percentage of the original radioactive nuclei will remain after 150 years?
- b) In how many years will 75% of the original number remain?

Formulation of the problem

Let x be the amount of radioactive nuclei present after t years and $\frac{dx}{dt}$ represents the rate at which the nuclei decay. Since the nuclei decay at a rate proportional to the amount present, $\frac{dx}{dt}$ is proportional to x .

The amount x is positive and x is decreasing.
Hence the constant of proportionality is negative

$$\text{Therefore } \frac{dx}{dt} = -kx \quad (1)$$

Solving using the method of variables separable

$$\frac{dx}{x} = -kdt$$

$$\ln x + \ln a = -kt$$

$$\ln ax = -kt$$

$$\ln ax = e^{-kt}$$

$$ax = e^{-kt}$$

$$x = ce^{-kt}$$

Let m denote the amount initially present (when $t = 0$)

$$m = ce^0 = c$$

Hence the solution of the equation is

$$x = me^{-kt} \quad (2)$$

$$\text{It is further given that when } t = 50, \quad x = \frac{m}{2} \quad (3)$$

using (3) in (2)

$$\frac{m}{2} = m e^{-k(50)}$$

$$e^{-50k} = \frac{1}{2}$$

$$(e^{-k})^{50} = \frac{1}{2}$$

$$e^{-k} = \left(\frac{1}{2}\right)^{\frac{1}{50}}$$

Hence (2) becomes

$$x = m(e^{-k})^t$$

$$x = m \cdot \left(\frac{1}{2}\right)^{\frac{t}{50}} \quad (4)$$

This is the relation between x, m and t.

- a) If $t = 150$, we must find x
substitute this in (4)

$$\begin{aligned} x &= m \cdot \left(\frac{1}{2}\right)^{\frac{150}{50}} \\ &= m \left(\frac{1}{8}\right) \\ &= 0.125 m \end{aligned}$$

Thus 12.5% of the original number remain after 150 years.

- b) If $x = 75\%$ of m. we must find t from (4)

$$\begin{aligned} 0.75m &= m \left(\frac{1}{2}\right)^{\frac{t}{50}} \\ \left(\frac{1}{2}\right)^{\frac{t}{50}} &= 0.75 \\ \frac{t}{50} \ln 0.5 &= \ln 0.75 \\ t &= 50 \frac{\ln 0.75}{\ln 0.5} \\ &= 20.75 \text{ years.} \end{aligned}$$

Thus in 20.75 years 75% of the original number remains.

13.4 Mixture problems

Suppose a substance is allowed to flow into a container having a certain mixture at a certain rate and the mixture is kept uniform by stirring, sometimes this uniform mixture may simultaneous flow out of the container at a different rate.

Our aim is to determine the quantity of the substance m present in the mixture at any time t .

The basic equation to form here is

$$\frac{dx}{dt} = \text{Rate In} - \text{Rate out}$$

The following example will illustrate the problems of such kind

Example 5

A tank initially contains 200 litres of pure water. Starting at time $t = 0$ a brine containing 0.2 kg of salt dissolved per litre flows into the tank at the rate of 10 litres per minute. The mixture is kept uniform by stirring and the well stirred mixture simultaneously flows out of the tank at the same rate.

- How many kg of salt enters into the tank per minute?**
- How many kg of salt flows out of the tank per minute?**
- How much salt is in the tank at any time t ?**
- How much salt is present at the end of 25 minutes?**
- How much salt is present after a very long time?**

Formulation of the problem

Let x denote the amount of salt (in kg) in the tank at a general time $t > 0$. Then Rate of change of $x = \text{IN} - \text{OUT}$

$$\text{or } \frac{dx}{dt} = \text{IN} - \text{OUT}$$

- To find IN

The brine contains 0.2 kg per litre. It flows in at the rate of 3 litres per minute. Hence in 1 minute 3 litres $\times$ 0.2 kg = 0.6 kg of salt is going into the tank.

- To find OUT

At time t , x kg of salt is present in 50 litres of water.

Hence 1 litre of the brine has $\frac{x}{50}$ kg

3 litres of brine goes out.

That is $3 \frac{x}{50} = \frac{3x}{50}$ kg goes out per minute.

Hence $\frac{dx}{dt} = \text{IN} - \text{OUT}$ gives

$$\frac{dx}{dt} = 0.6 - \frac{3x}{50} \quad (1)$$

The initial condition is that

when $t = 0$, no of kg of sat in the tank is 0 kg

Then when $t = 0$, $x = 0$ or $x(0) = 0$

From the equation (1) we have

$$\frac{dx}{dt} = \frac{30 - 3x}{50}$$

$$\frac{dx}{dt} = \frac{30-3x}{50} = \frac{3(10-x)}{50}$$

or $\frac{dx}{10-x} = \frac{3dt}{50}$

Integrating both sides we have

$$\int \frac{dx}{10-x} = \int \frac{3dt}{50}$$

$$\text{or } -\ln(10-x) + \ln c = \frac{3}{50}t$$

$$\ln(10-x) - \ln c = -\frac{3}{50}t$$

$$\text{or } \frac{(10-x)}{c} = e^{-\frac{3}{50}t}$$

$$10-x = ce^{-\frac{3}{50}t}$$

$$\text{or } x = 10 - ce^{-\frac{3}{50}t} \quad (3)$$

using the initial condition that

when $t = 0$, $x = 0$ we have

$$0 = 10 - c \text{ or } c = 10$$

substituting $c = 10$ in (3) we have

$$x = 10 - 10e^{-\frac{3t}{50}}$$

$$\text{or } x = 10 \left(1 - e^{-\frac{3t}{50}} \right) \quad (4)$$

c) If $t = 25$, $x = 10 \left(1 - e^{-\frac{15}{50}} \right)$

13.5 Electric Circuit problems (First order equations)

In an electric circuit in series there are

- i. Inductance L, measured in henries
- ii. The resistance R, measured in Ohms
- iii. The current in the circuit I measured in amperes and
- iv. The electromotive force EMF, E measured in Volts

The basic differential equation in an electric circuit is given by Kirchhoff's Law.

$$L \frac{di}{dt} + Ri + \frac{1}{C}q = E \quad (1)$$

Let us consider R and L constants. The following is the Electric circuit.

we can solve the equation (1) which is a linear of first order in i (current) and t (time)

The following examples illustrate the method of solving such problems:

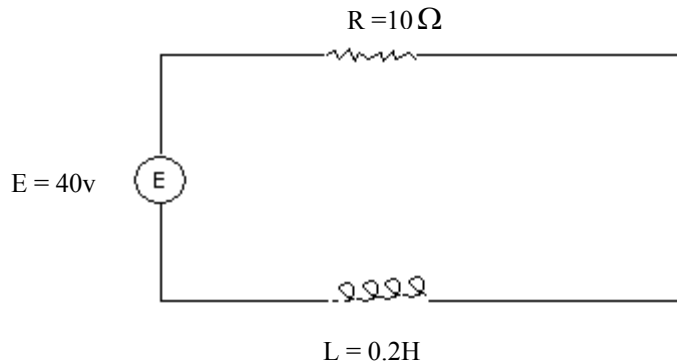
Example 6

A circuit has in series a constant electro motive force 40v, a resistor of 10Ω and an inductance 0.2 Henry the initial current is 0,

- Sketch the circuit.
- State the basic equation in electric circuit.
- By solving the basic equation find the current i on the circuit at time $t > 0$.
- What is the steady current after a long time?

Solution

- Circuit diagram showing the Electromotive force E , V from Battery, Resistor $R\Omega$ and the inductance L Henry



- The basic differential equation is

$$L \frac{di}{dt} + Ri = E \quad (1)$$

- Let $L = 0.2H$, $R = 10\Omega$ and $E = 40V$
Substituting these values in the equation (1) we have

$$0.2 \frac{di}{dt} + 10i = 40$$

$$\text{or } \frac{di}{dt} + 50i = 200 \text{ (Divide by 0.2)} \quad (2)$$

This is a linear equation of first order in i and t .

The integrating factor is $e^{\int 50 dt} = e^{50t}$

The solution of $\frac{dy}{dx} + py = \phi$ is

$$ye^{\int p dx} = \int \phi e^{\int p dx} dx \text{ (see lesson 5)}$$

Hence the solution of (2) is

$$ie^{50t} = \int 200e^{50t} dt$$

$$\text{or } ie^{50t} = 200 \frac{e^{50t}}{50} + c$$

$$ie^{50t} = 4e^{50t} + c$$

$$\text{Then } i = \frac{4e^{50t}}{e^{50t}} + \frac{c}{e^{50t}}$$

$$\text{or } i = 4 + ce^{-50t}$$

using the initial condition that $i = 0$ when $t = 0$, we have

$$0 = 4 + c \text{ and hence } c = -4.$$

putting $c = -4$ in 3 we have

$$i = 4 - 4e^{-50t}$$

$$\text{Hence } i = 4(1 - e^{-50t})$$

This equation gives the value of current at any time $t > 0$.

d) when $t \rightarrow \infty$ $e^{-50t} \rightarrow 0$

Hence $i = 4$ ampere, after a long time.

13.6 Application of Differential Equations of second order in electrical circuit.

If an electric circuit contains the capacitors c in addition to Electromotive force E , resistance R and inductance L we have the Basic equation

$$L \frac{di}{dt} + Ri + \frac{1}{c}q = E \quad (1)$$

since current I is the rate of change of coulomb q we have

$$i = \frac{dq}{dt} \quad (2)$$

substituting (2) in (1) we have

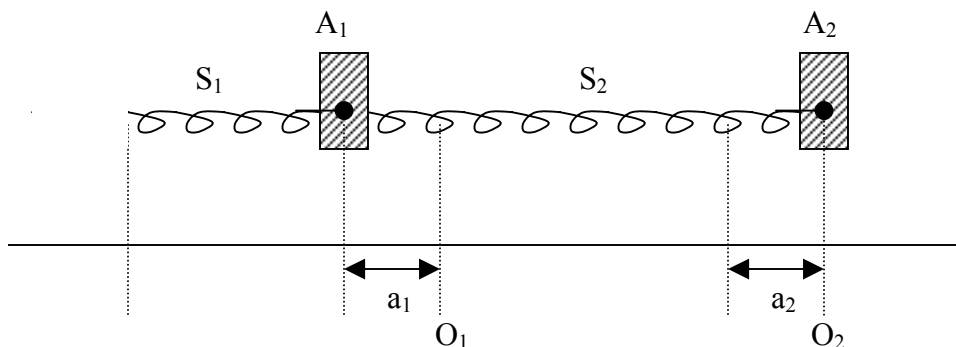
$$L \frac{d}{dt} \left(\frac{dq}{dt} \right) + R \left(\frac{dq}{dt} \right) + \frac{q}{c} = E$$

$$L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{c} = E \quad (3)$$

Equation (3) is a second order linear equation in the dependent variable q from which the charge q can be determined.

13.7 Application of system of equations on elastic springs

On a smooth horizontal table-top an object A_1 is connected to a fixed point P by a mass-less spring S_1 of natural length L_1 . An object A_2 is then connected to A_1 by a mass-less spring S_2 of natural length L_2 in such a way that the fixed point P and the centers of gravity A_1 and A_2 all lie in a straight line (see figure below)



The Object A_1 is then displaced a distance a_1 to the left of its equilibrium position O_1 the object A_2 is displaced a distance a_2 to the right of its equilibrium position O_2 and at time $t = 0$, the two objects are released. To find the position of the two objects at any further time.

Problems of this kind can be solved using the system of simultaneous situations as seen from the following examples.

Example 7

Two masses M_1 and M_2 suspended from two springs as shown in the figure below. The coefficients of stiffness of the springs are K_1 and K_2 respectively. The displacements of the masses from their position of equilibrium are x and y . It can be established that the following equations must hold.

$$M_2 \frac{d^2 y}{dt^2} = -k_2 (y - x)$$

$$M_1 \frac{d^2 x}{dt^2} = k_2 (y - x) - k_1 x$$

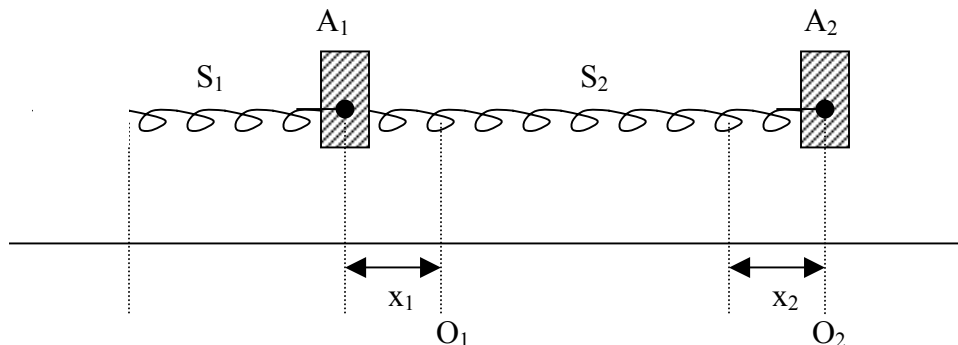
Formulation of the problem

Let us assume that no external forces act upon the system and the masses of the system A_1 and A_2 be m_1 and m_2 . Let the spring constant of S_1 and S_2 be K_1 and K_2 respectively.

Let x_1 denote the displacement of A_1 from its equilibrium position O_1 at time $t \geq 0$. Let x_1 is positive when A_1 is to the right of O_1 .

In the same way let x_2 denote the displacement of A_2 from its equilibrium position O_2 at time $t \geq 0$.

Let x_2 be the positive when A_2 is to the right of O_2



Apply Newton's

Consider the forces acting on A_1 at time $t \geq 0$. There are two forces, F_1 exerted by spring S_1 with magnitude $K_1(x_1)$ (by Hook's Law) and F_2 exerted by S_2 with magnitude

$$K_2(x_2 - x_1).$$

Now applying Newton's Second Law to the object A_1 we have,

$$m_1 \frac{d^2 x_1}{dt^2} = -k_1 x_1 + k_2 (x_2 - x_1)$$

in the same way the force F_3 exerted by spring S_2 is given by $F_3 = -k(x_2 - x_1)$ and by Newton's Second Law

$$m_2 \frac{d^2 x_2}{dt^2} = -k_2 (x_2 - x_1)$$

The initial conditions are

$$x_1(0) = a \quad \text{and} \quad x_1'(0) = 0; \quad x_2(0) = a_2 \quad \text{and} \quad x_2'(0) = 0$$

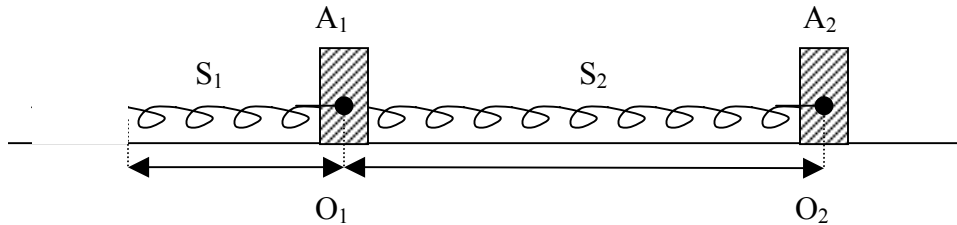
Thus the mathematical formulation of the problem consists of the differential equations

$$m_1 \frac{d^2 x_1}{dt^2} + (k_1 + k_2)x_1 - k_2 x_2 = 0$$

$$m_2 \frac{d^2 x_2}{dt^2} - k_2 x_1 + k_2 x_2 = 0$$

They form a system of homogeneous linear differential equations with constant coefficients.

Example 8



Suppose two masses A_1 and A_2 are connected by two springs S_1 and S_2 and placed on a smooth table as in the figure. Let A_1 and A_2 are displaced to the right with distances x_1 and x_2 respectively.

Let k_1 and k_2 be the spring constants of S_1 and S_2 respectively. Then the differential equations of the system at the time t is given by

$$m_1 \frac{d^2 x_1}{dt^2} + (k_1 + k_2)x_1 - k_2 x_2 = 0 \quad (1)$$

$$m_2 \frac{d^2 x_2}{dt^2} - k_2 x_1 + k_2 x_2 = 0 \quad (2)$$

If the masses of A_1 and A_2 are one unit each, and the spring constants $k_1 = 3$ and $k_2 = 2$, write down the system of two equations.

Taking the initial conditions

$$x_1(0) = -1, \quad x_1'(0) = 0, \quad x_2(0) = 2, \quad x_2'(0) = 0$$

find the displacements x_1 and x_2 at time $t > 0$

Solution

The system of equations (1) and (2) became

$$\frac{d^2 x_1}{dt^2} + 5x_1 - 2x_2 = 0$$

$$\frac{d^2 x_2}{dt^2} - 2x_1 + 2x_2 = 0$$

$$\text{or} \quad (D^2 + 5)x_1 - 2x_2 = 0 \quad (3)$$

$$-2x_1 + (D^2 + 2)x_2 = 0 \quad (4)$$

Applying the operator $(D^2 + 2)$ to (3) and multiplying the equation (4) by 2 and adding we get

$$[(D^2 + 2)(D^2 + 5) - 4]x_1 = 0$$

$$\text{or} \quad (D^4 + 7D^2 + 6)x_1 = 0$$

The auxiliary equation of (5) is

$$(m^2 + 6)(m^2 + 1) = 0$$

$$m = \pm\sqrt{-6} \quad \text{or} \quad m = \pm\sqrt{-1}$$

Hence the general solution is

$$x_1 = c_1 \sin t + c_2 \cos t + c_3 \sin \sqrt{6}t + c_4 \cos \sqrt{6}t \quad (5)$$

In the same way multiplying equation (3) by 2, applying the operator $(D^2 + 5)$ to the equation (4) and adding we obtain $(D^4 + 7D^2 + 6)x_2 = 0$ whose general solution is

$$x_2 = k_1 \sin t + k_2 \cos t + k_3 \sin \sqrt{6}t + k_4 \cos \sqrt{6}t \quad (6)$$

Here there are 8 constants but the determinant of the operator coefficient

$$\begin{vmatrix} D^2 + 5 & -2 \\ -2 & D^2 + 2 \end{vmatrix} = D^4 + 7D^2 + 6$$

is of order 4 since D^4 is the highest power of D . There must be certain relationships between the C 's and K 's.

Substituting the values of x_1 and x_2 in the equations (5) and (6) in the given equations we have

$$k_1 = 2c_1, \quad k_2 = 2c_2, \quad k_3 = -\frac{1}{2}c_3 \quad \text{and} \quad k_4 = -\frac{1}{2}c_4$$

Hence we have

$$x_1 = c_1 \sin t + c_2 \cos t + c_3 \sin \sqrt{6}t + c_4 \cos \sqrt{6}t$$

$$x_2 = 2c_1 \sin t + 2c_2 \cos t - \frac{1}{2}c_3 \sin \sqrt{6}t - \frac{1}{2}c_4 \cos \sqrt{6}t$$

If we apply the conditions

$$x_1 = \frac{3}{5} \cos t - \frac{8}{5} \cos \sqrt{6}t$$

$$x_2 = \frac{5}{5} \cos t + \frac{4}{5} \cos \sqrt{6}t$$

$$x_1 = -1, \quad \frac{dx_1}{dt} = 0$$

$$\text{we have } \left. \begin{array}{l} c_2 + c_4 = -1 \\ c_1 + \sqrt{6} c_3 = 0 \end{array} \right\} \quad (7)$$

if we apply $x_2 = 2$ and $\frac{dx_2}{dt} = 0$ we have

$$\left. \begin{array}{l} 2c_2 - \frac{1}{2}c_4 = 2 \\ 2c_1 - \frac{\sqrt{6}}{2}c_3 = 0 \end{array} \right\} \quad (8)$$

From (7) and (8) we get

$$c_1 = 0, \quad c_2 = \frac{3}{5}, \quad c_3 = 0, \quad c_4 = -\frac{8}{5}$$

Thus the particular solution of this problem is

$$x_1 = \frac{3}{5} \cos t - \frac{8}{5} \cos \sqrt{6}t$$

$$x_2 = \frac{5}{5} \cos t + \frac{4}{5} \cos \sqrt{6}t$$

Note that these system of equations can also be solved using matrix method shown in the previous lesson.

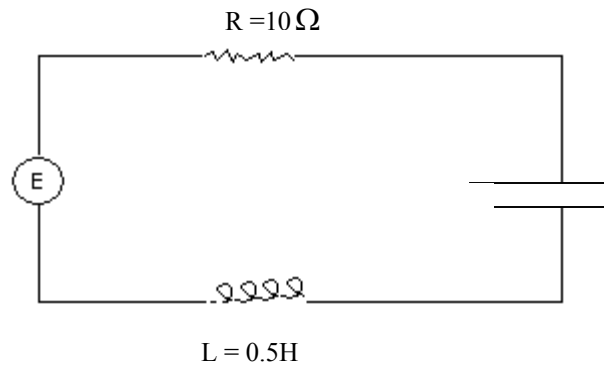
Example 9

A circuit has in series an electromotive force given by $E = 100 \sin 40t$, a resistor of 10Ω and an inductor of 0.5 Henry. If the initial current is zero

- i. Draw the circuit
- ii. Write down the equation
- iii. Solve the equation and find the current i at time $t > 0$.

Solution

- i. The circuit diagram



Volt = Current x Resistance.
Or $E = iR$ gives $E = 10i$

$$E = L \frac{di}{dt} = 0.5 \frac{di}{dt}$$

$$L \frac{di}{dt} + Ri + \frac{1}{C}q = E \quad (\text{from equation 1})$$

$$\text{Hence } \frac{1}{2} \frac{di}{dt} + 10i + 0 = 100 \sin 40t$$

$$\text{or } \frac{di}{dt} + 20i = 200 \sin 40t \quad (2)$$

The initial condition is $i = 0$ when $t = 0$. Equation (2) is of the form

$$\frac{dy}{dx} + Py = Q \quad (\text{first order linear equation})$$

whose solution is $ye^{\int P dx} = \int Qe^{\int P dx}$

Hence the solution of the equation (2) is

$$ie^{\int 20 dt} = \int 200 \sin 40t \cdot e^{20t} dt$$

$$ie^{20t} = 200 \sin 40t \int t \cdot e^{20t} dt$$

Integrating and simplifying we have

$$i = 2(\sin 40t - 2 \cos 40t) + Ce^{-20t}$$

Applying the condition that $i = 0$ when $t = 0$ we have

$$i = 2\sqrt{5} \sin(40t + \phi) + 4e^{-20t}$$

where ϕ is determined from

$$\phi = \cos^{-1} \frac{1}{\sqrt{5}} = 1.11 \text{ radian}$$

Hence the current at time t is given by

$$i = 4.47 \sin(40t - 1.11) + 4e^{-20t}$$

If the initial current is zero and the initial charge on the capacitor is 0.01, find the current at any time $t > 0$, assuming the differential equation

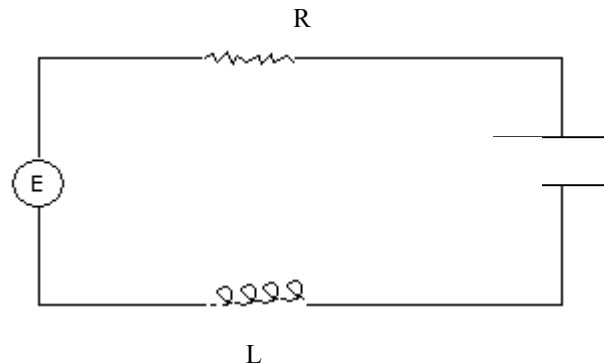
$$L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{1}{C} i = \frac{dE}{dt}$$

Exercise 13

1. The population of a country doubles in 50 years. Assume that the rate of increase is proportional to the number of inhabitants.
 - a. State this information in a differential equation, if x denotes the population at time t .
 - b. Solve the differential equation using that the constant of proportionality is k
 - c. Show that $x = Ce^{kt}$
 - d. At time $t = 0$, $x = x_0$. show that $x = x_0 e^{kt}$
 - e. Show that in 79 years later the population will become three times the original population
2. The population of a town doubles in 30 years. In how many years will it treble under the assumption that the rate of increase is proportional to the number of inhabitants?
3. In a certain culture of yeast the amount of active ferment grows at a rate proportional to the amount present. If the amount doubles in 1 hour, how many times the original amount may be anticipated at the end of $2\frac{3}{4}$ hours.
4. In a certain culture of Bacteria the rate of increase is proportional to the number present.
 - a. It is found that the number doubles in 2 hours. How many bacteria may expected at the end of 12 hours?
5. The human population of a certain island satisfies the logistic law $\frac{dx}{dt} = kx - \lambda x^2$ where k and λ are given by $k = 0.03$ and $\lambda = 3 \times 10^{-8}$
 - a. If the population in 1980 is 200,000 find a formula for the population in future years.
 - b. What will be the population expected in the year 2010?
 - c. What is the limiting value of the population as $t \rightarrow \infty$?
6. A radio active element decomposes or decays at the rate proportional to the amount present. If half the original amount disappears in 1600 years find the percentage lost in 100 years.
7. A large tank initially contains 50 litres of brine in which there is dissolved 1.2 kg of salt. Brine containing 0.2 kg of dissolved salt per litre flows into the tank at the rate of 5 litres per minute. The mixture is kept uniform by stirring and the stirred mixture simultaneously flows out at the slower rate of 3 litres per minute. Let x be the amount of salt in the tank at time $t > 0$. How much salt is in the tank at any time $t > 0$
8. A circuit has in series a constant electromotive force of 40v, a resistor of 10Ω and an inductor of 0.2 Henry. If the initial current is zero, find the current at time $t > 0$.

9. A circuit has in series a constant electromotive force of 100v, a resistor of $10\ \Omega$ and capacitor of 2×10^{-4} farads. The switch is closed at time $t = 0$ and the charge on the capacitor at this instant is zero. Find the charge and the current at time $t > 0$.

10.



In the figure given above

$$E = 100 \sin 200t \text{ volts}$$

$$R = 40\ \Omega$$

$$L = 0.25 \text{ Henry}$$

$$C = 4 \times 10^{-4} \text{ farad.}$$

If the initial current is zero and the initial charge on the capacitor is 0.01 coulomb

find the current at any time $t > 0$. (Assume the equation $L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{1}{C} i = \frac{dE}{dt}$)

11. A circuit has in series an electromotive force given by $E = 100 \sin 60t$ volts, a resistor of $2\ \Omega$, an inductor of 0.1 H and a capacitor of $\frac{1}{260}$ farads. If the initial current and the initial charge on the capacitor are both zero, find the charge on the capacitor at any $t > 0$.

(Hint: Using the formula, $L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E$ we have

$$\frac{d^2 q}{dt^2} + 20 \frac{dq}{dt} + 2600q = 1000 \sin 60t \text{ the complementary function is } q = e^{-10t}$$

$(c_1 \sin 50t + c_2 \cos 50t)$. The particular integral is $q = A \sin 60t + B \cos 60t$. Apply the condition to the general solution to get

$$q = 0.77e^{-10t} \cos(50t - 0.88) - 0.64 \cos(60t - 0.69)$$

12. A circuit has in series a constant electromotive force 100v, a resistor of $10\ \Omega$ and a capacitor of 2×10^{-4} farads. The switch is closed at time $t = 0$ and the charge on the capacitor at this instant is zero.

- sketch the circuit diagram
- write down the basic equation containing i.
- derive the second order equation in q.
- solve the equation to find q at time $t > 0$
- Hence find i at time $t > 0$ using $i = \frac{dq}{dt}$

Summary

We have learnt the following from this lesson.

- i). Application of differential equations for
- ii). First order equations for growth of population and decay of radio active materials.
- iii). For mixture problems of liquids.
- iv). Second order equations for electrical circuits.

Further Reading

7. Differential Equations

By Frank Ayres, Ph. D

Schaum's outlines series

Mc Graw Hill Book company

New York, Toronto, Sydney.

8. Differential Equations

By Shepley L. Ross

John Wiley and sons

New York. Toronto. Singapore

9. Differential Equations

By Dr. Sengottaiyan Ph. D

Oxford Publications,

Harrow. London. Nairobi.



Microsoft Equation

3.0